Numerical Analysis of the Aerodynamic Characteristics of NACA-4312 Airfoil

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1 Introduction

Aerodynamics is a critical field of study concerning the motion of air and its effects on bodies, which is fundamental to the design of aircraft, wind turbines, and other vehicles[1]. A key component in this study is the airfoil, the cross-sectional shape of a wing, which is primarily responsible for generating the aerodynamic forces required for flight.

In the early 20th century, the National Advisory Committee for Aeronautics (NACA) began a systematic investigation that led to the development of families of standardized airfoils that are still in common use today. While extensive research and data have been accumulated for common profiles such as the NACA-0012 and NACA-4412, a substantial numerical database is not available for the NACA-4312 airfoil. This study is therefore motivated by the need to complement the presently limited body of knowledge for this specific airfoil profile.

This project aims to investigate the aerodynamic characteristics of a NACA-4312 airfoil using a computational approach. The primary objectives are to determine the lift coefficient (C_L and drag coefficient (C_D at various angles of attack ($\propto$). This analysis is conducted using Computational Fluid Dynamics (CFD), a powerful tool that is widely used in aerodynamics because it can be more cost-effective and yield more accurate results than traditional experimental processes. The commercial software ANSYS Fluent is used to solve the Reynolds-Averaged Navier-Stokes (RANS) equations that govern the fluid flow.

The simulations are performed on a 2D model of the airfoil at a fixed Reynolds number (Re) 5×10^5. To account for the effects of turbulence, two common and distinct turbulence models are employed for analysis and validation: the Standard KE model and the SST kw model. The subsequent sections of this report will detail the simulation methodology, mesh generation, a grid independent study, and a final validation of the results against the reference paper.

1.1 Governing equations used

The governing equations used for the numerical analysis of the aerodynamic characteristics of the NACA-4312 air foil are the Reynolds-Averaged Navier-Stokes (RANS) equations.

Since the flow is modelled as steady-state and incompressible, the RANS equations consist of the following forms:

- **Continuity equation**

$$\frac{\partial \rho}{\partial t} + div(\rho \boldsymbol{u}) = 0$$

- **Navier stokes equations**

$$\frac{\partial(\rho u)}{\partial t} + div(\rho u \boldsymbol{u}) = -\frac{\partial P}{\partial x} + div(\mu\ grad\ u) + S_{M_x}$$

$$\frac{\partial(\rho v)}{\partial t} + div(\rho v \boldsymbol{u}) = -\frac{\partial P}{\partial y} + div(\mu\ grad\ v) + S_{M_y}$$

$$\frac{\partial(\rho w)}{\partial t} + div(\rho w \boldsymbol{u}) = -\frac{\partial P}{\partial z} + div(\mu\ grad\ w) + S_{M_z}$$

- **Reynolds average continuity equation**

$$\frac{\partial \bar{\rho}}{\partial t} + div(\bar{\rho}\widetilde{\boldsymbol{U}}) = 0$$

- **Reynolds average Navier stokes equation**

$$\frac{\partial(\bar{\rho}\tilde{U})}{\partial t} + div\left(\bar{\rho}\tilde{U}\widetilde{\boldsymbol{U}}\right) = -\frac{\partial \tilde{P}}{\partial x} + div\left(\mu\ grad\ \tilde{U}\right) + \left[-\frac{\partial\left(\overline{\bar{\rho}u'^2}\right)}{\partial x} - \frac{\partial(\overline{\bar{\rho}u'v'})}{\partial y} - \frac{\partial(\overline{\bar{\rho}u'w'})}{\partial z}\right] + S_{M_x}$$

$$\frac{\partial(\bar{\rho}\tilde{V})}{\partial t} + div\left(\bar{\rho}\tilde{V}\widetilde{\boldsymbol{U}}\right) = -\frac{\partial \tilde{P}}{\partial y} + div\left(\mu\ grad\ \tilde{V}\right) + \left[-\frac{\partial(\overline{\bar{\rho}u'v'})}{\partial x} - \frac{\partial\left(\overline{\bar{\rho}v'^2}\right)}{\partial y} - \frac{\partial(\overline{\bar{\rho}v'w'})}{\partial z}\right] + S_{M_y}$$

$$\frac{\partial(\bar{\rho}W)}{\partial t} + div\left(\bar{\rho}\widetilde{W}\widetilde{\boldsymbol{U}}\right) = -\frac{\partial \tilde{P}}{\partial z} + div\left(\mu\ grad\ \widetilde{W}\right) + \left[-\frac{\partial(\overline{\bar{\rho}u'w'})}{\partial x} - \frac{\partial(\overline{\bar{\rho}v'w'})}{\partial y}\frac{\partial\left(\overline{\bar{\rho}w'^2}\right)}{\partial z}\right] + S_{M_z}$$

- **Standard K-E model follows below equations**

$$\frac{\partial(\rho k)}{\partial t} + div(\rho k\boldsymbol{U}) = div\left[\frac{\mu_t}{\sigma_k}\ grad\ k\right] + 2\mu_t S_{ij} \cdot S_{ij} - \rho\varepsilon$$

$$\frac{\partial(\rho\varepsilon)}{\partial t} + div(\rho\varepsilon\boldsymbol{U}) = div\left[\frac{\mu_t}{\sigma_\varepsilon}\ grad\ \varepsilon\right] + C_{1\varepsilon}\frac{\varepsilon}{k}2\mu_t S_{ij} \cdot S_{ij} - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k}$$

- **SST-KW model follows below equation**

$$\frac{\partial(\rho\omega)}{\partial t} + \mathrm{div}(\rho\omega\mathbf{U}) = \mathrm{div}\left[\left(\mu + \frac{\mu_t}{\sigma_{\omega,1}}\right)\mathrm{grad}(\omega)\right] + \gamma_2\left(2\rho S_{ij} . S_{ij} - \frac{2}{3}\rho\omega\frac{\partial U_i}{\partial x_j}\delta_{ij}\right) - \beta_2\rho\omega^2 + 2\frac{\rho}{\sigma_{\omega,2}\omega}\frac{\partial k}{\partial x_k}\frac{\partial \omega}{\partial x_k}$$

2 Assumptions and Simplifications

To establish a feasible computational model, several assumptions and simplifications were made, based on the reference study.

1. Nature of the Problem

The fluid flow is modelled as two-dimensional (2D), steady-state, and incompressible. The 2D simplification is valid as the analysis focuses on the air foil's cross-sectional performance, assuming a uniform flow along an infinite span. A steady-state approach was chosen because the objective is to determine the time-averaged aerodynamic forces at fixed angles of attack, rather than analysing transient flow phenomena. The flow is considered incompressible because the inlet velocity of 7.5 m/s results in a very low Mach number, where changes in air density due to velocity are negligible.

2. Fluid Properties

The working fluid is air, with constant properties as follows:

1. Density: 1.225 kg/m^3
2. Viscosity:1.7894$\times 10^{-5}$ kg/ms

Table 1

Property	Value
Density-constant	1.225 kg/m^3
Viscosity-constant	1.7894$\times 10^{-5}$ kg/ms
Temperature	288.16K
Pressure-Gauge	0 Pa

Specific heat ratio	1.4

Operating Conditions and Geometry

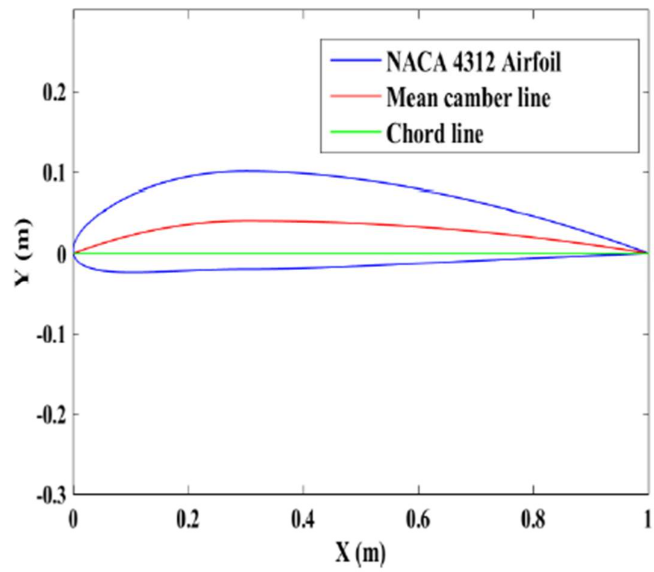

Figure 1 -NACA 4312 Aerofoil

All simulations were conducted at a fixed Reynolds Number (Re) of 5×10^5 for a range of angles of attack from 0° to 16°. This dimensionless number corresponds to the specified fluid properties and the geometry, which is a NACA-4312 airfoil with a chord length of 1000 mm. This Reynolds number dictates an inlet velocity of 7.5 m/s for the simulation

To solve the Reynolds-Averaged Navier-Stokes (RANS) equations for turbulent flow, two different and widely used models were employed for this analysis[44]:

- **Standard K-E model**: A robust two-equation model widely used for general turbulent flow simulations.

- **SST (Shear Stress Transport) KW model**: Another two-equation model known for its accuracy in predicting flows with adverse pressure gradients and separation, which are common in airfoil simulations.

The reason for using both models is to allow for a direct comparison of their performance and to validate the results by assessing the consistency between their predictions for this specific aerodynamic application

3 Flow Domain and Computational Mesh

3.1 Computational domain

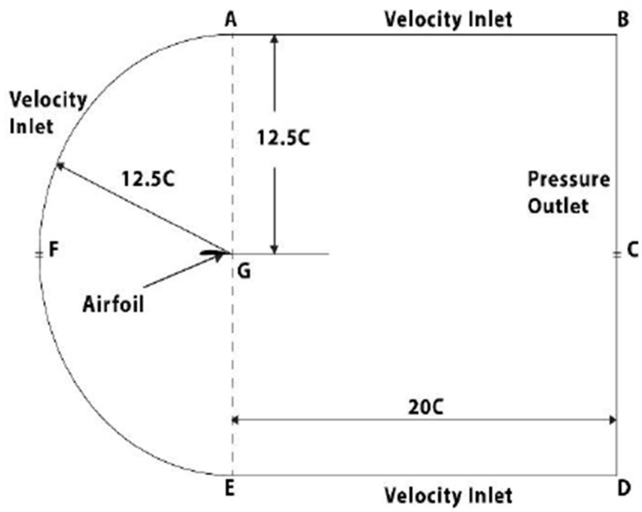

Figure 2 - Computational domain and boundary conditions on research paper

3.2 Boundaries and boundary conditions

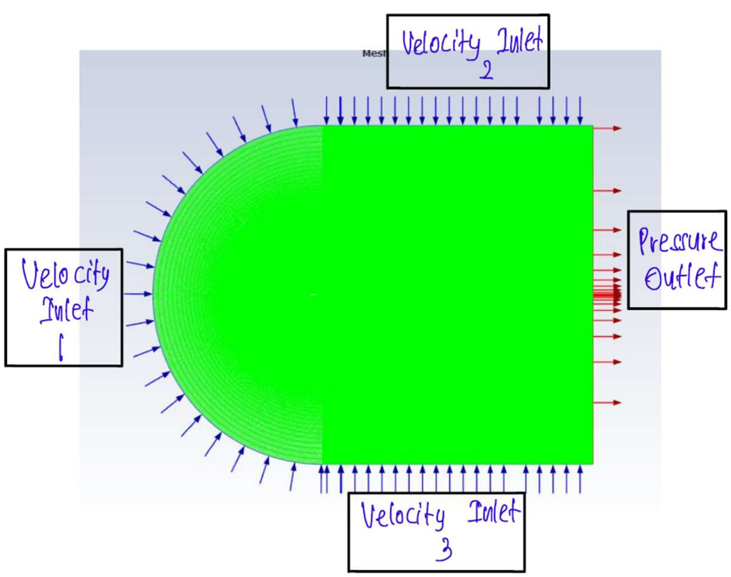

Figure 3 -Boundaries of the domain

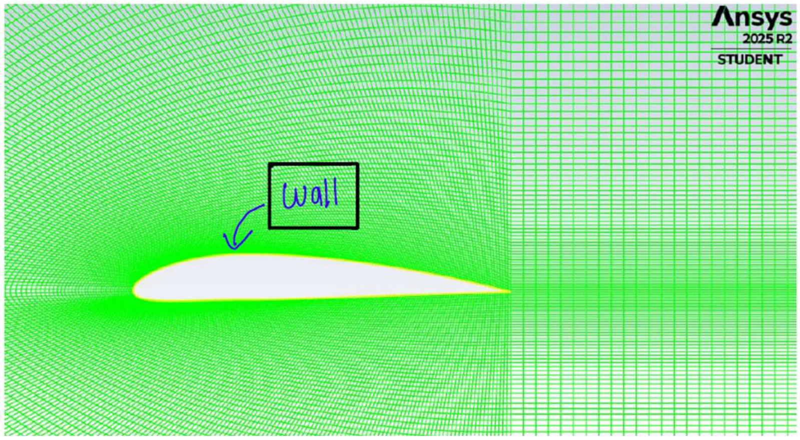

Figure 4 -Boundaries of the domain -2

3.3 Boundary condition values

Inlet: For the inlet, a Velocity Inlet was chosen as the boundary condition. This is used to define uniform, steady airflow entering the domain. The free-stream velocity magnitude was set to 7.5, a value taken from the referenced research paper to ensure valid comparison.

This In the ANSYS Fluent setup, this velocity vector was decomposed into its Cartesian components (V_x and V_y). This method allows the angle of attack ($\propto$) to be precisely.

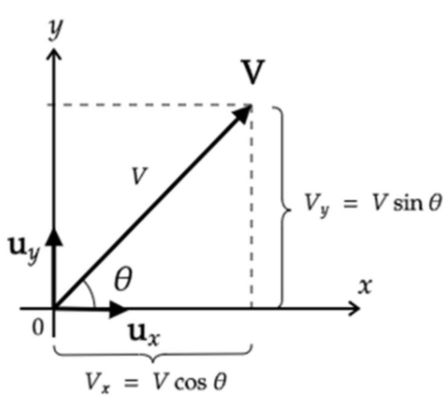

Figure 5-Velocity components

Table 2 Velocity components for various angle of attacks

AOA	Velocity_X	Velocity_Y
0	7.5	0
2	7.495431203	0.261746225
4	7.481730377	0.523173553
6	7.458914215	0.783963475
8	7.427010516	1.043798257
10	7.386058148	1.302361333
12	7.336107006	1.559337681
14	7.277217947	1.814414217
16	7.20946272	2.067280169

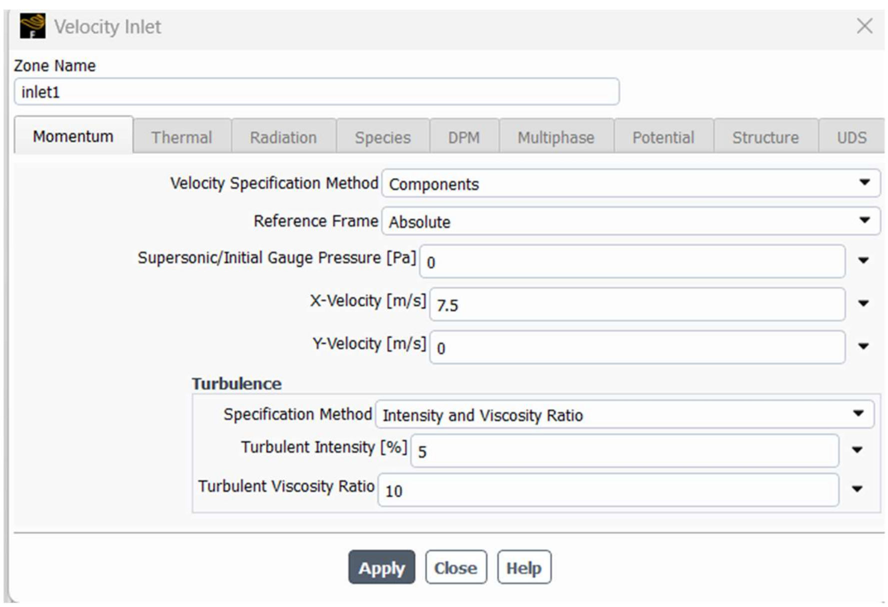

Figure 6 - Example Inlet 1 boundary values

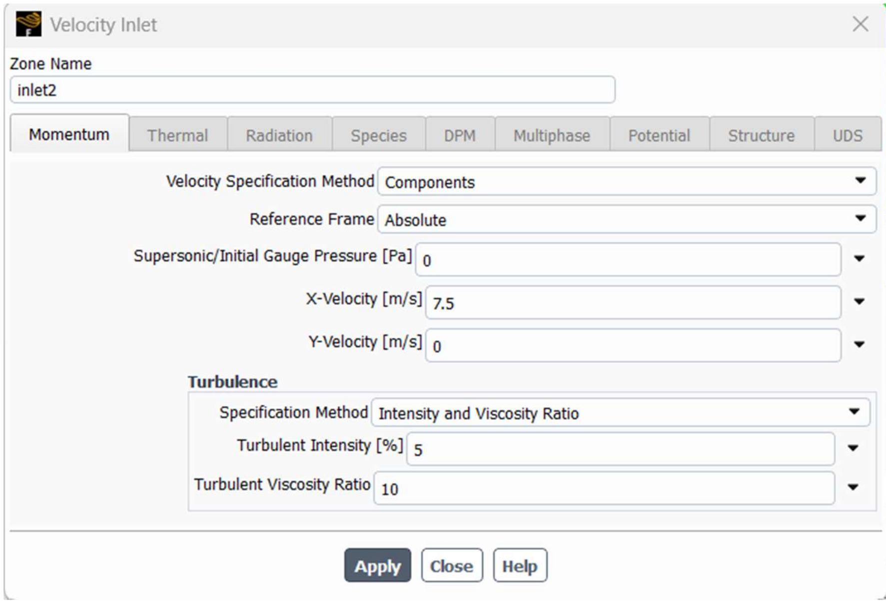

Figure 7 - Inlet 2 Boundary values

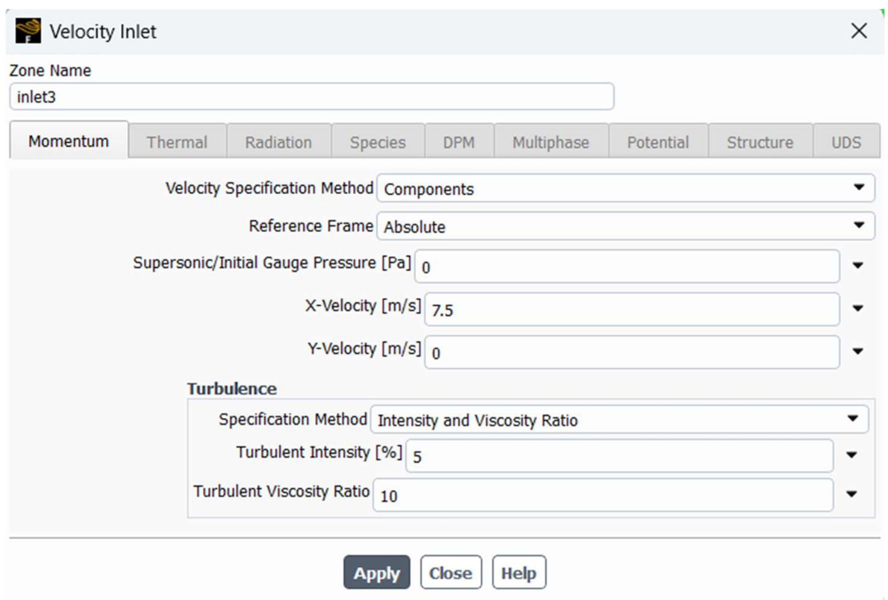

Figure 8- Inlet 3 -Boundary values

Outlet: For the outlet, pressure outlet boundary condition was used. It allows air to exit the domain smoothly and prevents reflection of disturbances back into the computational domain. The gauge pressure was set to 0 Pa

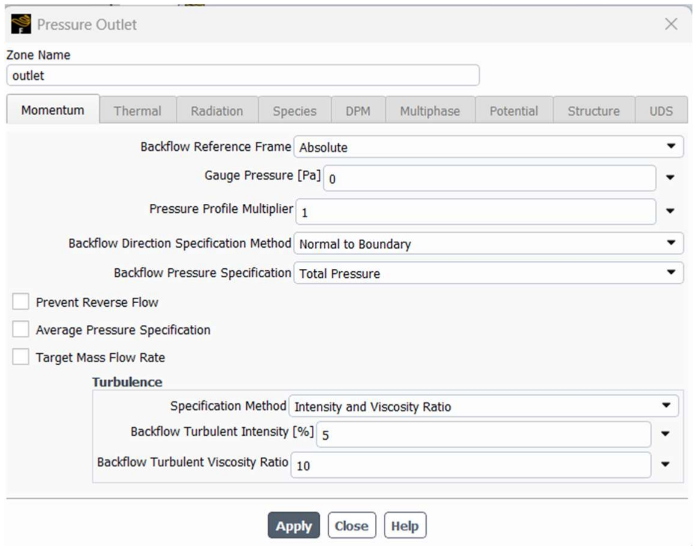

Figure 9- Pressure outlet- boundary values

Aerofoil: For the aerofoil, no slip wall condition was selected. This condition allow zero velocity at the aerofoil surface, capturing viscous shear stresses and boundary layer effects accurately

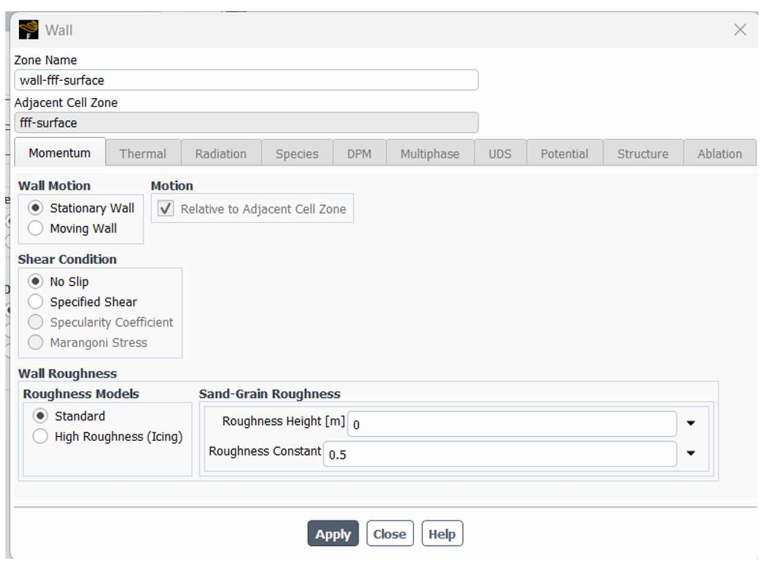

Figure 10-Aerofoil wall- boundary values

3.4 Mesh

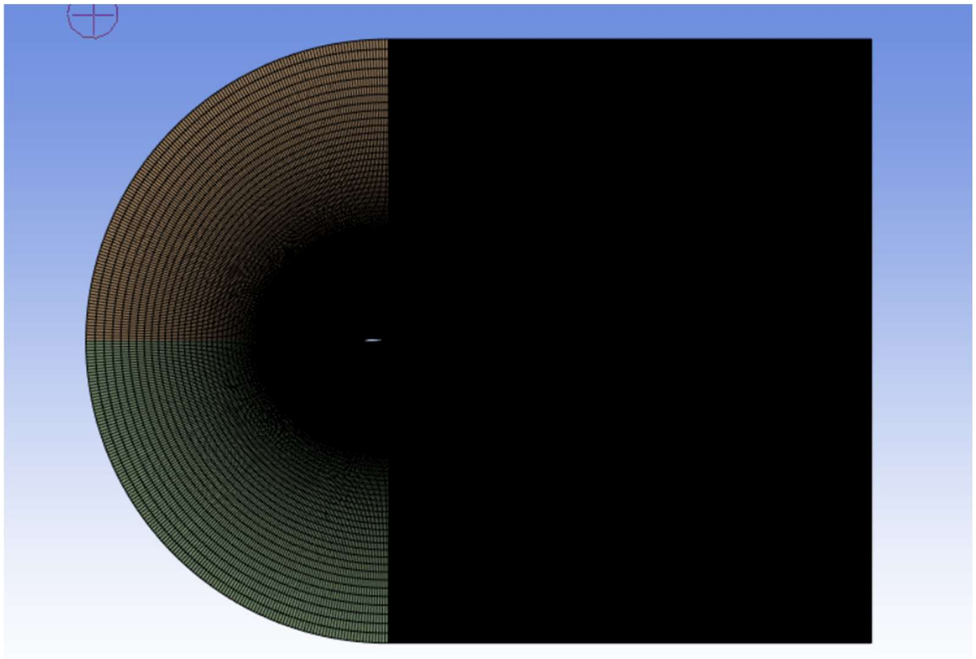

Figure 11 -Mesh

For this analysis, a C-type structured mesh was generated using ANSYS Meshing. This mesh topology is shaped like the letter 'C', wrapping around the air foil’s leading edge and extending far into the downstream wake region, as shown in above figure. This structure is particularly effective for air foil simulations as it allows for a high density of quadrilateral cells in the critical wake area and provides better convergence and control for the wall functions used at the air foil surface. The computational domain was extended 12.5C upstream and 20C downstream from the air foil to minimize the influence of far-field boundary conditions on the results.

3.4.1 Near-Wall Treatment and Y+ Value

Accurate modelling of the near-wall region is critical for capturing aerodynamic forces. This simulation employs a wall function approach — a computationally efficient method that models the flow within the boundary layer rather than fully resolving it. The validity of this approach depends on the placement of the first mesh cell relative to the wall, quantified by the non-dimensional distance $y+$. The Standard k–ε turbulence model requires the first cell to lie within the turbulent log-law region of the boundary layer, corresponding to $y+$ values greater than 30. Since the inflation layer method could not be applied with face meshing, a bias factor was used instead to control the mesh height at the air foil surface. This approach ensured that the resulting mesh-maintained $y+$ values between 30 and 60 for all simulations.

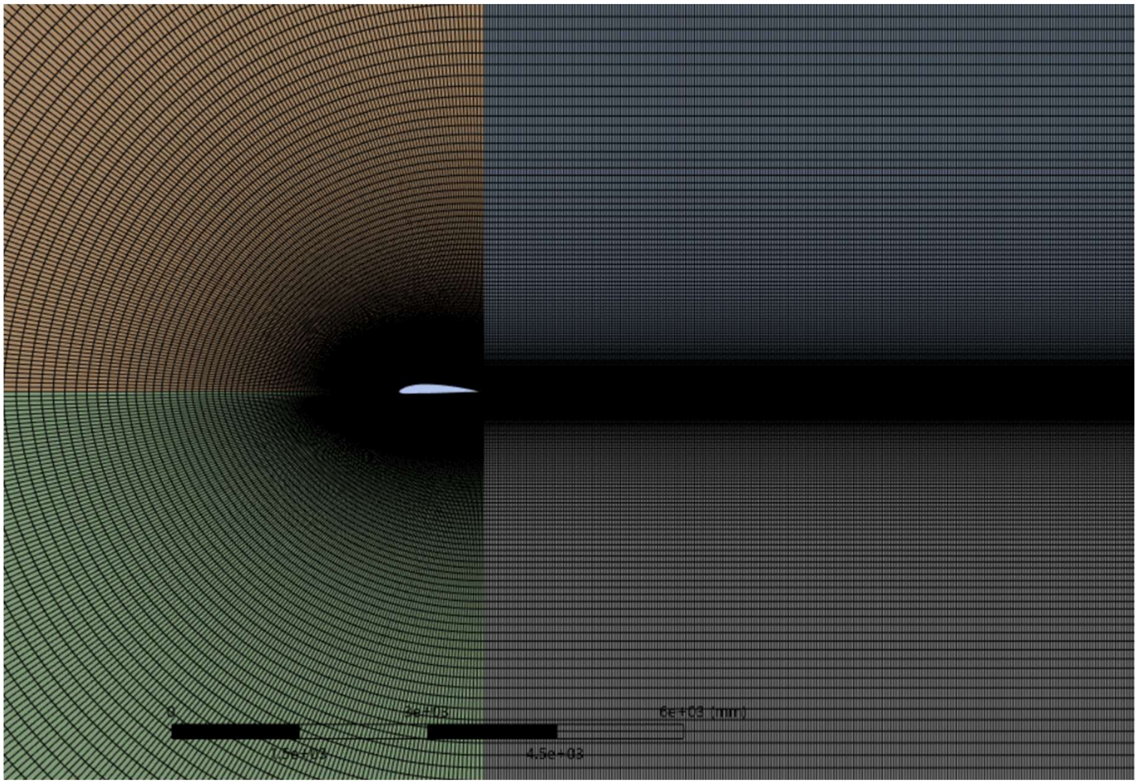

Figure 12-Mesh

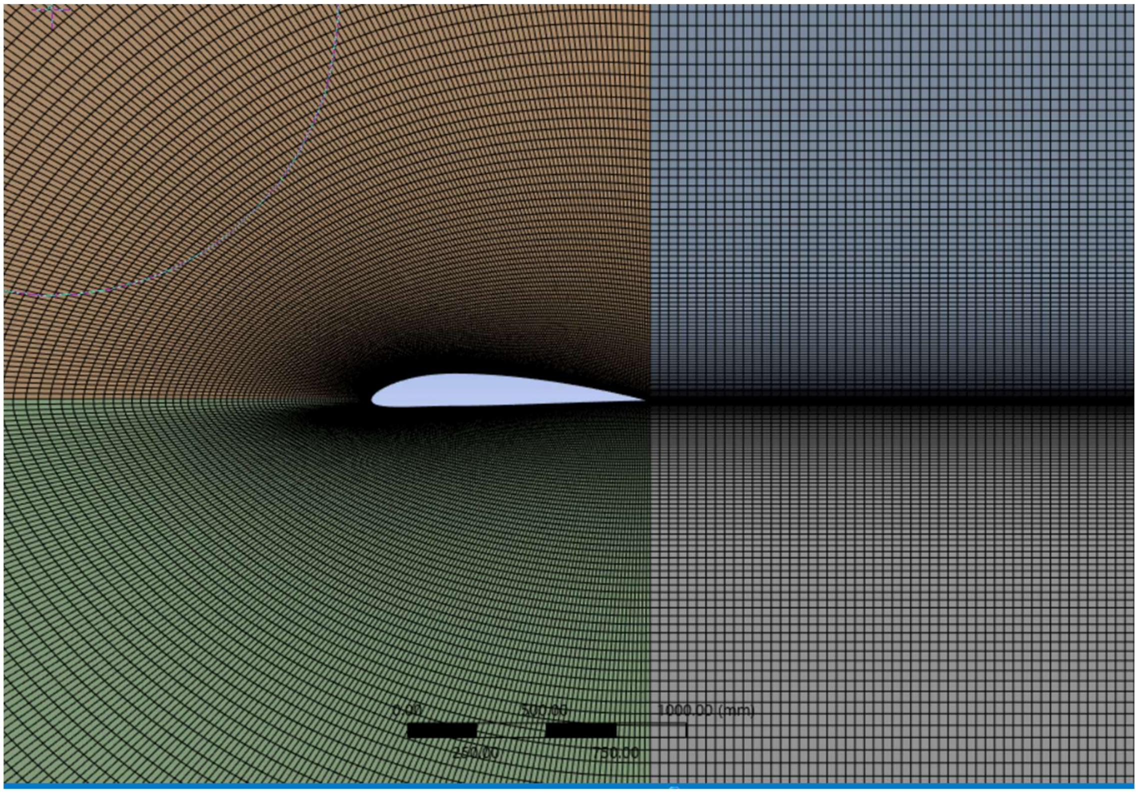

Figure 13-Near wall surface mesh

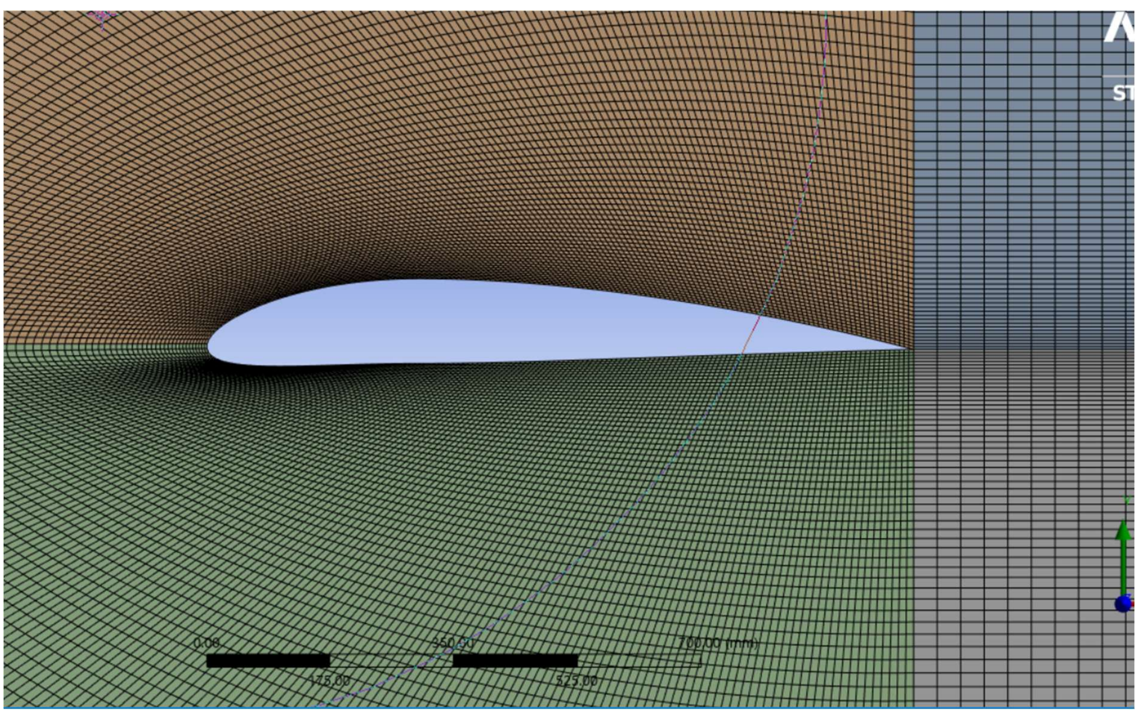

Figure 14-Near wall surface mesh

3.4.2 The "Apples-to-Apples" Comparison

When comparing the performance of two different turbulence models, the goal is to isolate the model itself as the only major variable. To achieve this, the same mesh was used for both the Standard *k–ε* and SST *k–ω* simulations, ensuring a fair and direct comparison.

The Standard *k–ε* model with wall functions requires a mesh where the first cell has a $y+$ value greater than 30. To create a fair comparison, this same high-$y+$ mesh (with $y+$ values between 30 and 60) was also used for the SST *k–ω* model. Using two different meshes (for example, a fine $y+ \approx 1$ mesh for SST and a coarse $y+ > 30$ mesh for *k–ε*) would introduce additional variables, making it unclear whether any differences in the results were due to the turbulence model or the mesh itself.

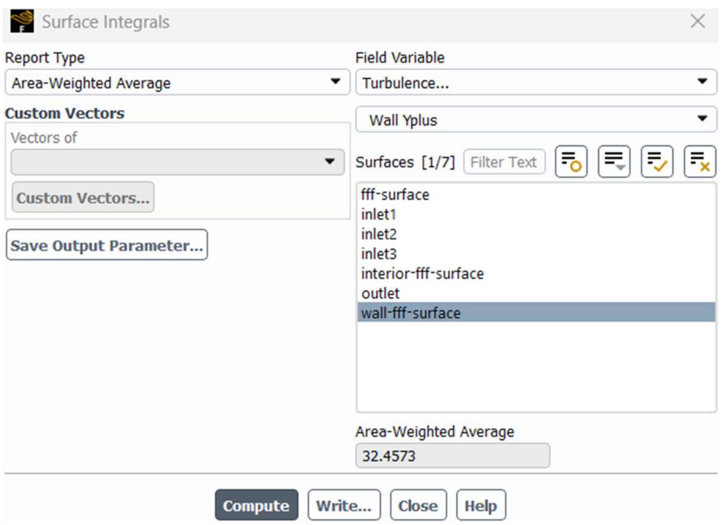

Figure 15- Wall Y+ Value

3.4.3 How the SST *k–ω* Model Adapts

This strategy is effective because the SST *k–ω* model is designed to be robust and flexible. Unlike the Standard *k–ε* model, the SST formulation incorporates blending functions that allow it to automatically adapt to different near-wall resolutions. When the first mesh cell is located far from the wall (i.e., has a high $y+$ value), the SST model transitions from its boundary-layer-resolving mode to a wall function approach.

Although the SST *k–ω* model typically performs best with $y+ \approx 1$, it can still produce valid and consistent results on coarser, high-$y+$ meshes. This adaptability makes it highly suitable for comparative aerodynamic studies where maintaining an identical mesh across different turbulence models is necessary

3.4.4 Mesh statistics

- Total number of nodes =211040
- Total number of elements = 210000

Statistics	
Nodes	211040
Elements	210000
Show Detailed Statistics	No

Figure 16-Mesh statistics

3.4.5 Orthogonal quality

Details of "Mesh"	
Physics Preference	CFD
Solver Preference	Fluent
Element Size	Default (2050.2 mm)
Export Format	Standard
Export Preview Surface Mesh	No
Sizing	
Quality	
Check Mesh Quality	Mesh Quality Worksheet
Target Skewness	Default (0.9)
Smoothing	Medium
Mesh Metric	Orthogonal Quality
Min	0.1825
Max	1.
Average	0.98167
Standard Deviation	7.5242e-002

Figure 17- Orthogonal quality

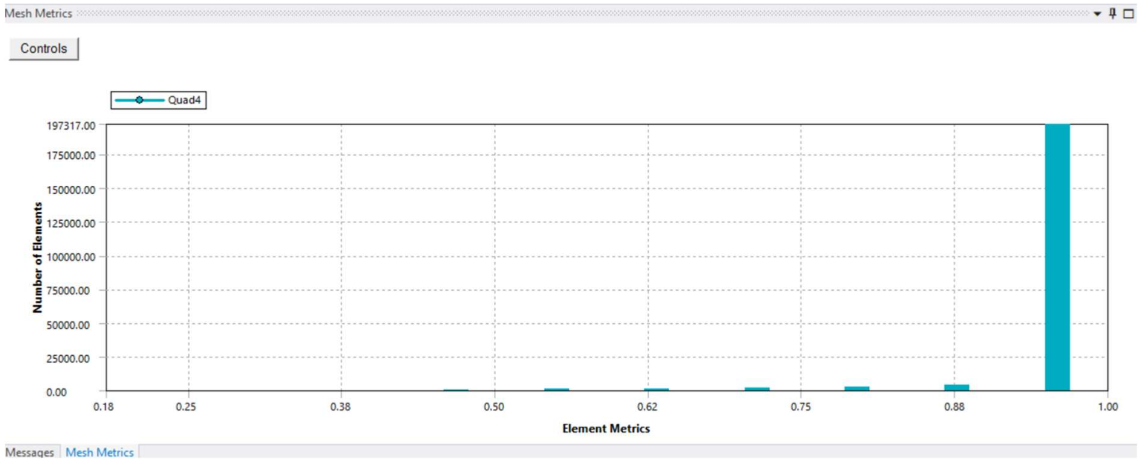

Figure 18-Orthogonal quality

3.4.6 Skewness

Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	CFD
Solver Preference	Fluent
Element Size	Default (2050.2 mm)
Export Format	Standard
Export Preview Surface Mesh	No
Sizing	
Quality	
Check Mesh Quality	Mesh Quality Worksheet
Target Skewness	Default (0.9)
Smoothing	Medium
Mesh Metric	Skewness
Min	1.3057e-010
Max	0.88553
Average	3.9705e-002
Standard Deviation	0.12015

Figure 19 -Skewness

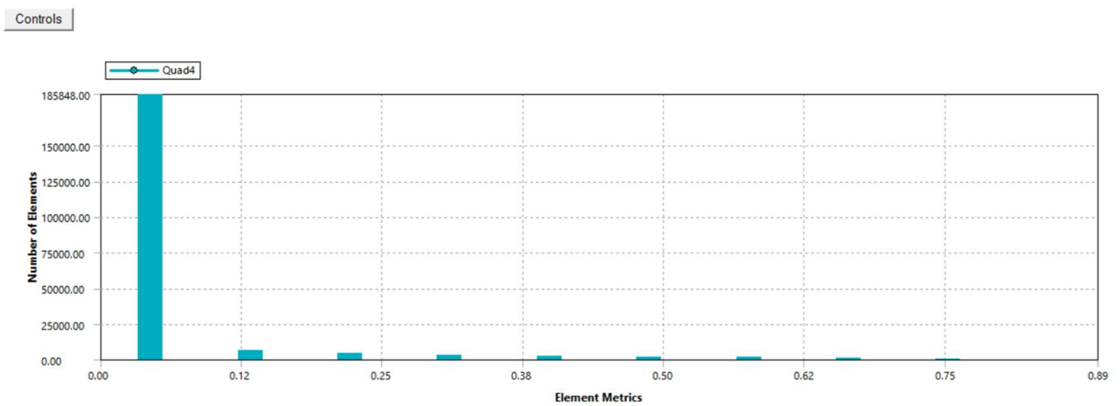

Figure 20-Skewness

3.4.7 Aspect ratio

Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	CFD
Solver Preference	Fluent
Element Size	Default (2050.2 mm)
Export Format	Standard
Export Preview Surface Mesh	No
Sizing	
Quality	
Check Mesh Quality	Mesh Quality Worksheet
Target Skewness	Default (0.9)
Smoothing	Medium
Mesh Metric	Aspect Ratio
Min	1.0043
Max	12.34
Average	3.7311
Standard Deviation	2.5747

Figure 21 - Aspect ratio

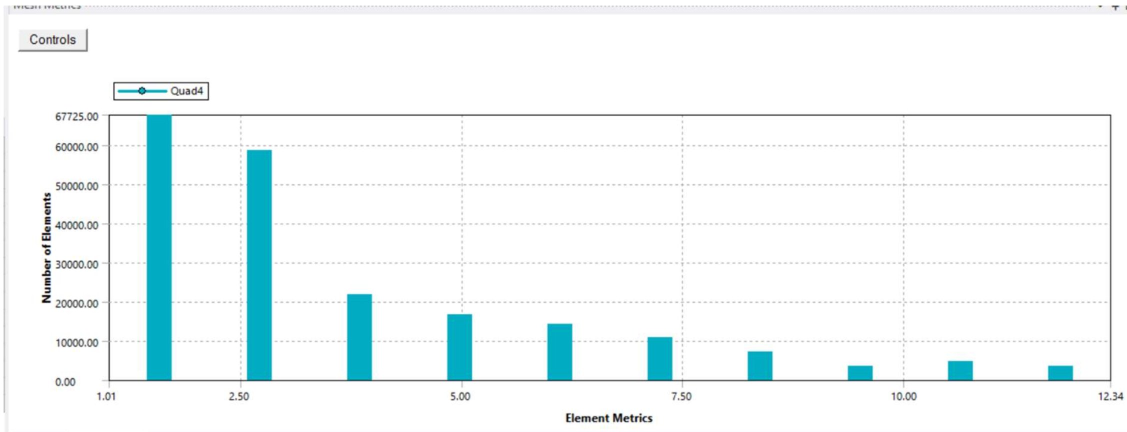

Figure 22-Aspect ratio

Table 3 - Recommended values

Parameter	**Value**	**Recommended value**
Number of nodes	211040	Not mentioned in the research paper
Number of elements	210000	In the referenced study, a grid size of 105,000 elements was selected based on a grid independence test. However, in the present work, a finer grid consisting of 210,000 elements was chosen as the optimal value following an independently conducted grid independence analysis
Orthogonal quality	Minimum = 0.1825	0.95 -1: Excellent

	Maximum = 1 Average = 0.98167	
Skewness	Minimum = 1.3057e-010 Maximum = 0.88553 Average = 3.9705e-002	0-0.25: Excellent
Aspect ratio	Minimum = 1.0043 Maximum = 12.34 Average = 3.7311	1 to 10 overall

4 Simulation Settings

The simulation was solved using the Reynolds-Averaged Navier-Stokes (RANS) equations in ANSYS Fluent. The specific numerical schemes and models were selected to ensure a stable and accurate solution for this aerodynamic analysis.

4.1 Convective flux calculation scheme

This defines the numerical scheme for solving the convection terms (how properties move with the flow).

- **Momentum equation (Second order upwind)**

  The Second Order Upwind scheme (Order 2) was selected for the Momentum equation because it provides:

  - **Higher Accuracy:** It delivers results with lower numerical diffusion compared to a first-order scheme.
  - **Flow Gradient Resolution:** It is critical for correctly resolving complex flow gradients, such as those found in wakes or boundary layers.
  - **Accurate Prediction:** It ensures accurate prediction of key aerodynamic forces (e.g., lift and drag).

- **Turbulence equations (First Order Upwind)**

  The First Order Upwind scheme (Order 1) was selected for the turbulence equations (TKE and TDR) primarily for solver stability

  - **Robustness:** The scheme is highly robust, making it less susceptible to numerical oscillations and instabilities.
  - **Stable Convergence:** It is chosen to ensure a stable and steady convergence of the overall CFD solution, which is essential when solving the inherently stiffer turbulence transport equations.

4.2 Diffusive flux calculation scheme

This defines the schemes for handling diffusion and pressure, ensuring the entire diffusive flux term maintains **second-order accuracy**.

- **Gradient:** The Green-Gauss Cell Based method was used. This is a standard and accurate method for computing gradients from cell centers to faces, achieving an order of accuracy equivalent to the second order. It is well-suited for the structured C-mesh used in this project.

- **Pressure:** A Second Order scheme was employed for pressure calculation (pressure interpolation). This choice is critical to ensure a high degree of accuracy in the pressure field, which is essential for the final calculation of lift and drag

4.3 Turbulence modelling

- To account for the effects of turbulence, two different two-equation RANS models were employed: the Standard KE Model and the SST (Shear Stress Transport) k-w model. Both models were simulated to allow for a direct comparison of their predictive capabilities

- **Comparison of Turbulence Models**

Table 4 - Comparison of turbulence models

Feature	**Standard $k-\epsilon$ (KE)**	**SST $k-\omega$**
Core Advantage	**Robust** and widely used, especially for free stream flows.	**High Accuracy** in complex aerodynamic flows and adverse pressure gradients.

Separation Prediction	Tends to under-predict separation (over-predicts eddy viscosity).	Specifically designed to **accurately predict flow separation** and adverse pressure gradients.
Computational Overhead	**Lower** overhead, particularly when paired with wall functions.	**Higher** when resolving the sublayer, but **comparable** when using wall functions.

- **Suitability and Wall Function Requirement**

The comparison strategy was specifically designed to be highly controlled. While the SST kw model is often recommended for use with a low y+ mesh (y+ = 1) to fully resolve the viscous sublayer, this study instead maintained a consistent high y+ approach for all simulations:

- **Wall Function Requirement:** The mesh for both models was designed to utilize the Standard Wall Function approach, sizing the first cell height to ensure the average y+ value was maintained in the range of $30 < y+ < 100$ across the wall surface.

- **Justification for High y+:** This consistent methodology was deliberately chosen to ensure an "apple-to-apple" comparison. By applying the same Wall Function treatment to both models, the analysis focused purely on the differences in the models' governing equations (e.g., their superior prediction of eddy viscosity and flow separation) rather than the effects of differing mesh resolutions.

- **Suitability:** This approach offered good stability and significantly reduced the computational overhead. The SST k-w model provided the superior result because, even with the wall function treatment, its formulation (a blend of k-w near the wall and k-e in the freestream) offers better accuracy in predicting critical aerodynamic features like adverse pressure gradients and separation compared to the Standard k-epsilon model.

4.4 Pressure velocity coupling

Comparison of Coupling Schemes

Several schemes exist for pressure-velocity coupling. For this steady-state flow simulation, the Coupled scheme was compared against the popular segregated algorithms: SIMPLE, SIMPLEC (SIMPLE-Consistent), and PISO (Pressure Implicit with Splitting of Operators).

Table 5 - Comparison of coupling schemes

Criteria	**Coupled Scheme**	**SIMPLE Scheme**	**SIMPLEC Scheme**	**PISO Scheme**
Computational Method	Solves the momentum and pressure equations simultaneously (single system).	Solves equations sequentially (momentum first, then pressure correction).	Sequential, similar to SIMPLE but with better pressure approximation.	Sequential, uses two or more corrector steps per iteration.
Convergence Speed	Generally, fastest for steady-state problems and structured meshes.	Slower convergence rate, widely used as a standard.	Often faster than SIMPLE due to improved consistency.	Excellent for transient (time-dependent) flow problems; slower for steady-state.
Computational Overhead	Higher (requires storing and solving a larger matrix).	Lower (solves smaller, sequential matrices).	Lower, similar to SIMPLE.	Lower, similar to SIMPLE.
Primary Use	High-performance steady-state or implicit transient solutions.	Standard for steady-state and initial solutions.	Standard for steady-state problems, less sensitive to relaxation.	Unsteady (transient) and time-marching simulations.

Final Selected Scheme

The **Coupled scheme** was ultimately selected to link the pressure and velocity equations. This choice was based on its superior **robustness** and its characteristic of providing the **fastest convergence** for the steady-state, incompressible flow problem analyzed in this project. While segregated schemes like SIMPLE, SIMPLEC, or PISO are common, the Coupled scheme's advantage in speed for large, steady-state problems outweighed its higher memory requirement.

5 Results and Validation

5.1 Grid independency analysis

A grid independency study was conducted to ensure that the simulation results were not dependent on the mesh resolution. The lift and drag coefficients were calculated for various mesh element counts while keeping all other parameters constant. In the research paper, they also have done a grid independency test, and they have chosen 105000 elements as their mesh size.

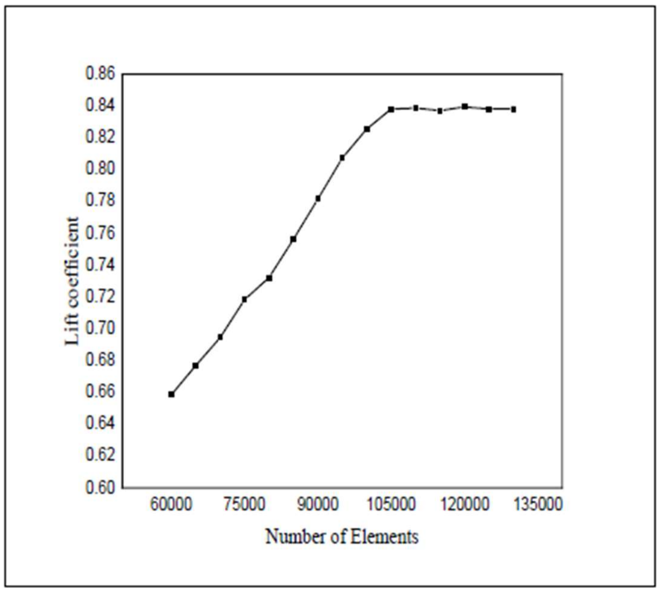

Figure 23 - Grid independency test done in the research paper

Similarly, for the present analysis, a grid independence test was conducted using the Standard K-E model at an angle of attack of 4. The results are presented in Figure [X] and Figure [Y] for the lift and drag coefficients, respectively. The plots demonstrate that as the mesh is refined, both coefficients begin to converge, with the rate of change diminishing significantly at higher element counts. Based on this convergent behavior, a mesh containing 210,000 elements was selected for all subsequent simulations. This mesh density was chosen because it provides a stable solution that is independent of the grid, ensuring a reliable balance between numerical accuracy and computational efficiency.

Table 6 -Lift coefficient and drag coefficient at angle of attack 4

Number of mesh elements	Lift coefficient	Drag coefficient
4800	0.72534573	0.039940962
16000	0.79701118	0.032437329
24000	0.81429823	0.029682463
72000	0.8427854	0.023435901
105600	0.84642429	0.022088415
210000	0.83717901	0.020777939
285600	0.839753	0.020146518
342000	0.84046327	0.019861652

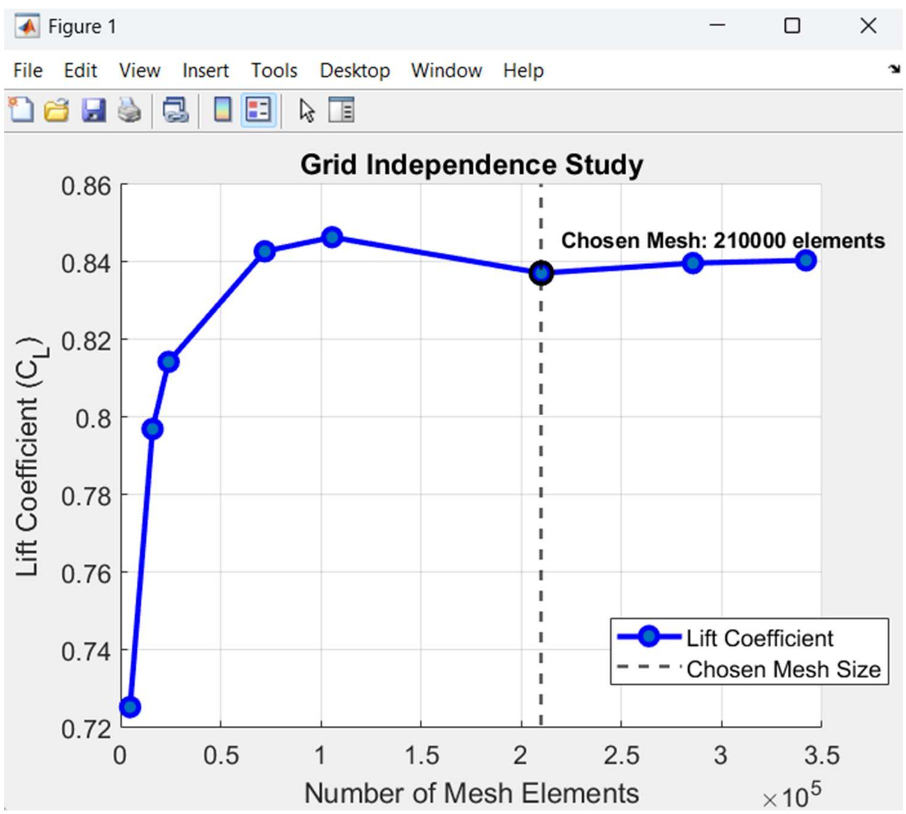

Figure 24- Number of mesh element vs Lift coefficient

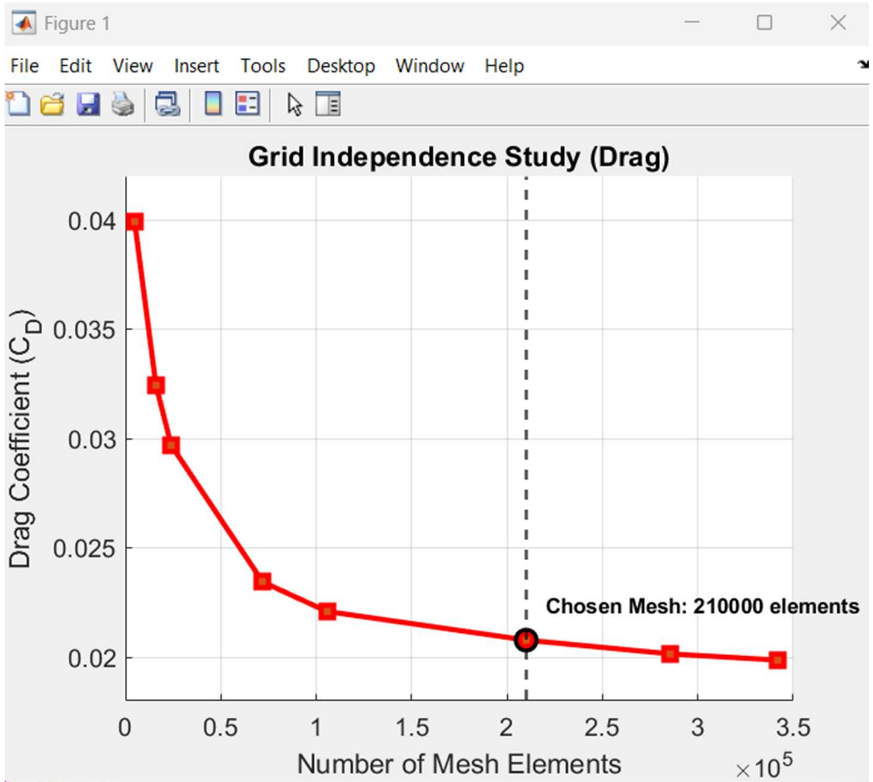

Figure 25- Number of mesh elements vs drag coefficient

As shown in the figures, both the lift and drag coefficients change significantly with coarser meshes. However, as the number of mesh elements increases, the values begin to converge and level off. The curves become notably flatter after approximately 210,000 elements, indicating that further increases in mesh density yield diminishing returns in terms of solution accuracy.

Based on these results, a mesh with 210,000 elements was chosen for the final analysis. This point lies in the stable, "grid-independent" region of the curve where the calculated lift and drag coefficients are no longer sensitive to mesh refinement. This choice provides an optimal balance, ensuring high solution accuracy while avoiding the unnecessary computational expense associated with an overly fine mesh.

5.2 Solution convergency study

To ensure the CFD results are accurate, we must first confirm the solution has converged. A converged solution is one that is stable and no longer changes with further iterations.

In this study, a dual-check method was used, based on the parameters displayed in the ANSYS solver.

1. **Equation Residuals**: The simulation was required to reduce the "parameter errors" (residuals) for the main flow equations (continuity, velocity, k, omega) to a specified low value, as set in the "Residual Monitors" window (1×10^{-4}).

2. **Physical Quantities:** Crucially, the simulation also monitored the key engineering values of Drag and Lift coefficients. These values had to stabilize and meet their own strict convergence criteria (1×10^{-4}) as defined in the "Convergence Conditions."

The solution was only considered complete when **"All Conditions are Met,"** meaning both the equation residuals and the force coefficients reached their target stability. This ensures the final data is both mathematically sound and physically stable.

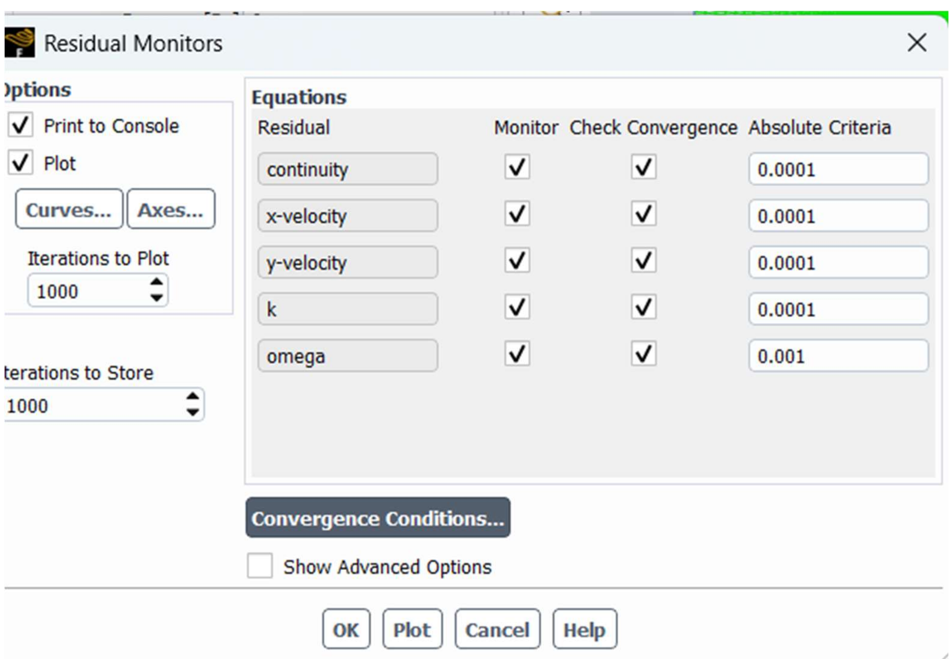

Figure 26 - Equation residuals convergence conditions

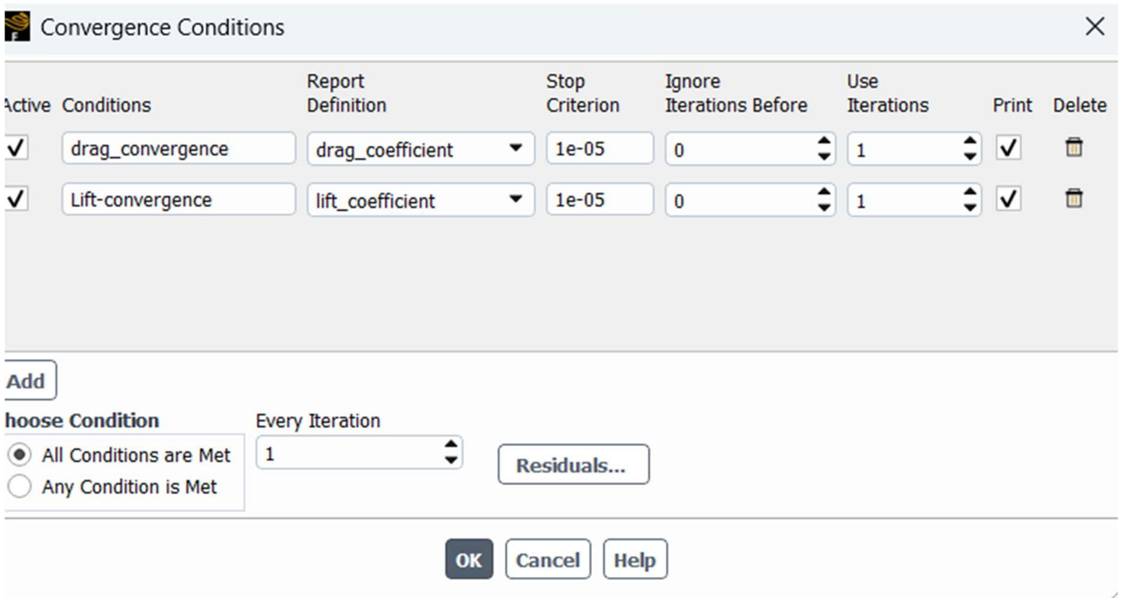

Figure 27 -Physical quantities convergence conditions

5.3 Convergence Analysis for Varying Angles of Attack

5.3.1 Angle of attack 0

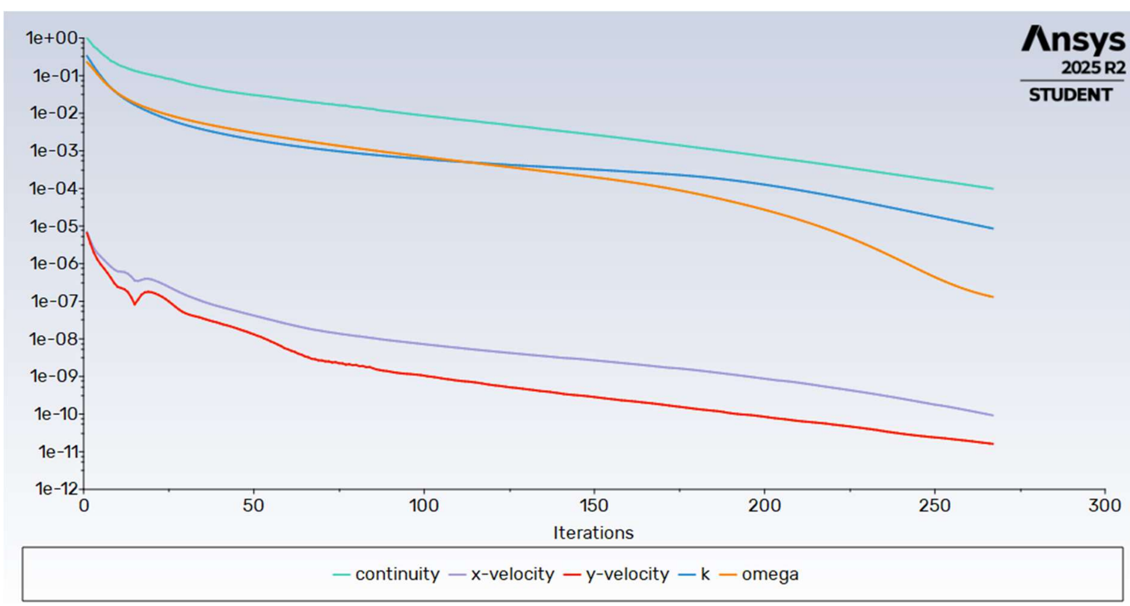

Figure 28 - Residual convergence

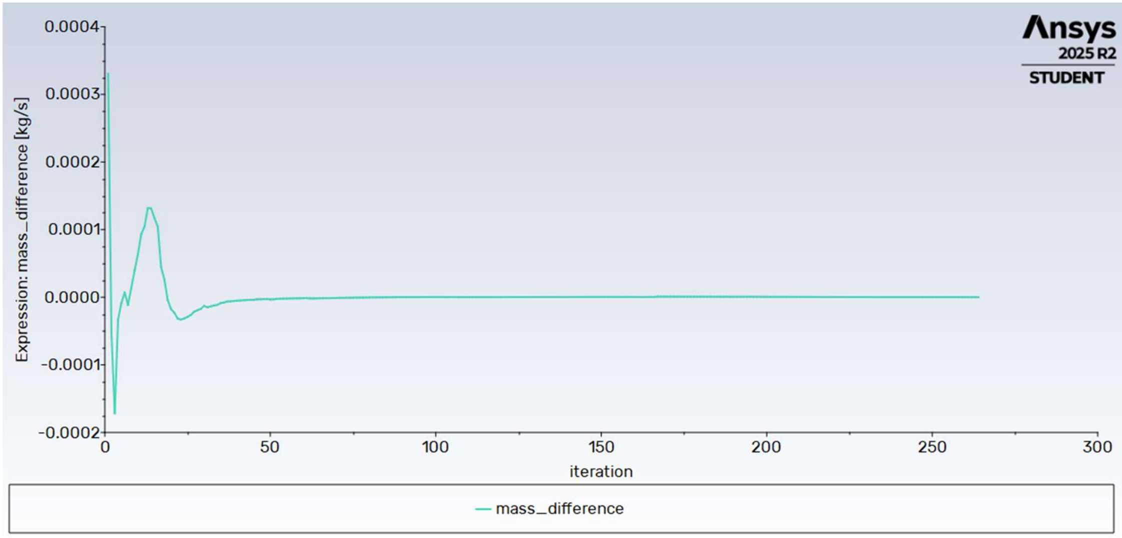

Figure 29 -Mass difference of the system

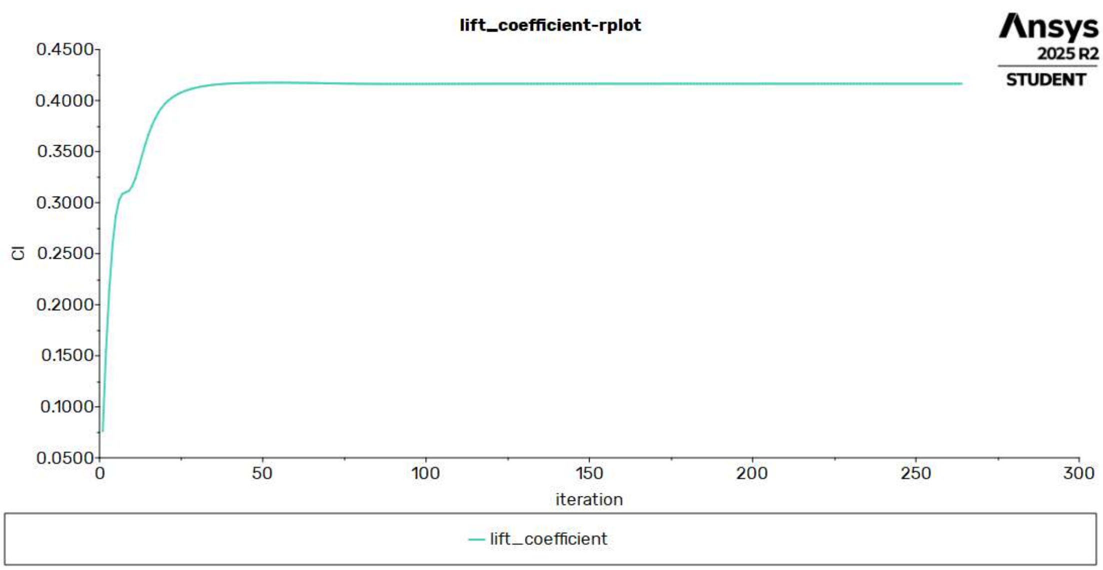

Figure 30 - Lift coefficient convergence

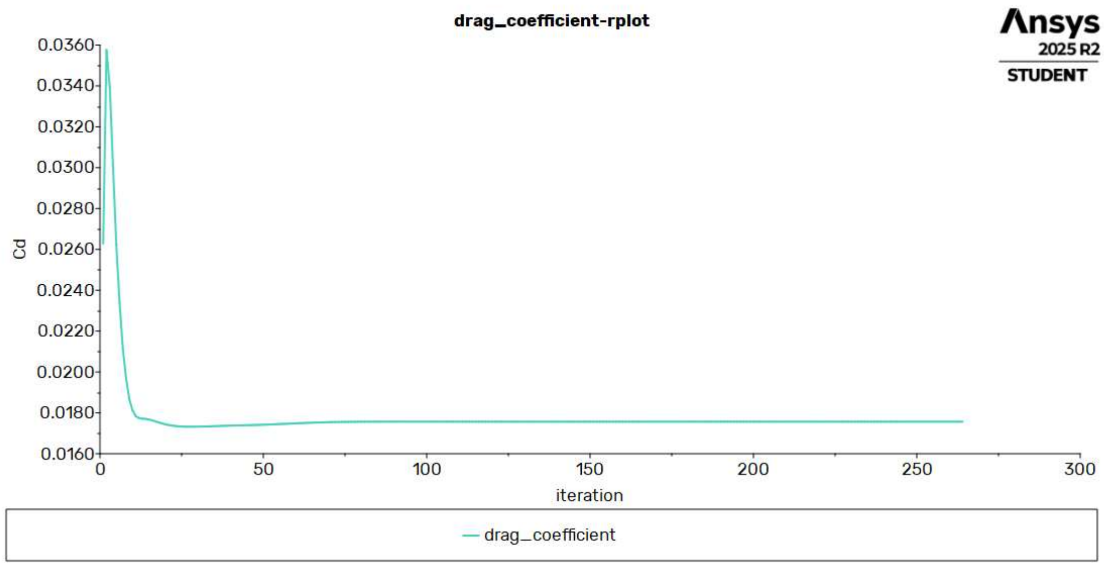

Figure 31 - Drag coefficient convergence

5.3.2 Angle of attack 2

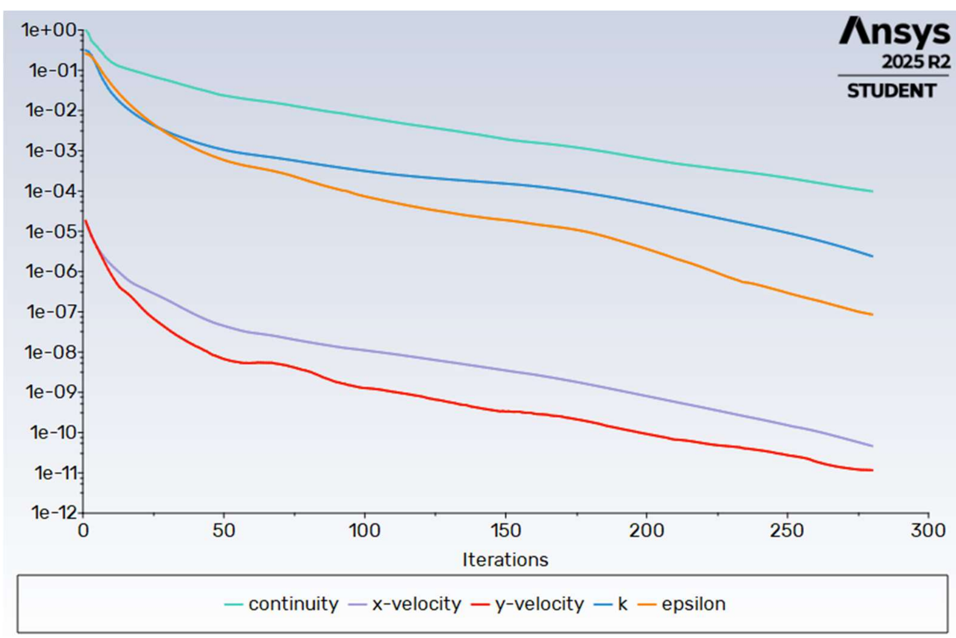

Figure 32 - Residual convergence

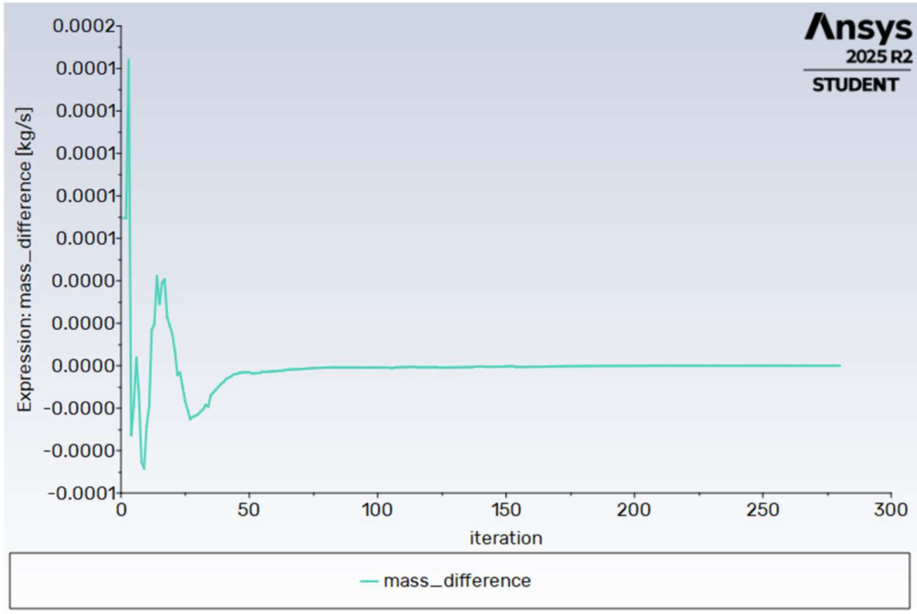

Figure 33- Mass difference of the system

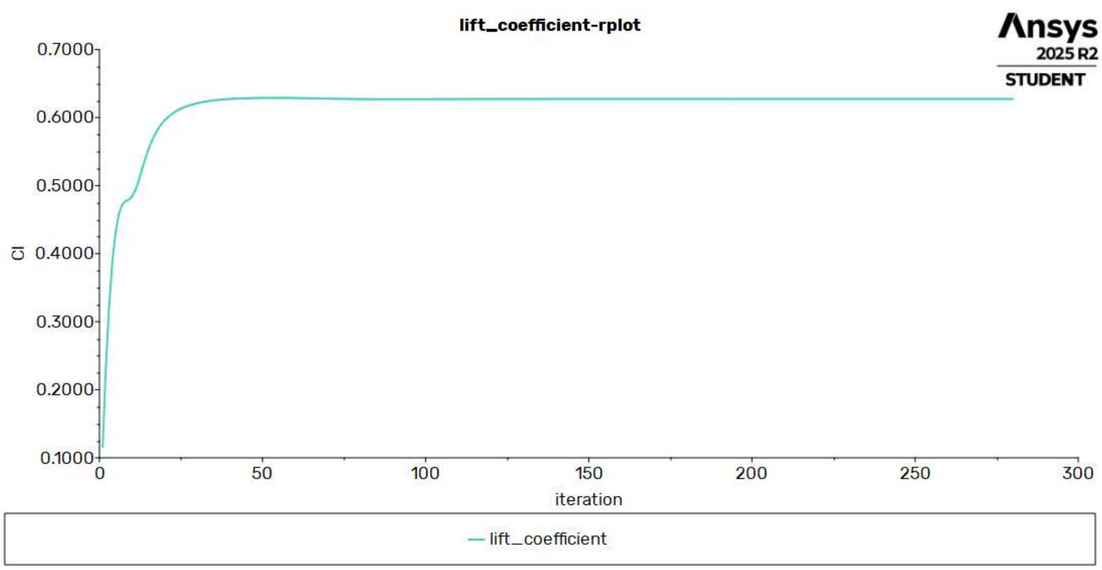

Figure 34- Lift coefficient convergence

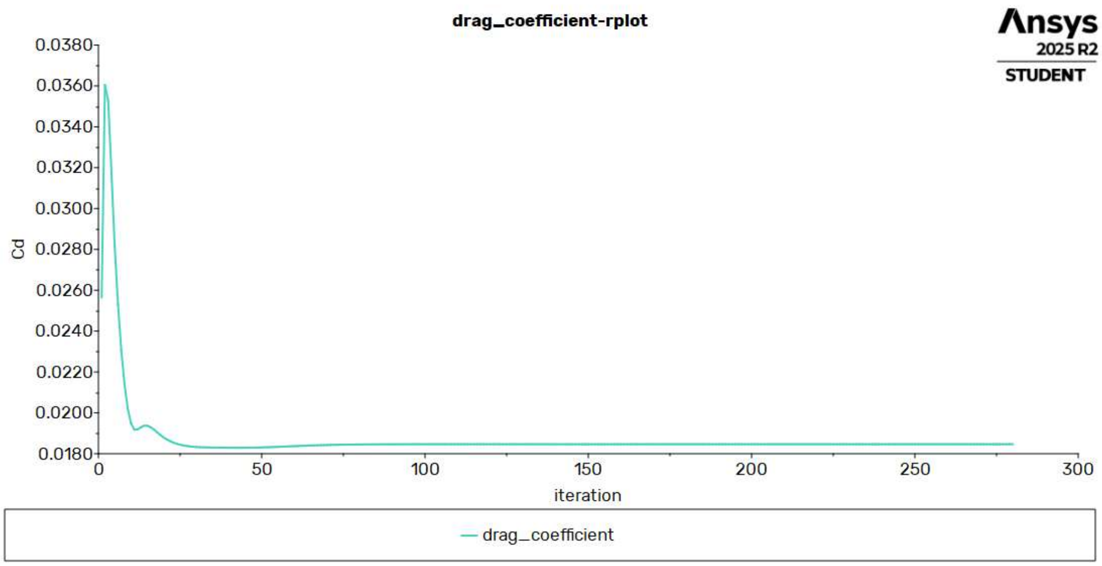

Figure 35- Drag coefficient convergence

5.3.3 Angle of attack 4

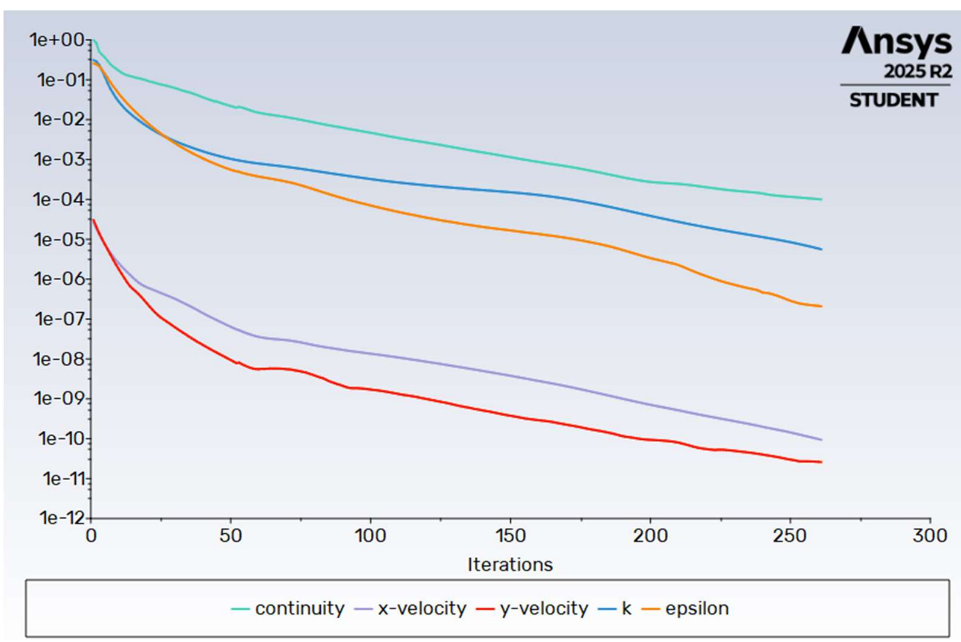

Figure 36-Residual convergence

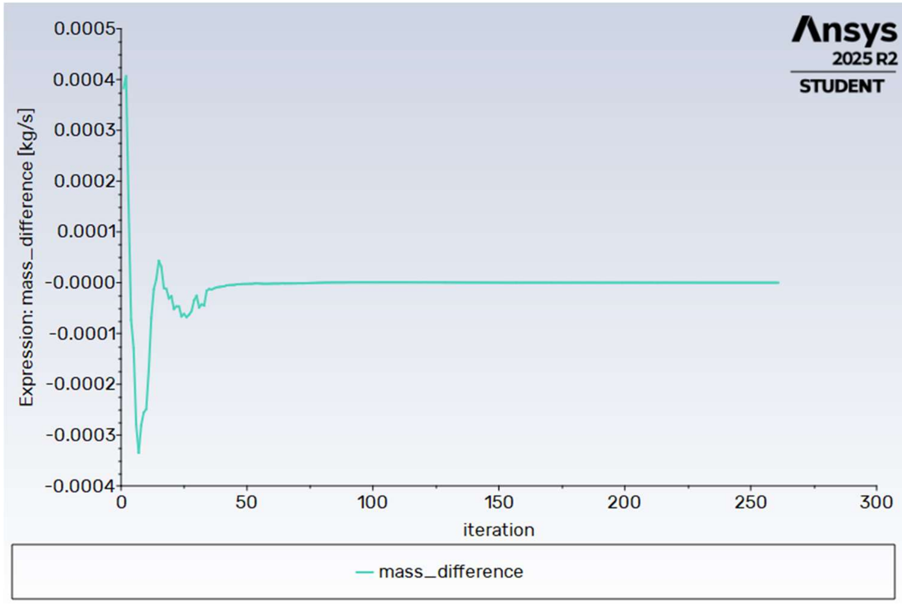

Figure 37 - Mass difference of the system

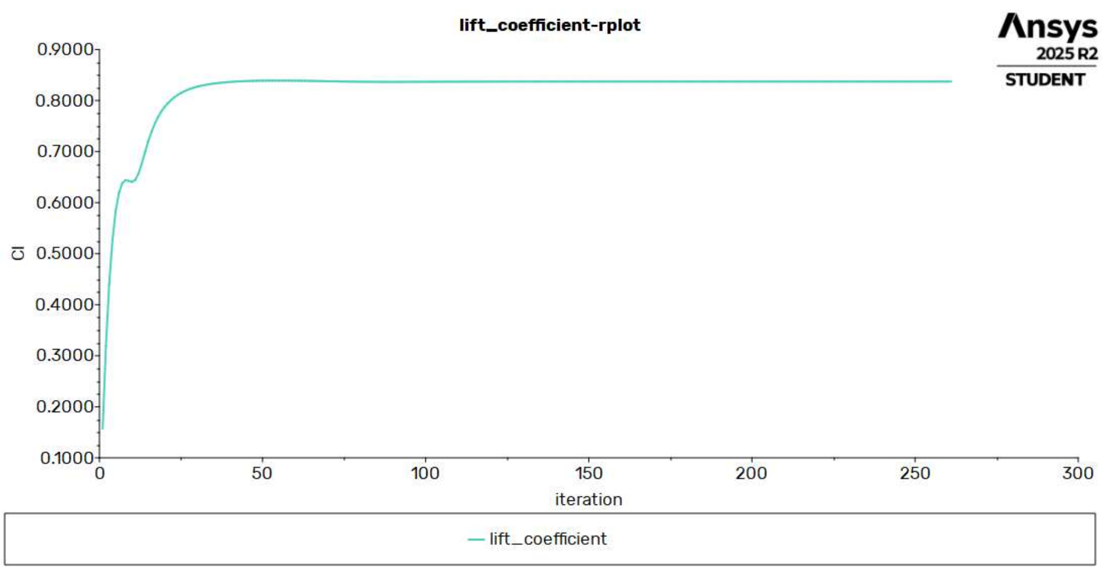

Figure 38 - Lift coefficient convergence

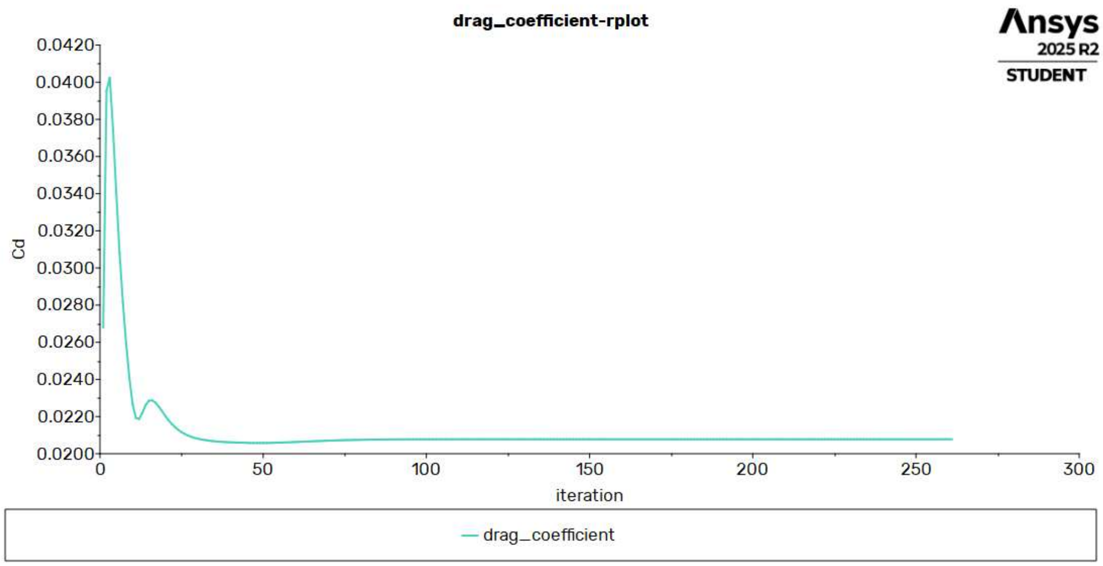

Figure 39 - Drag coefficient

5.3.4 Angle of attack 6

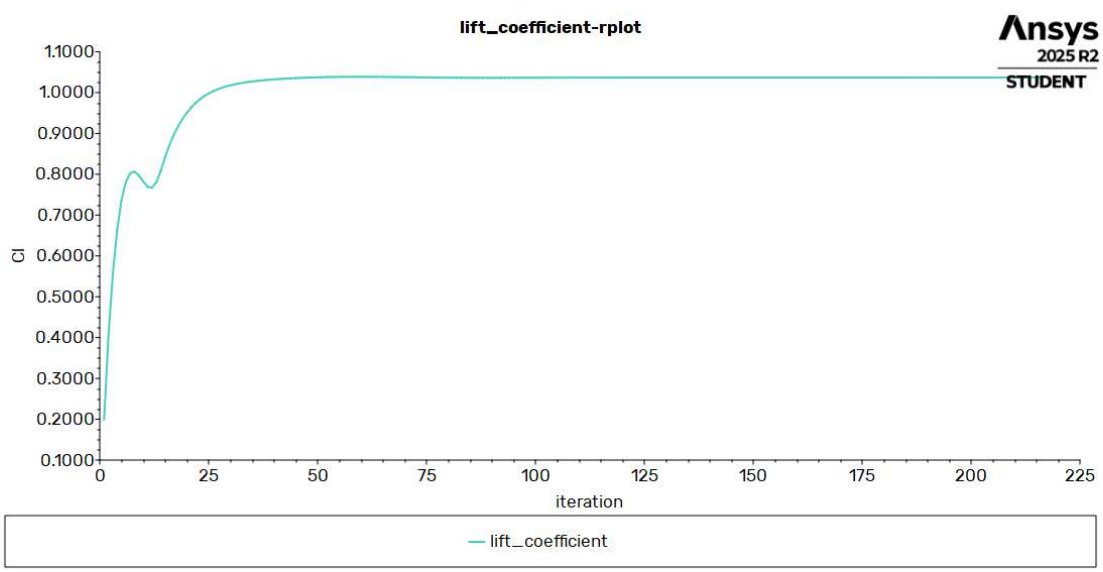

Figure 40 - Lift coefficient

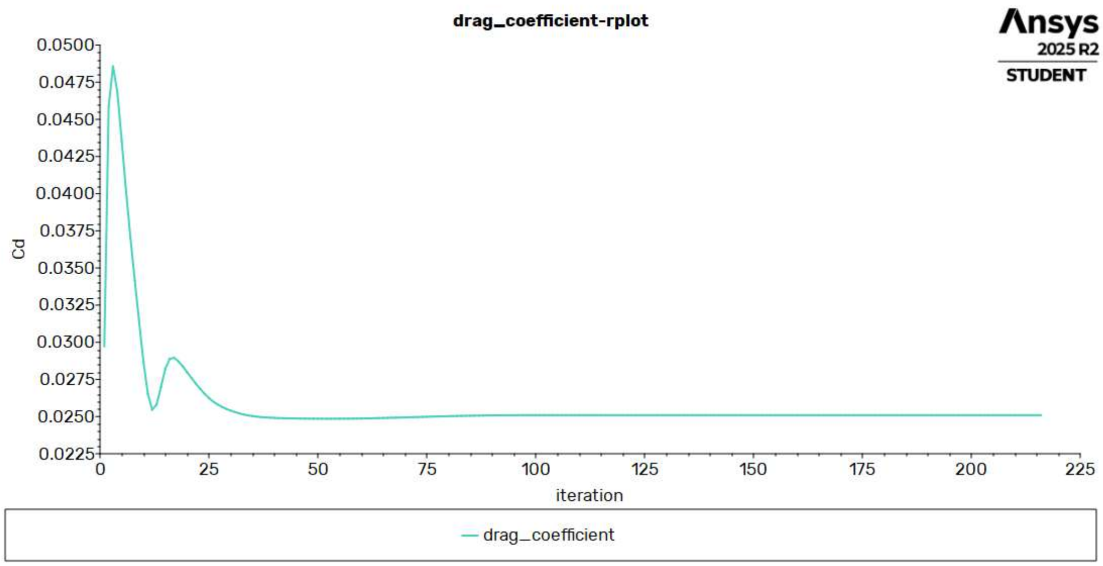

Figure 41 - Drag coefficient

5.3.5 Angle of attack 8

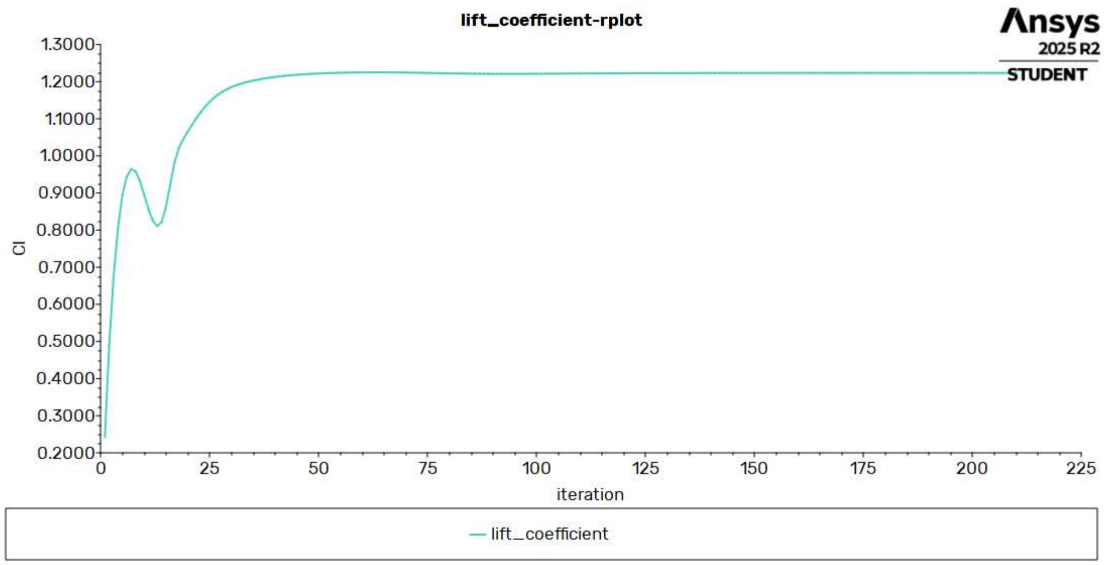

Figure 42 - Lift coefficient

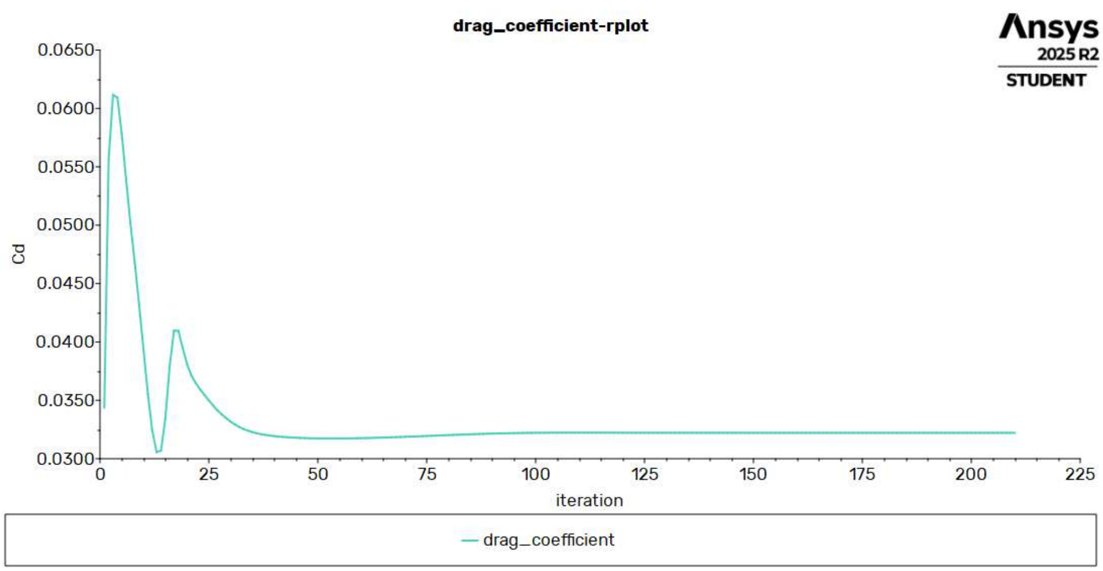

Figure 43 - Drag coefficient

5.3.6 Angle of attack 10

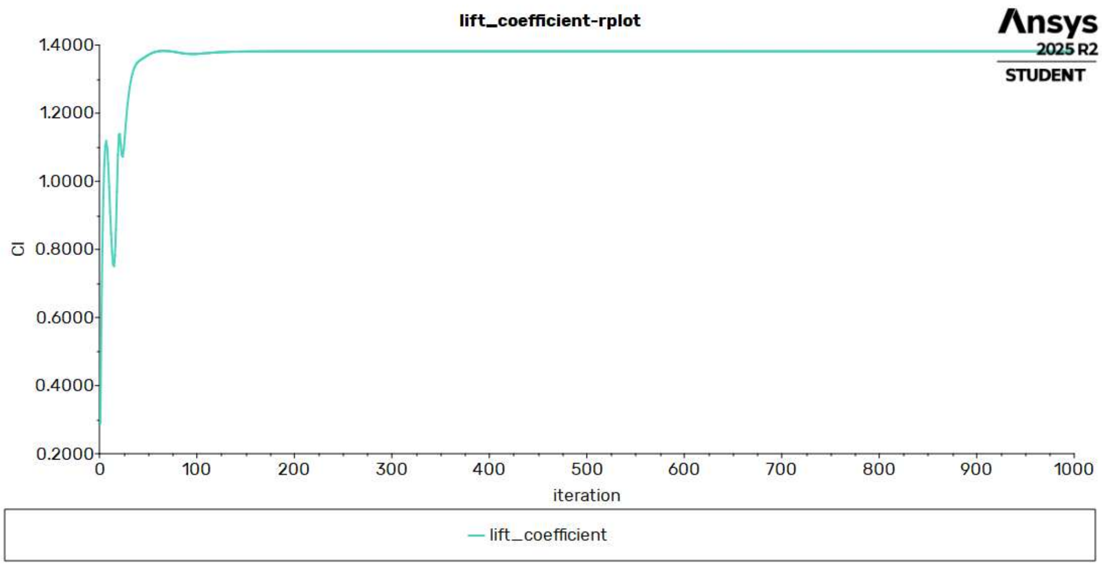

Figure 44 – Lift coefficient

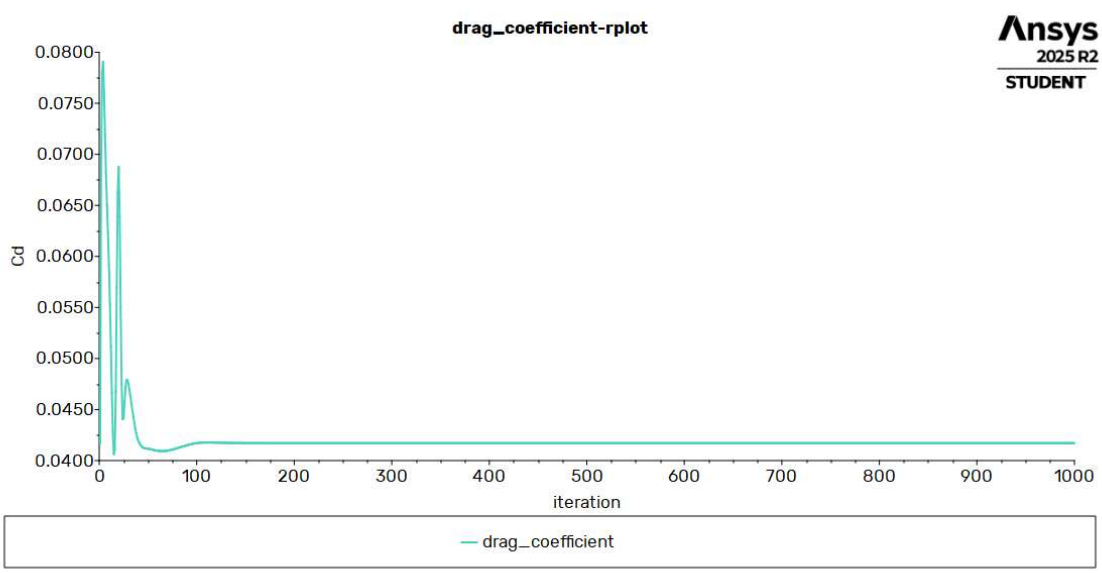

Figure 45 – Drag coefficient

5.3.7 Angle of attack 12

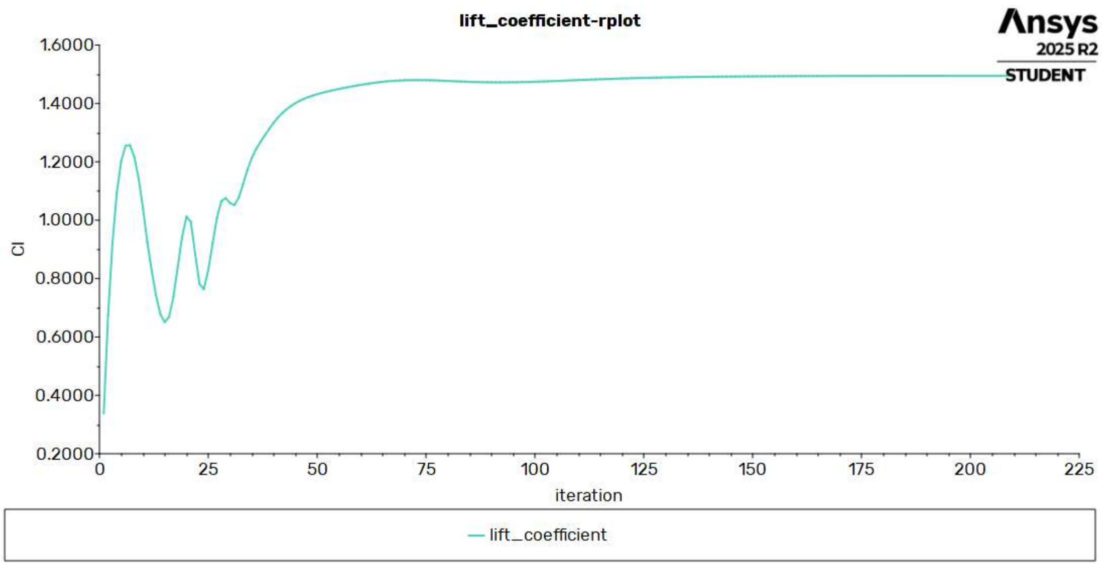

Figure 46 – Lift coefficient

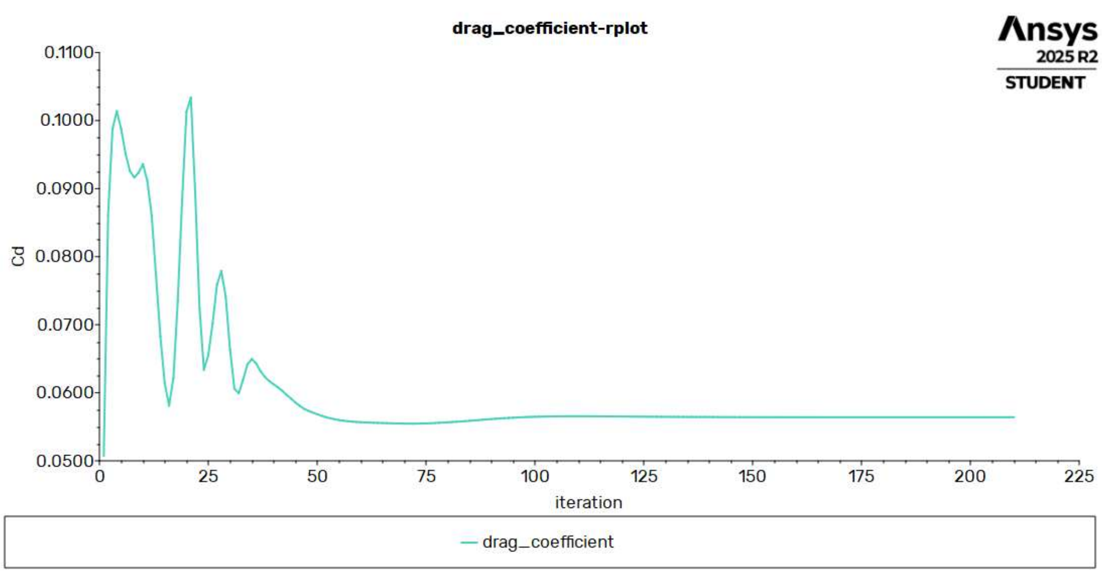

Figure 47 – Drag coefficient

5.3.8 Angle of attack 14

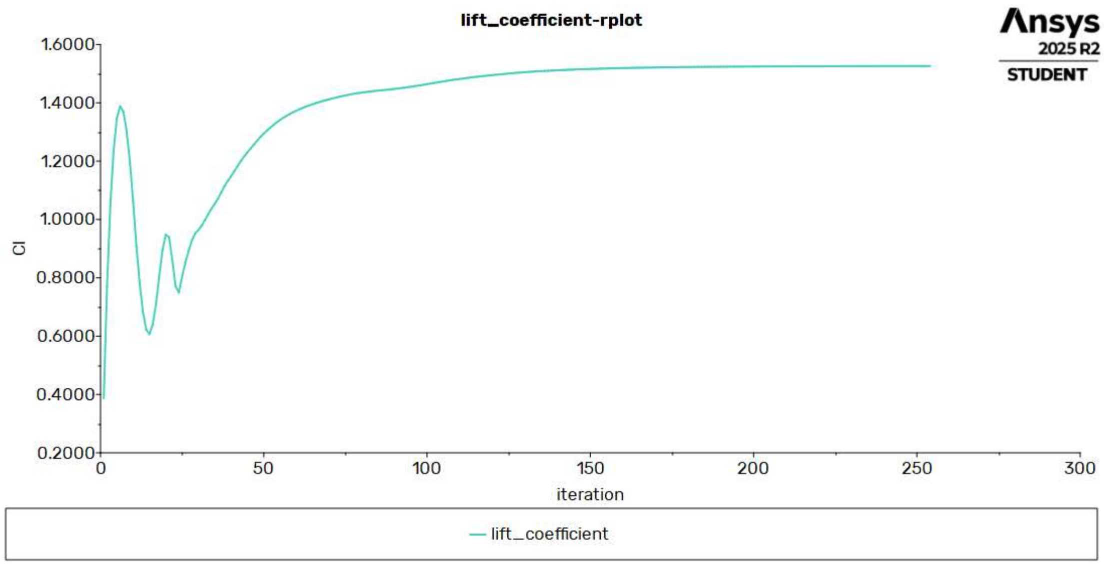

Figure 48 – Lift coefficient

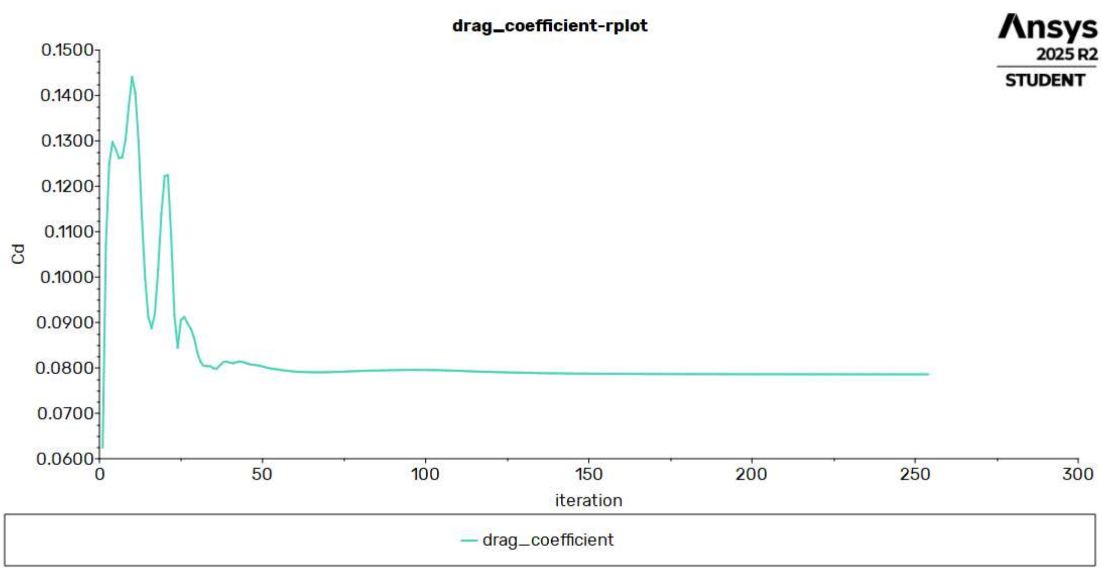

Figure 49 – Drag coefficient

5.3.9 Angle of attack 16

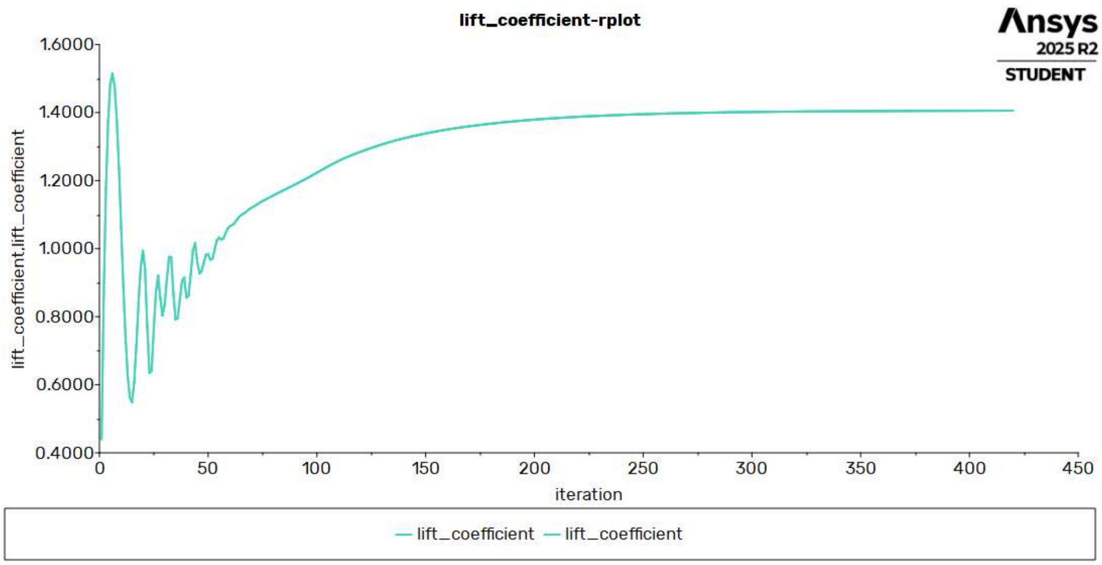

Figure 50 -Lift coefficient

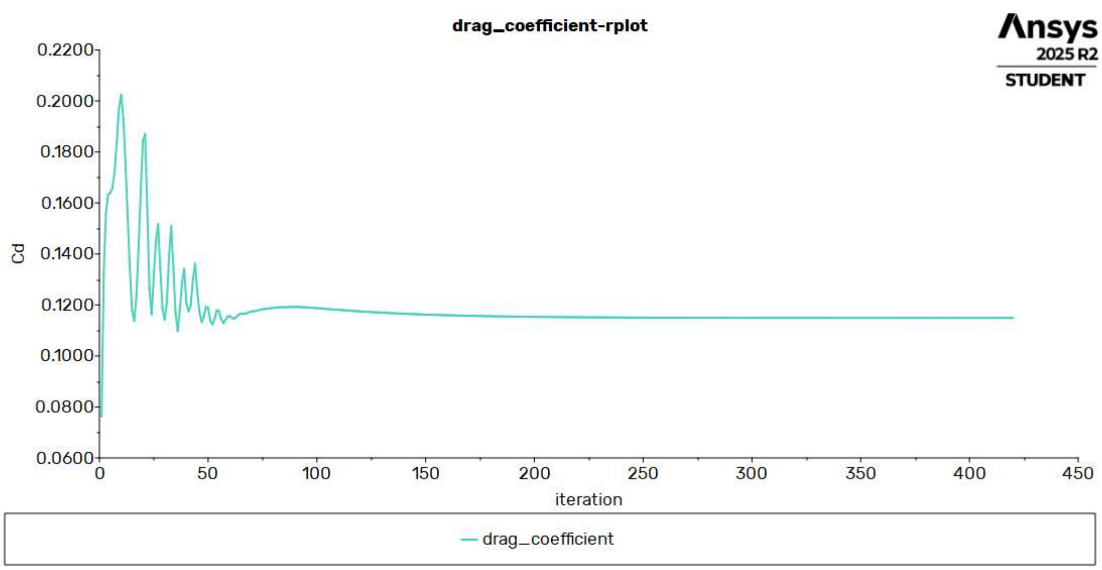

Figure 51 – Drag coefficient

5.4 SST KW model

5.4.1 Angle of attack 0

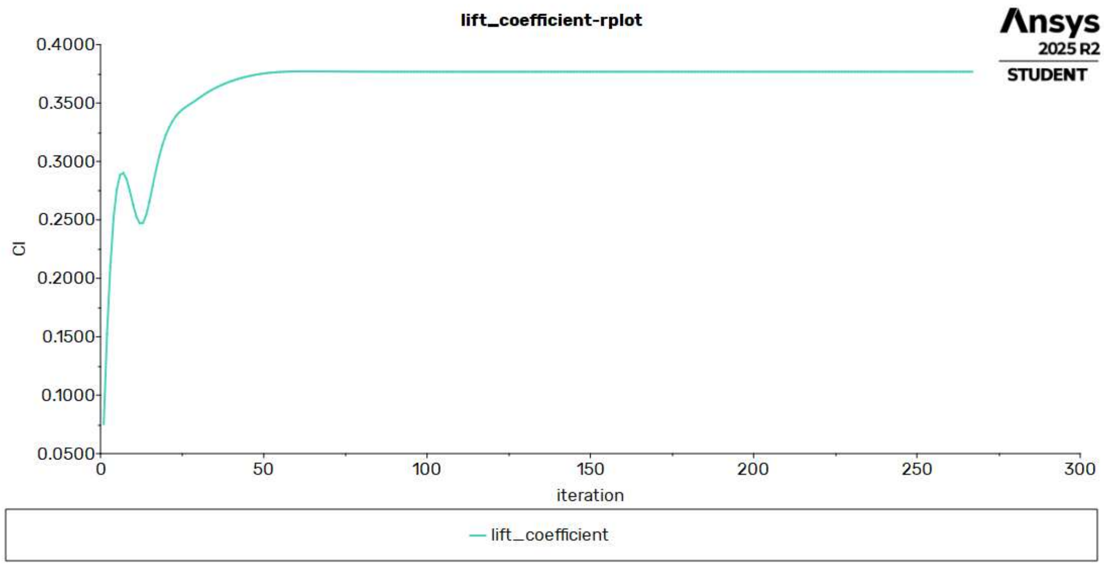

Figure 52 – Lift coefficient

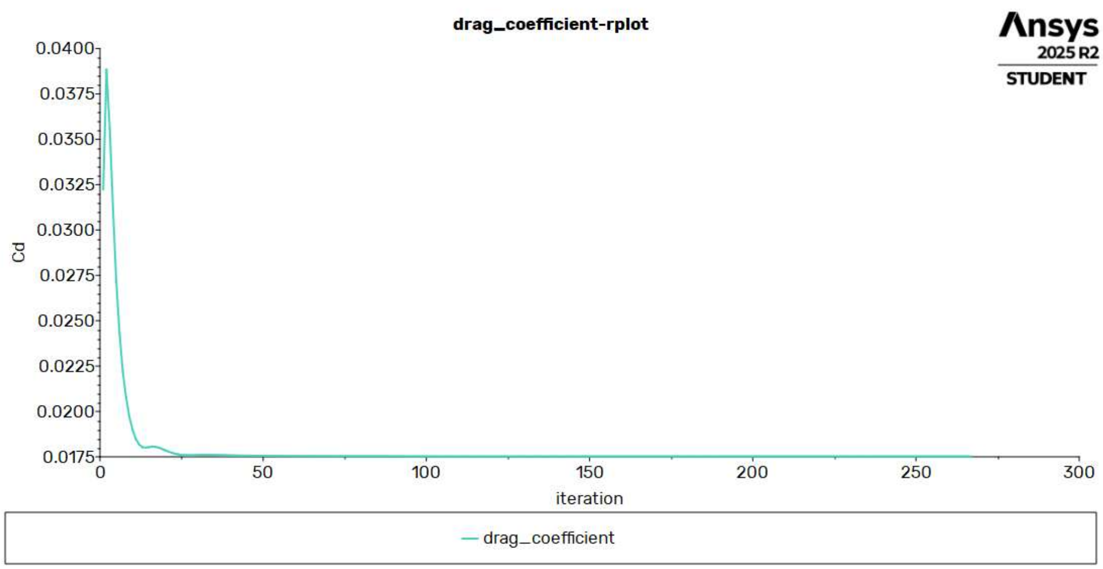

Figure 53- Drag coefficient

5.4.2 Angle of attack 2

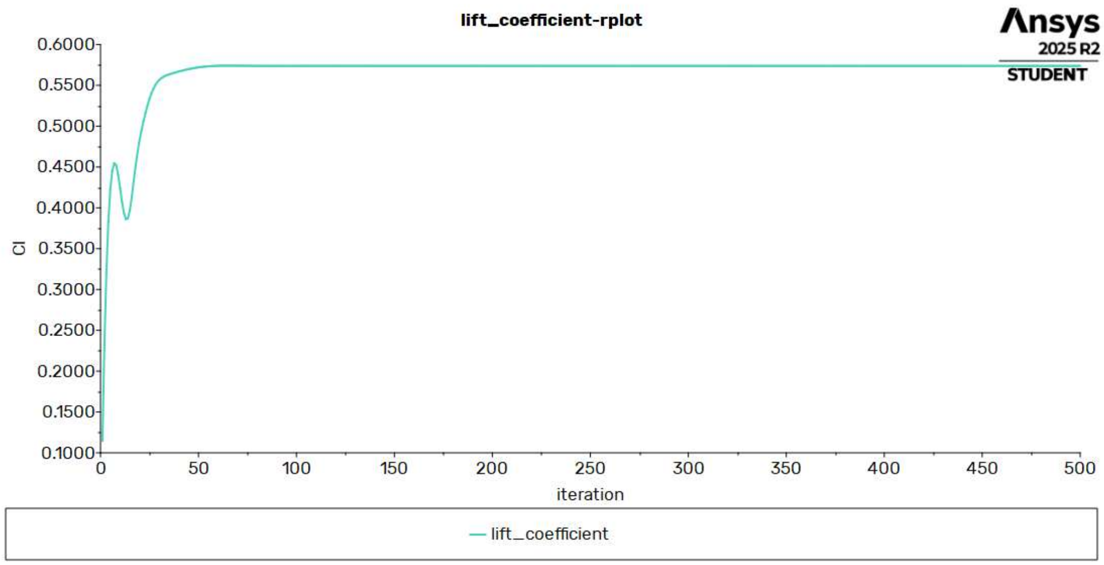

Figure 54 – Lift coefficient

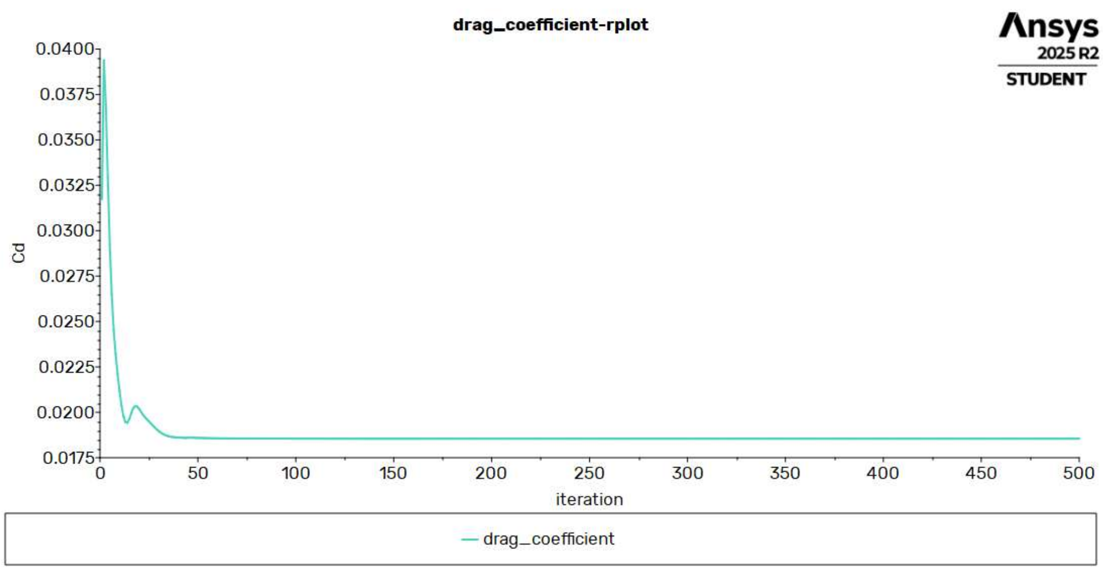

Figure 55 – Drag coefficient

5.4.3 Angle of attack 4

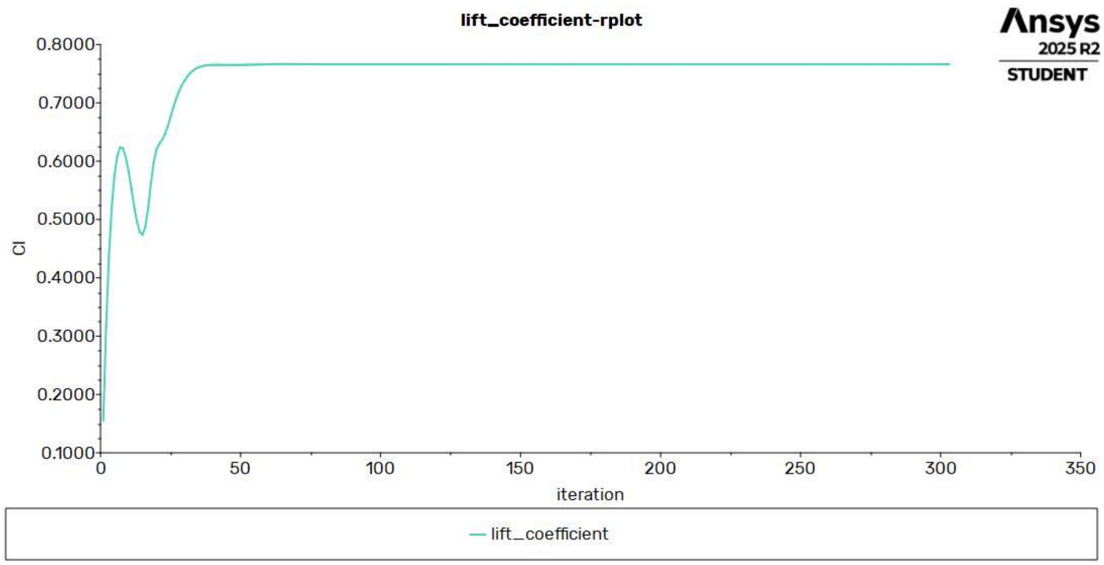

Figure 56 – Lift coefficient

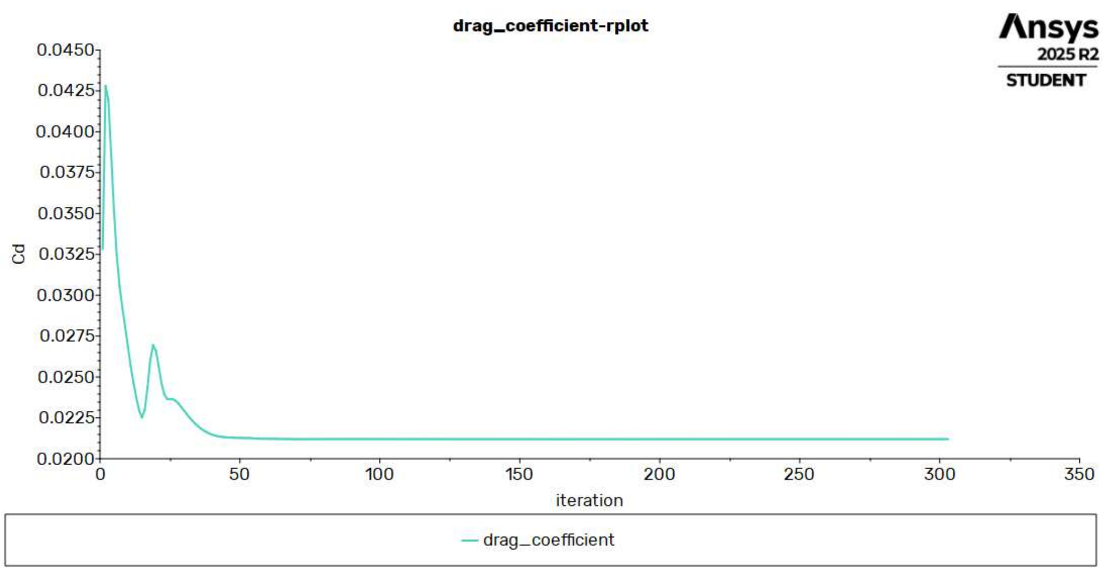

Figure 57 – Drag coefficient

5.4.4 Angle of attack 6

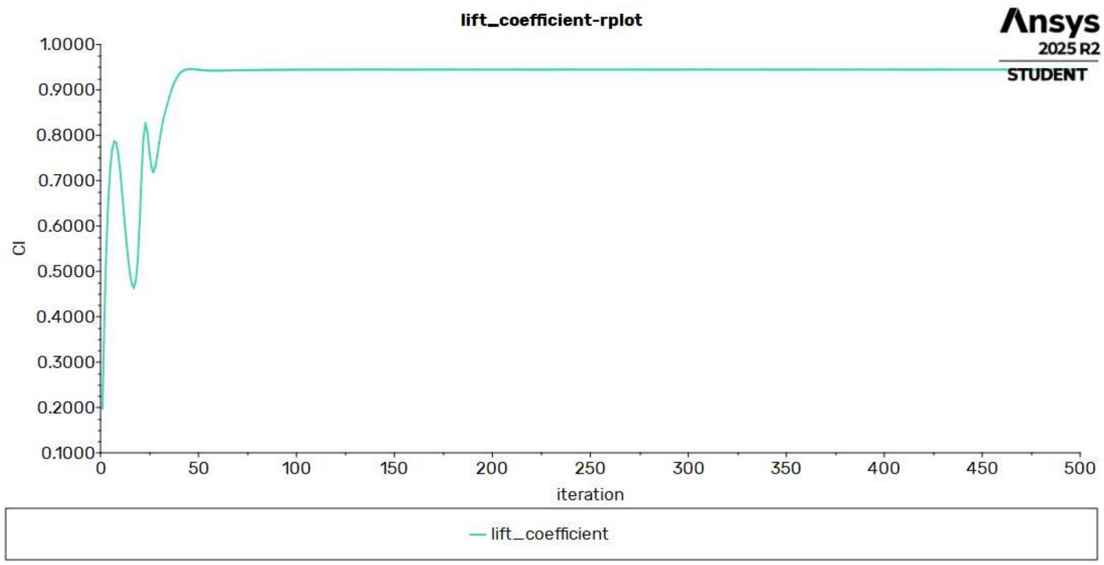

Figure 58 – Lift coefficient

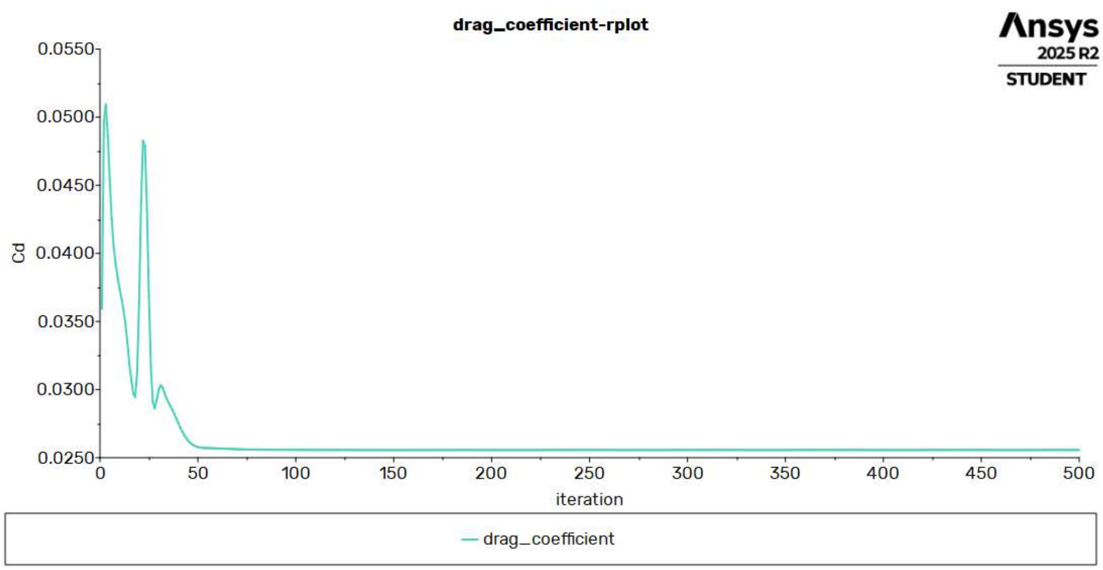

Figure 59 – Drag coefficient

5.4.5 Angle of attack 8

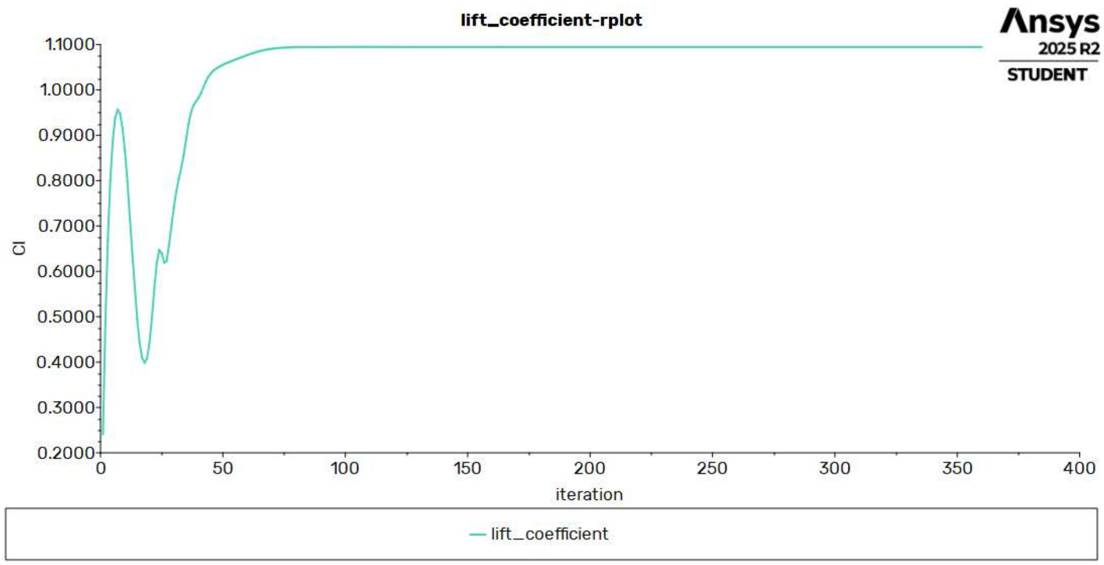

Figure 60 – Lift coefficient

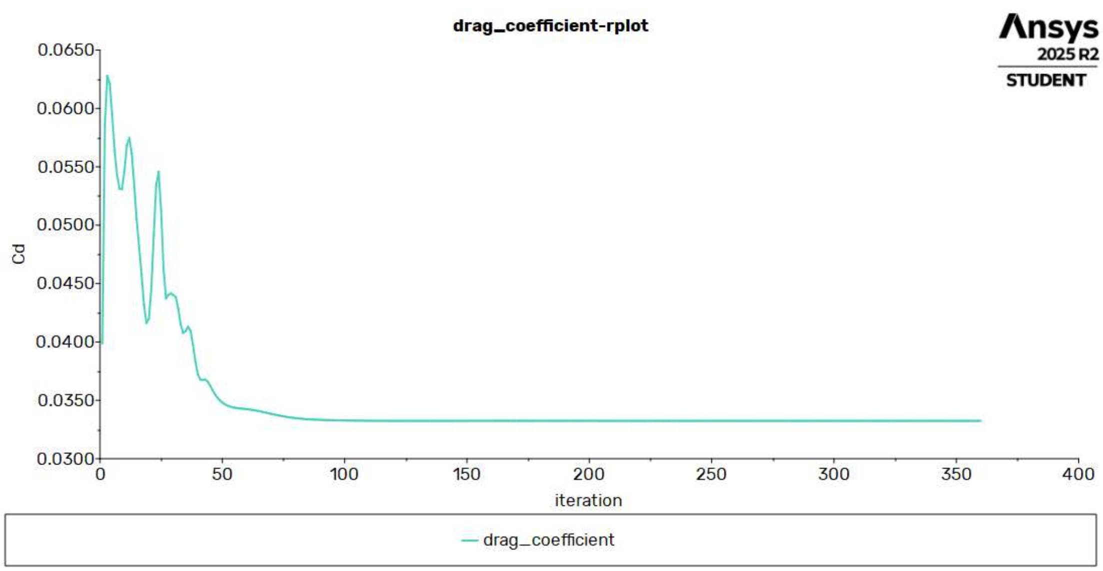

Figure 61 – Drag coefficient

5.4.6 Angle of attack 10

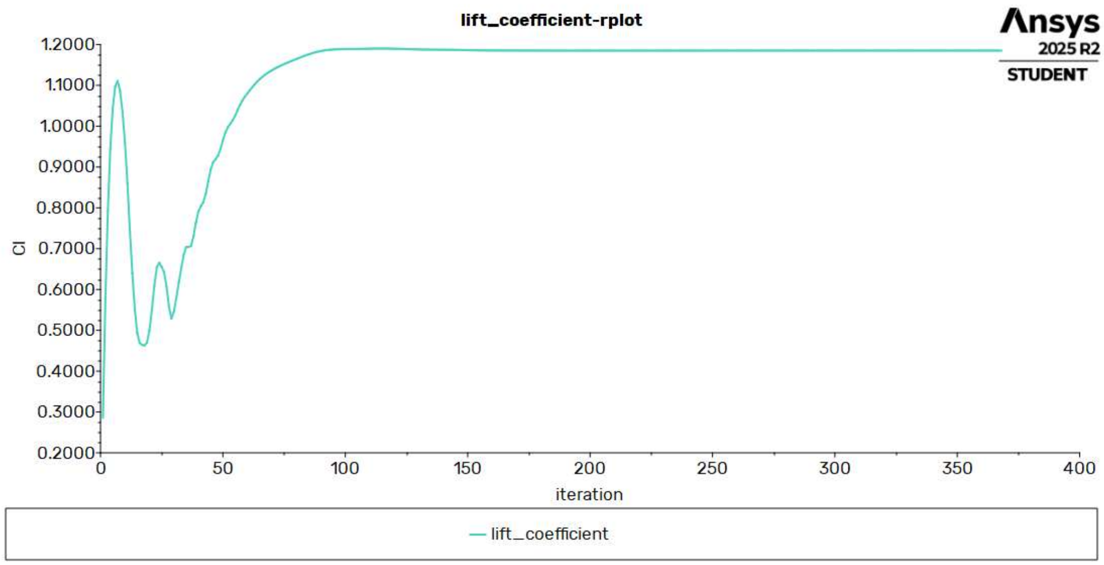

Figure 62 – Lift coefficient

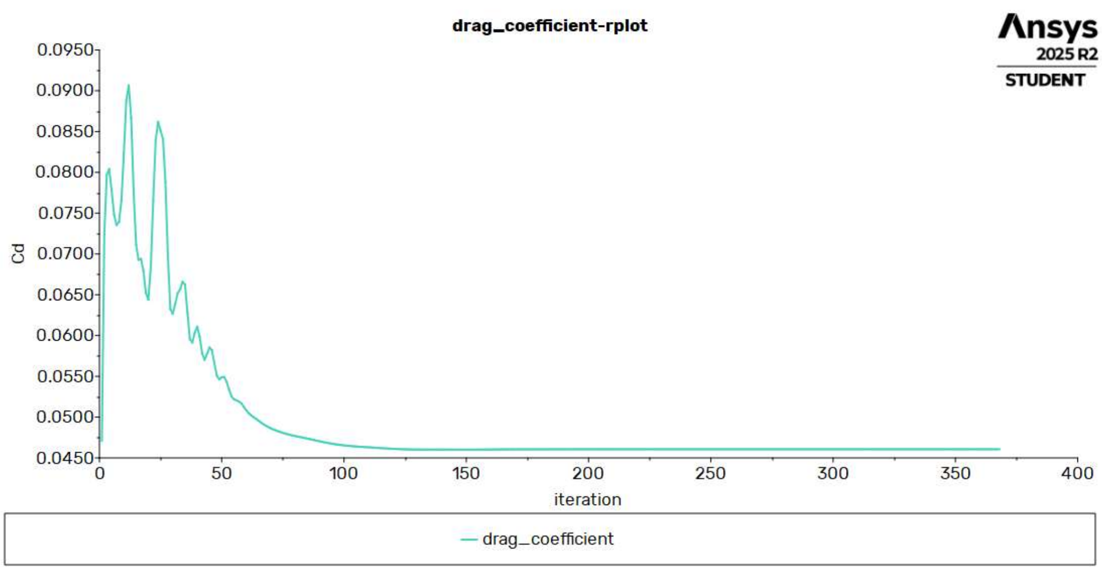

Figure 63 – Drag coefficient

5.4.7 Angle of attack 12

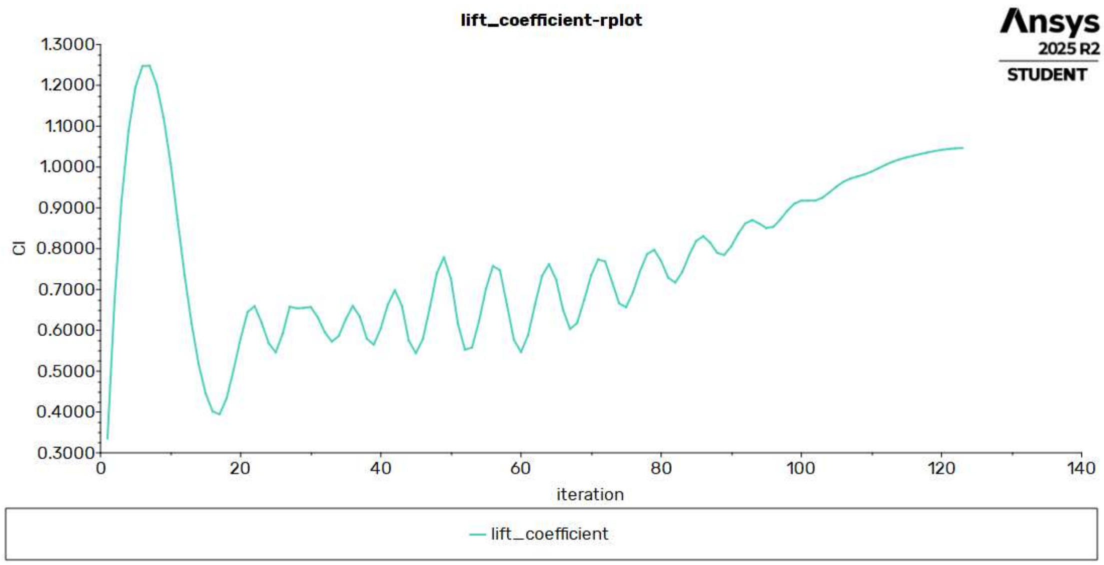

Figure 64 – Lift coefficient

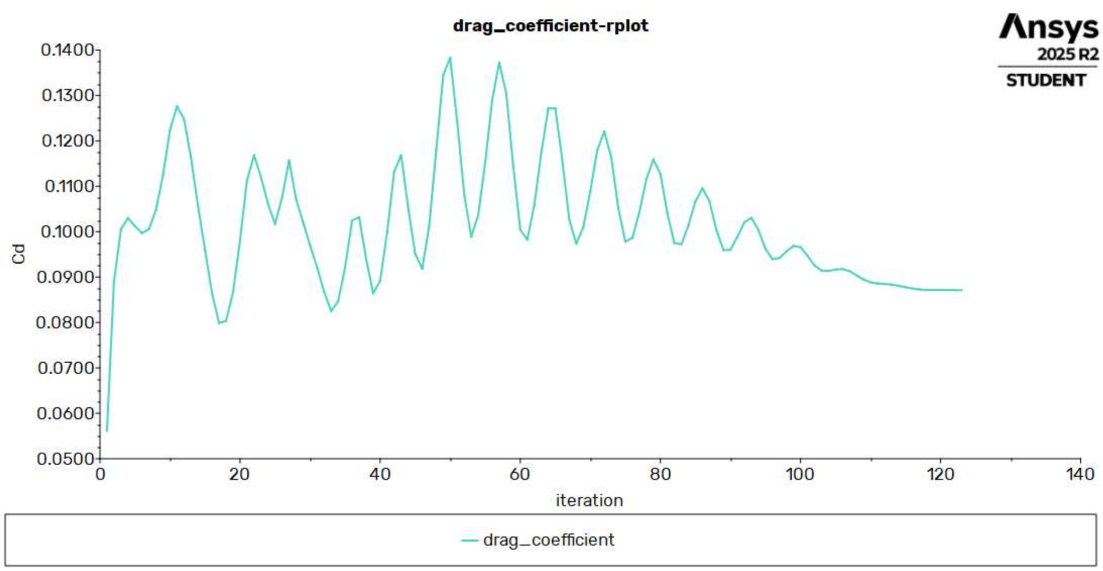

Figure 65 – Drag coefficient

5.4.8 Angle of attack 14

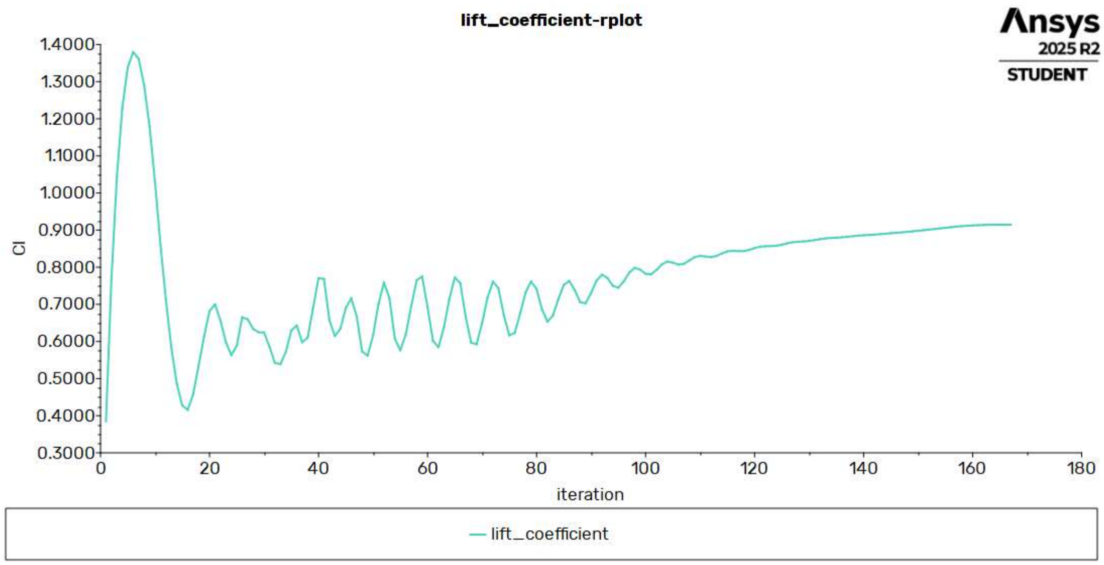

Figure 66 – Lift coefficient

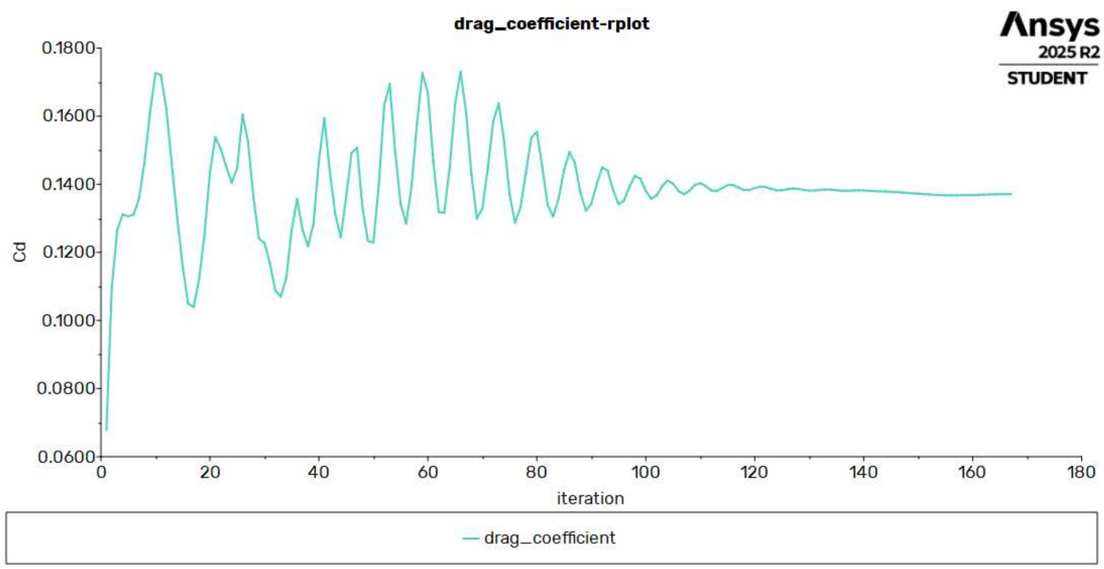

Figure 67 -Drag coefficient

5.5 Results

To investigate how turbulence modelling affects aerodynamic predictions, the air foil's performance was analysed using Computational Fluid Dynamics (CFD). Two RANS-based turbulence models—the Standard *k–ε* (KE) model and the Shear Stress Transport *k–ω* (SST KW) model—were applied for comparison. For both models, a series of simulations was conducted by systematically varying the **angle of attack from 0° to 16°**. The primary aerodynamic force coefficients—namely, the lift coefficient (C_L) and the drag coefficient (C_D)—were calculated at each angle of attack. This approach enabled a comprehensive comparison of the air foil's performance as predicted by each turbulence model across the operational range, including the onset of stall.

The initial phase of the computational analysis was conducted using the Standard k-epsilon (k-ε) turbulence model. For this model, a series of simulations was run by systematically varying the angle of attack from 0° to 16°. At each angle, the resulting lift coefficient (C_L) and drag coefficient (C_D) were calculated to quantify the airfoil's aerodynamic performance.

5.6 Lift Coefficient (C_L) Analysis

5.6.1 Standard KE model

Table 7 -Angle of attack vs Lift coefficient at Standard KE model

AOA	Velocity_X	Velocity_Y	Lift coefficient
0	7.5	0	0.41647076
2	7.495431203	0.261746225	0.62731829
4	7.481730377	0.523173553	0.83717901
6	7.458914215	0.783963475	1.0370278
8	7.427010516	1.043798257	1.2233617
10	7.386058148	1.302361333	1.381645
12	7.336107006	1.559337681	1.4947617
14	7.277217947	1.814414217	1.5263049
16	7.20946272	2.067280169	1.4056102

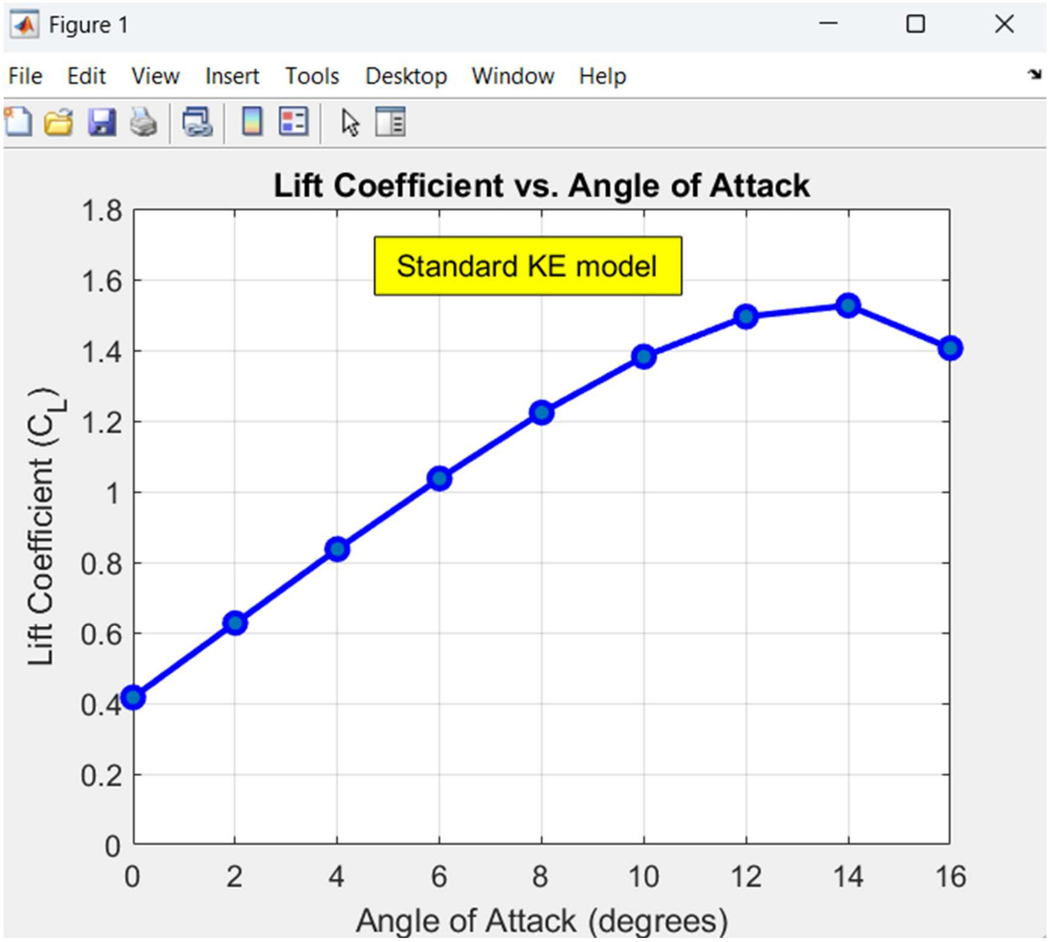

Figure 68 -Lift coefficient vs Angle of attack at Standard KE model

The graph of the lift coefficient (C_L) versus the angle of attack shows a distinct and expected trend.

- Linear Region: From an angle of attack of 0° up to approximately 12°, the lift coefficient increases in a nearly linear fashion. This indicates that in this range, the flow remains largely attached to the airfoil's upper surface, and lift is generated effectively.

- Stall Condition: Beyond 12°, the rate of increase in lift begins to slow. The maximum lift coefficient (C_L), with a value of approximately 1.54, is achieved at an angle of attack of 14°. After this peak, the lift coefficient drops to about 1.4 at 16°. This sharp drop in lift after reaching a maximum value is the definitive indicator of an aerodynamic stall. Based on these results, the critical angle of attack for this airfoil is approximately 14°.

5.6.2 SST KW model

Table 8 - Angle of attack vs Lift coefficient at SST KW model

AOA	Velocity_X	Velocity_Y	Lift coefficient
0	7.5	0	0.37675255
2	7.495431203	0.261746225	0.57382556
4	7.481730377	0.523173553	0.76620827
6	7.458914215	0.783963475	0.9447253
8	7.427010516	1.043798257	1.0945103
10	7.386058148	1.302361333	1.1851692
12	7.336107006	1.559337681	1.0464561
14	7.277217947	1.814414217	0.91456347
16	7.20946272	2.067280169	0.80176324

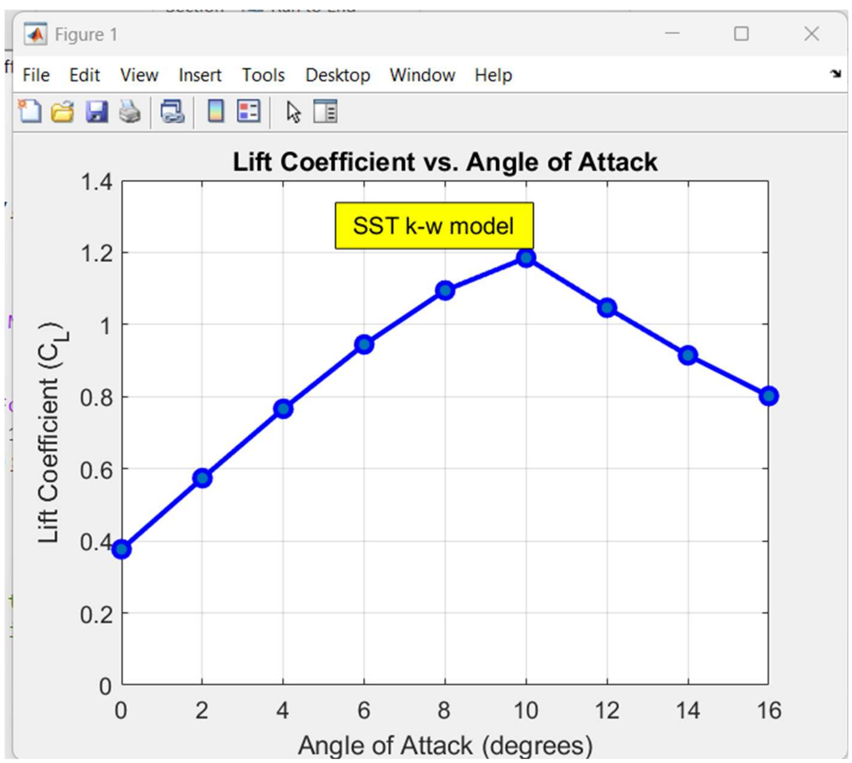

Figure 69 - Lift coefficient vs Angle of attack

The graph of the lift coefficient (C_L) versus the angle of attack for the SST k-w model shows a distinct and expected trend.

- **Linear Region:** From an angle of attack of 0° up to approximately 8°, the lift coefficient increases in a nearly linear fashion. This indicates that in this range, the flow remains largely attached to the airfoil's upper surface, and lift is generated effectively.

- **Stall Condition:** Beyond 8°, the rate of increase in lift begins to slow significantly. The maximum lift coefficient, with a value of approximately 1.19, is achieved at an angle of attack of 10°. After this peak, the lift coefficient drops sharply, falling to about 1.05 at 12° and continuing to decrease to 0.80 at 16°. This definitive drop in lift after reaching a maximum value indicates an aerodynamic stall. Based on these results, the critical angle of attack for this airfoil is approximately 10°

5.7 Drag Coefficient (C_D) Analysis

5.7.1 Standard KE model

Table 9 - Angle of attack vs Drag coefficient at Standard KE model

AOA	Velocity_X	Velocity_Y	Drag coefficient
0	7.5	0	0.017567372
2	7.495431203	0.261746225	0.018464407
4	7.481730377	0.523173553	0.020777939
6	7.458914215	0.783963475	0.025092899
8	7.427010516	1.043798257	0.032213806
10	7.386058148	1.302361333	0.041706486
12	7.336107006	1.559337681	0.056401842
14	7.277217947	1.814414217	0.078582261
16	7.20946272	2.067280169	0.11492205

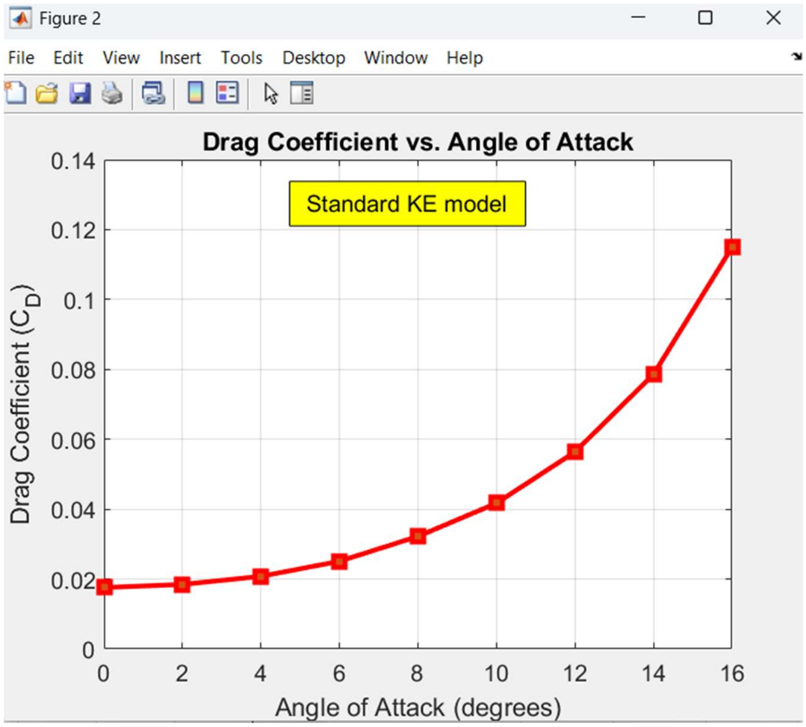

Figure 70 - Angle of attack vs Drag coefficient at Standard KE model

The drag coefficient (C_D) plot also demonstrates classic airfoil behavior.

- At low angles of attack (0° to 4°), the drag is minimal.
- As the angle of attack increases, the drag coefficient begins to rise at an increasingly steep rate. This is due to the combined effects of skin friction and growing pressure drag from flow separation near the trailing edge.
- Notably, the increase in drag becomes particularly aggressive between 12° and 16°, which corresponds to the same region where the airfoil approaches and enters stall. This sharp rise in drag is a direct consequence of the massive flow separation that causes the loss of lift.

In summary, the results from the Standard k-ε model clearly capture the stall phenomenon, identifying the critical angle of attack at approximately **14°**, which is characterized by the peak lift coefficient and a concurrent sharp increase in drag.

5.7.2 SST KW model

Table 10 - Angle of attack vs Drag coefficient

AOA	Velocity_X	Velocity_Y	Drag coefficient
0	7.5	0	0.017523173
2	7.495431203	0.261746225	0.018549105
4	7.481730377	0.523173553	0.021196927
6	7.458914215	0.783963475	0.025566404
8	7.427010516	1.043798257	0.033240192
10	7.386058148	1.302361333	0.046064688
12	7.336107006	1.559337681	0.087113577
14	7.277217947	1.814414217	0.13711361
16	7.20946272	2.067280169	0.18410973

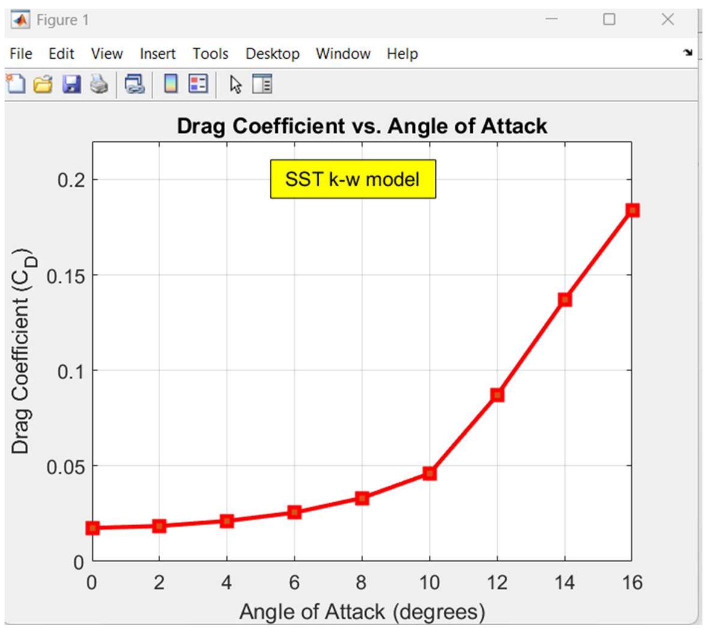

Figure 71 - Drag coefficient vs Angle of attack

The drag coefficient plot for the SST k-w model also demonstrates classic airfoil behavior.

- **At low angles of attack (0° to 8°),** the drag is minimal and increases at a slow, shallow rate.
- **As the angle of attack increases** beyond 8°, the drag coefficient begins to rise at an increasingly steep rate. This is due to the combined effects of skin friction and growing pressure drag as flow separation begins near the trailing edge.

- **Notably, the increase in drag becomes particularly aggressive between 10° and 16°**, which corresponds to the exact region where the airfoil enters stall (as identified in the lift plot). This sharp rise in drag is a direct consequence of the massive flow separation that causes the loss of lift.

5.7.3 Results using Standard K- E model

5.7.3.1 Angle of attack 0

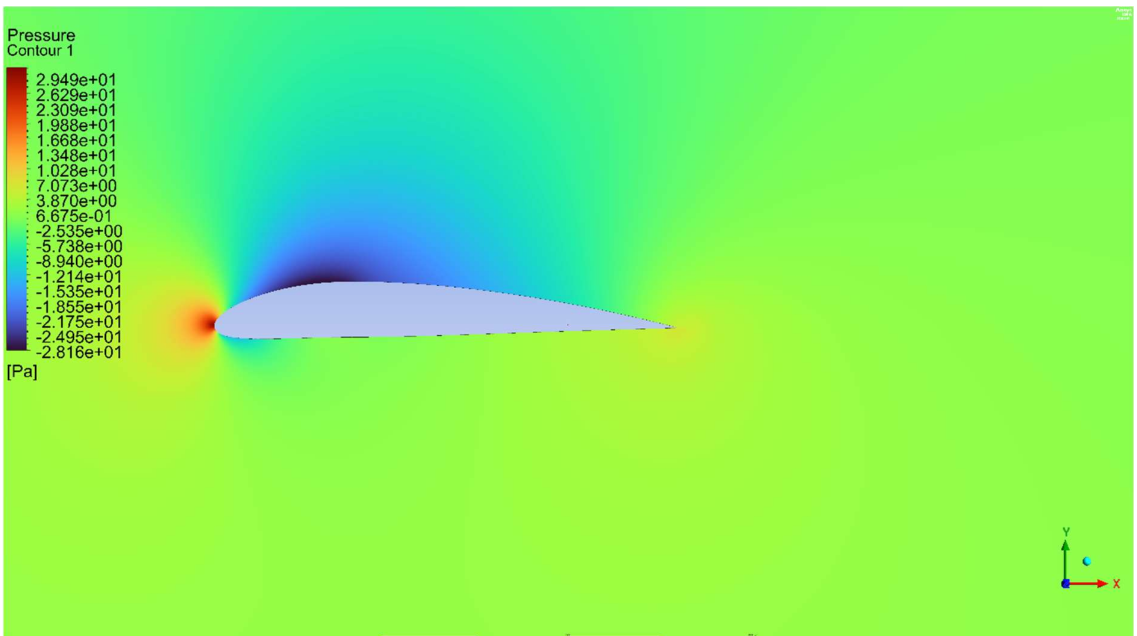

Figure 72 -Static pressure

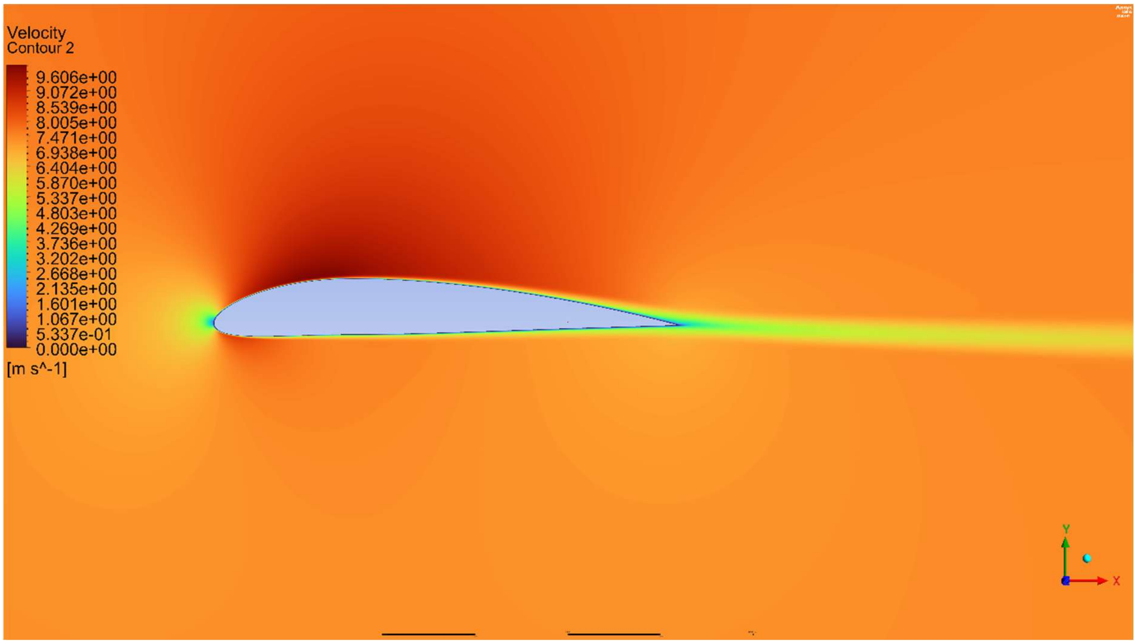

Figure 73 - Velocity distribution

5.7.3.2 Angle of attack 2

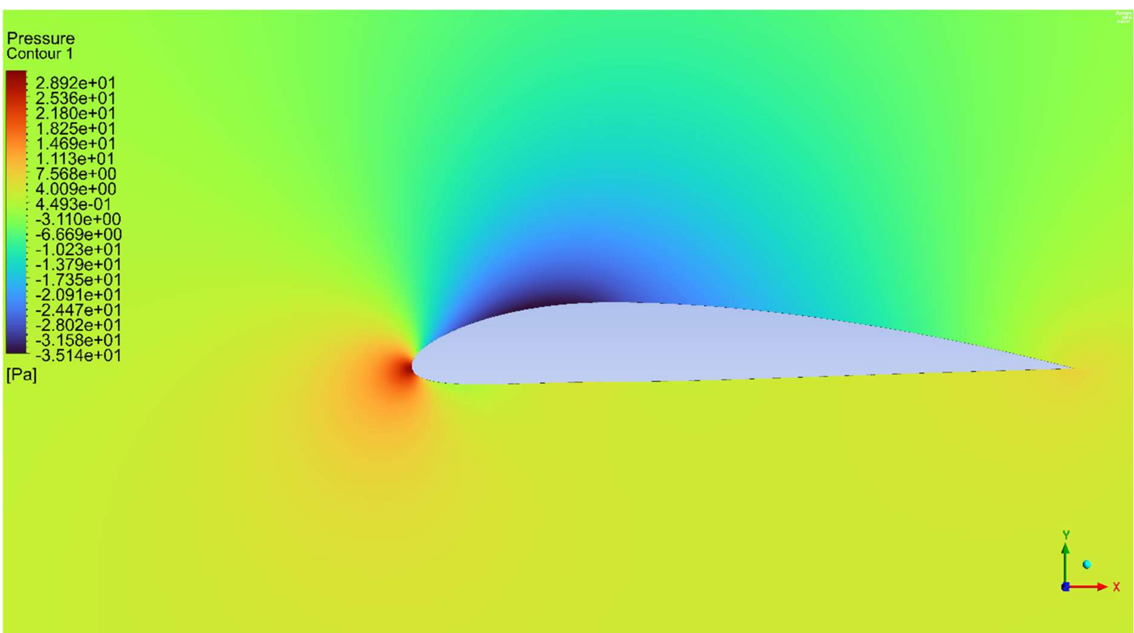

Figure 74 - Static pressure

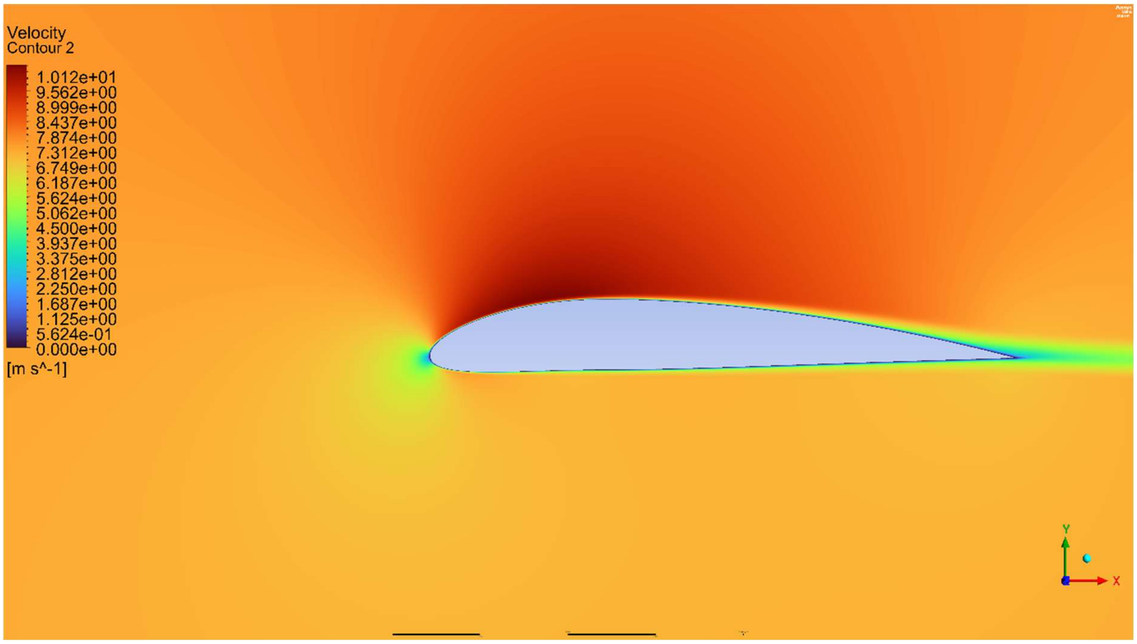

Figure 75- Velocity magnitude

5.7.3.3 Angle of attack 4

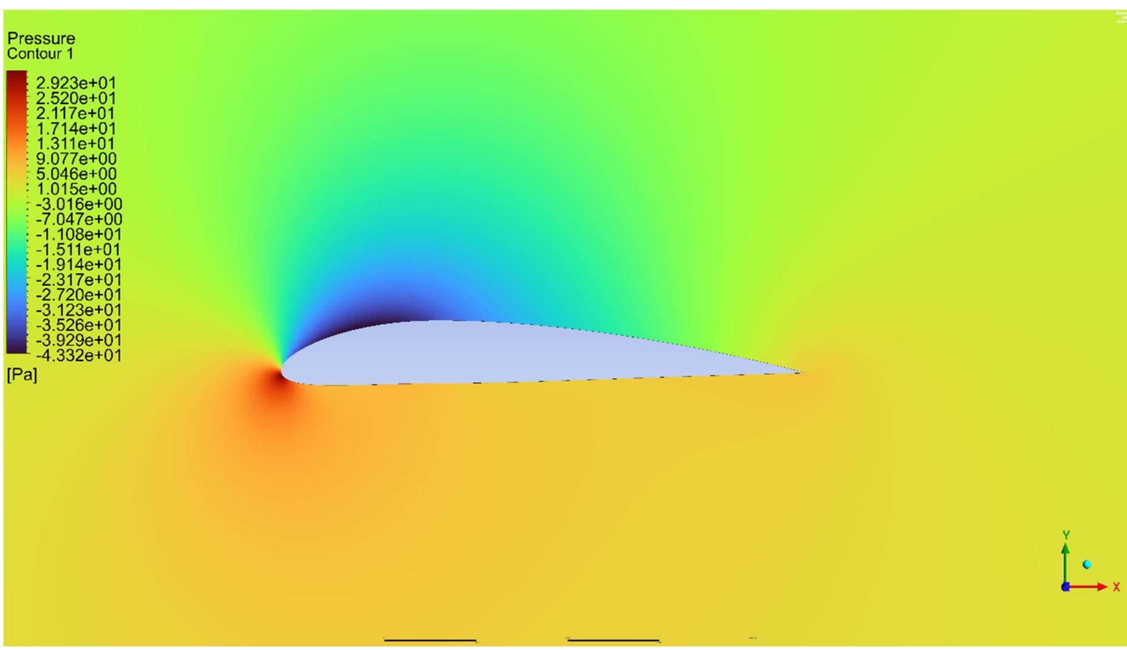

Figure 76-Static pressure

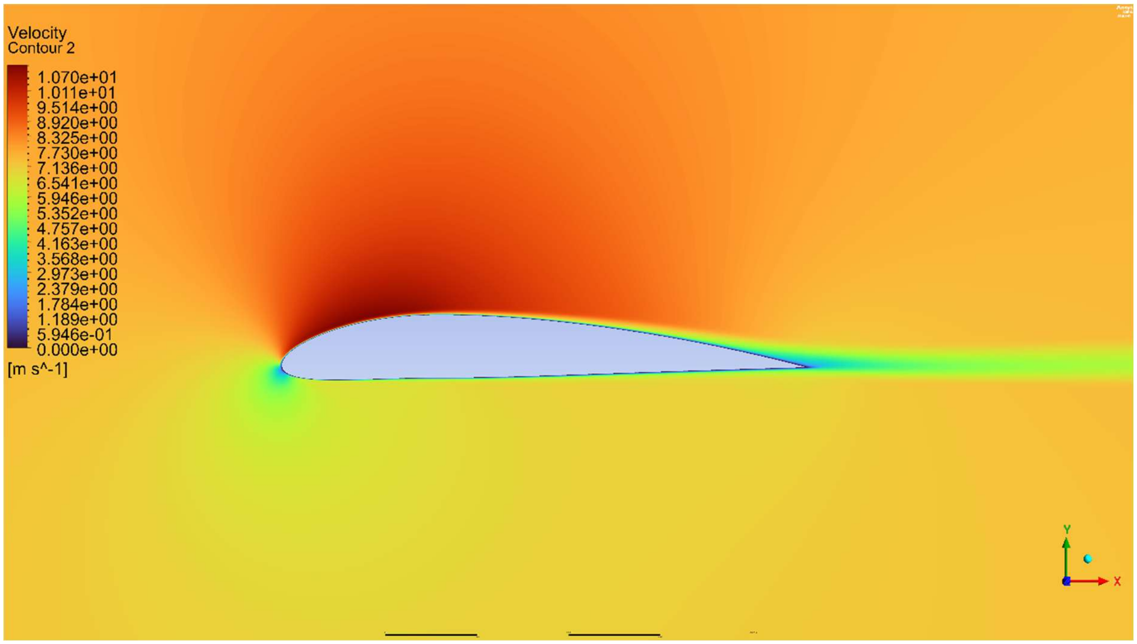

Figure 77 - Velocity magnitude

5.7.3.4 Angle of attack 6

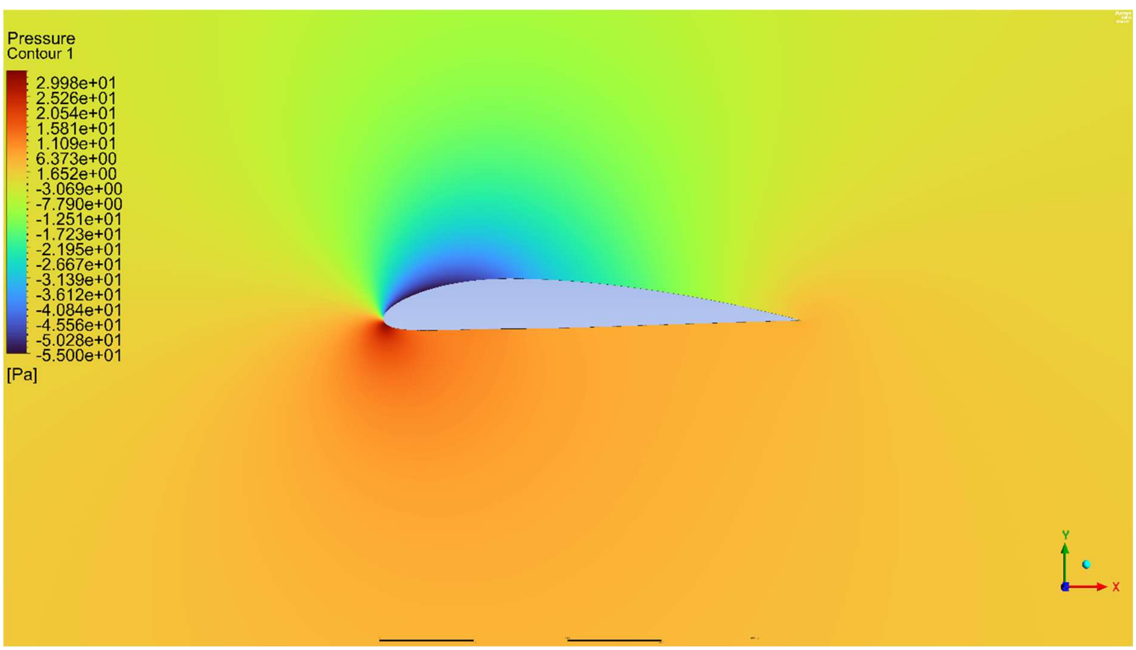

Figure 78-Static pressure

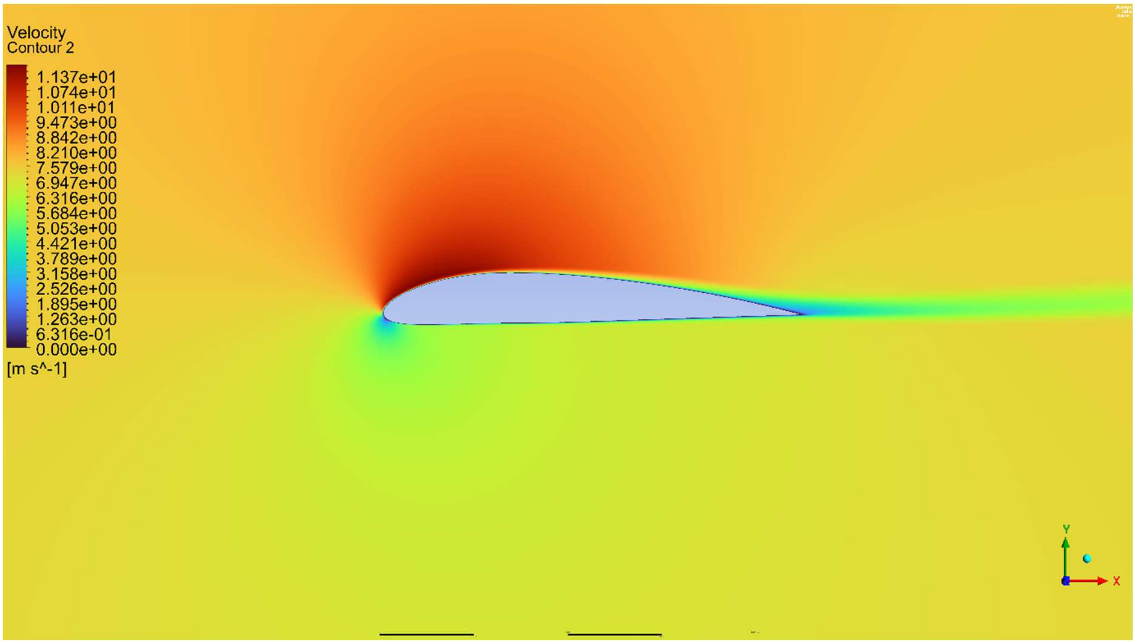

Figure 79-Velocity magnitude

5.7.3.5 Angle of attack 8

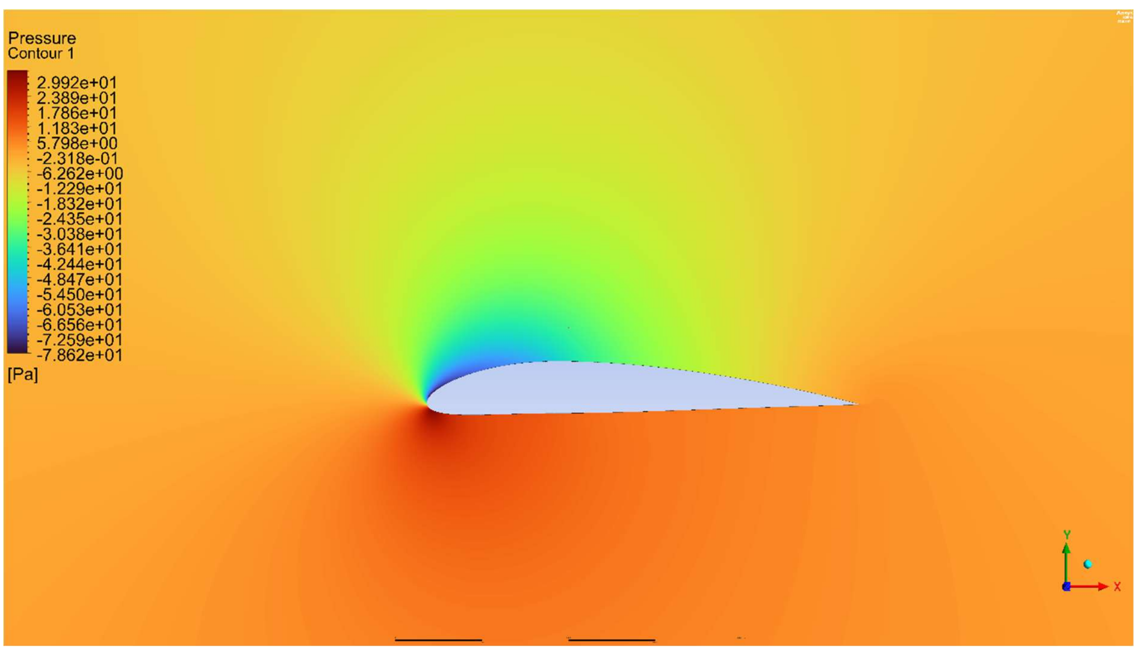

Figure 80 - Static pressure

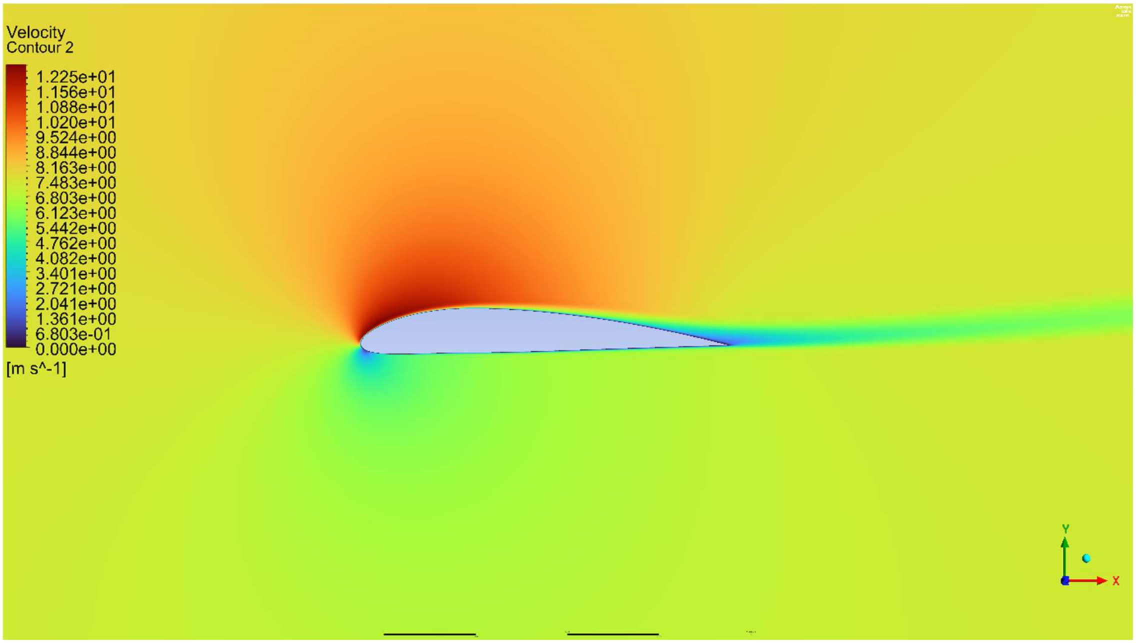

Figure 81 - Velocity magnitude

5.7.3.6 Angle of attack 10

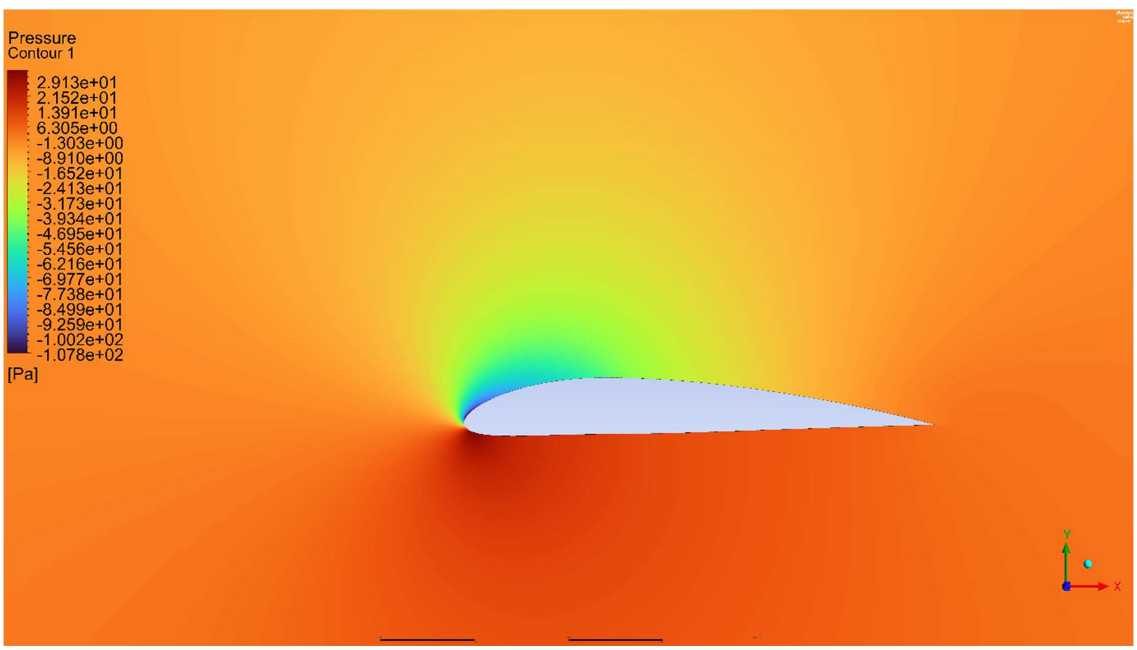

Figure 82 - Static pressure

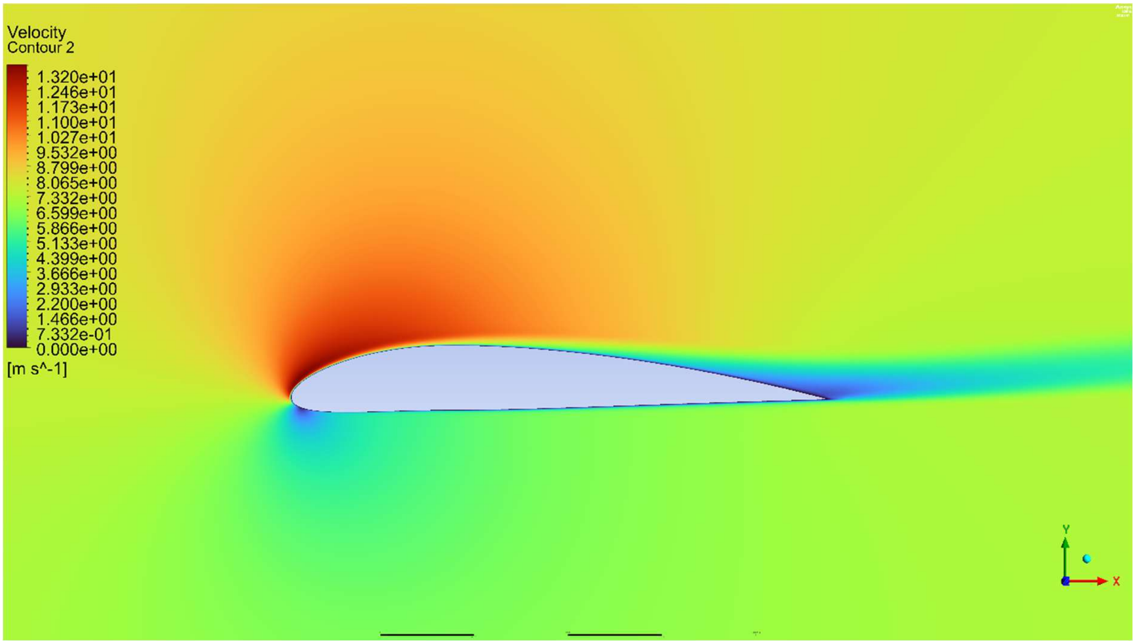

Figure 83- Velocity magnitude

5.7.4 Angle of attack 12

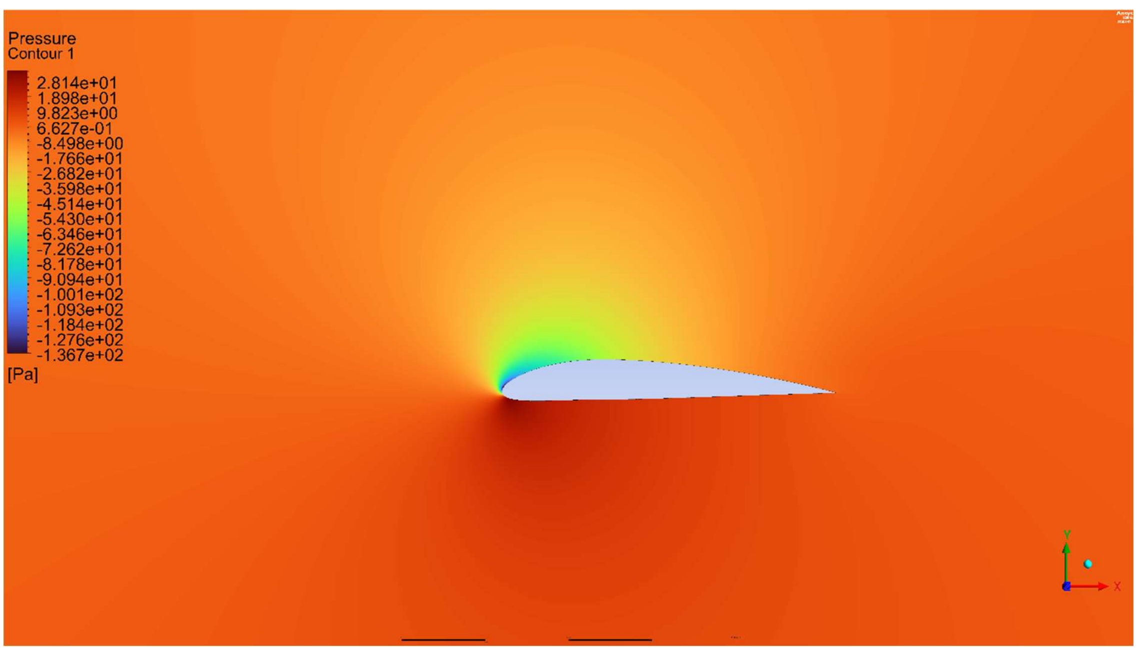

Figure 84-Static pressure

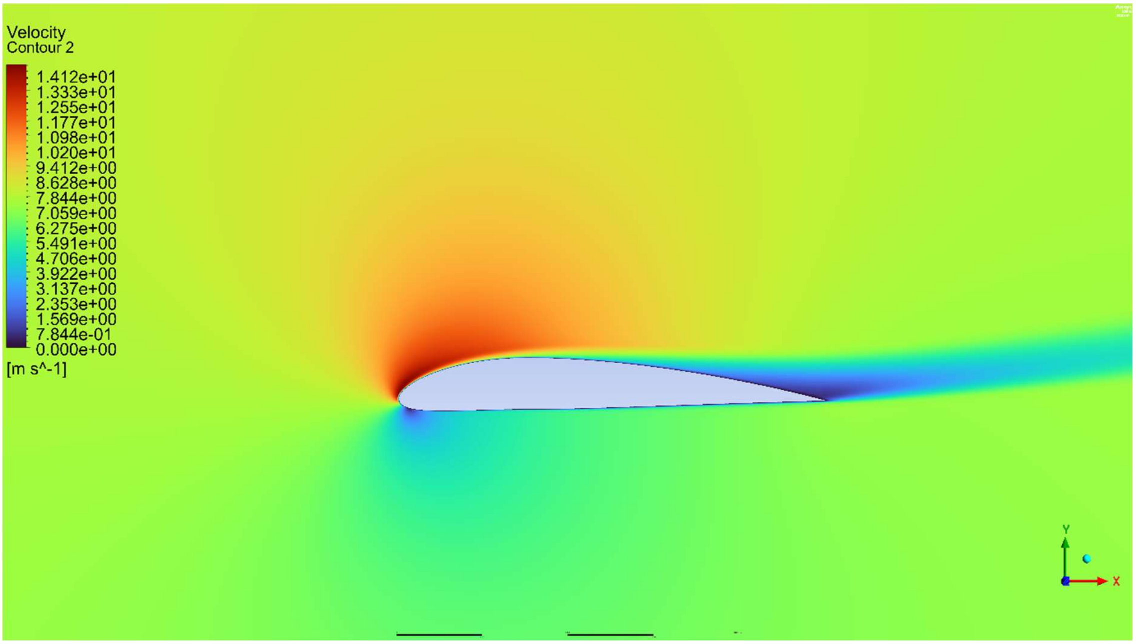

Figure 85 - Velocity magnitude

5.7.5 Angle of attack 14

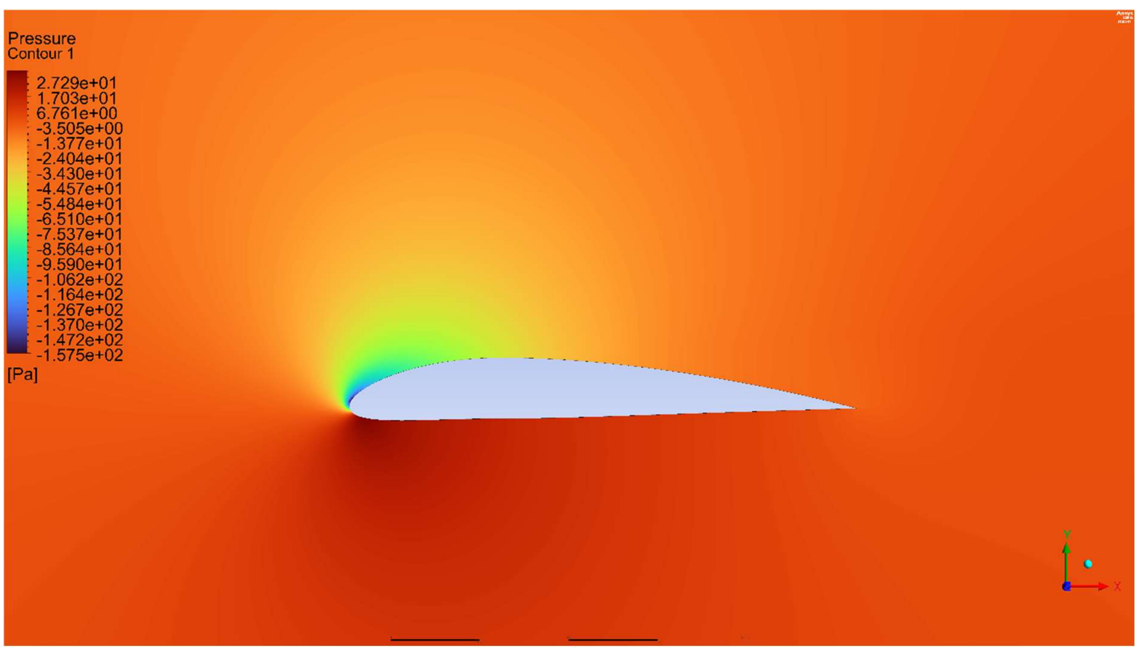

Figure 86-Static pressure

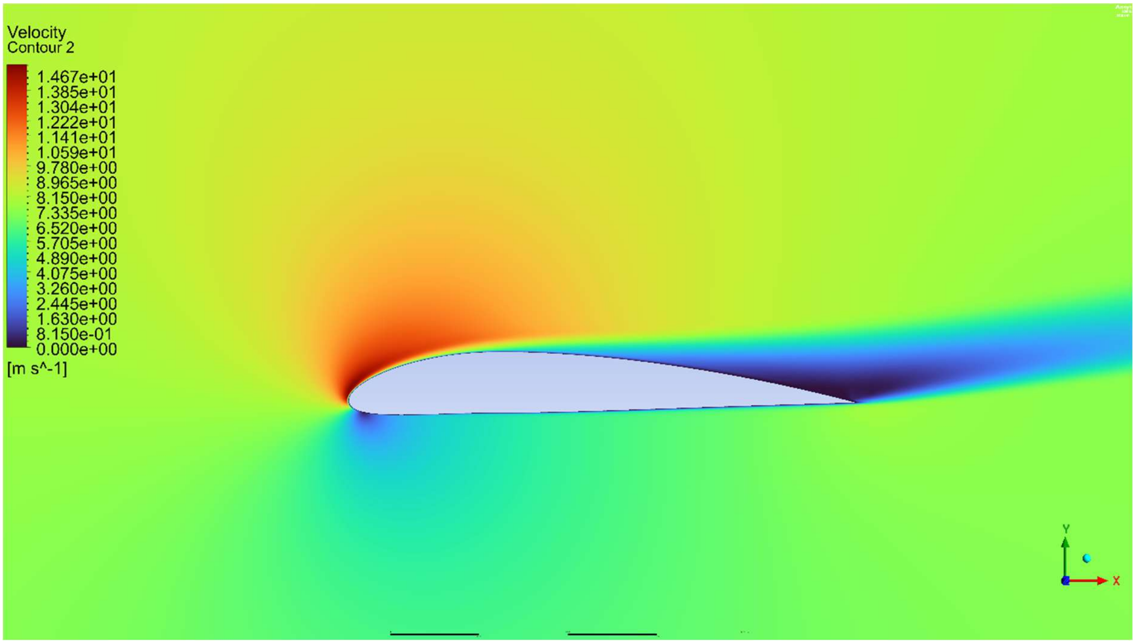

Figure 87 -Velocity magnitude

5.7.6 Angle of attack 16

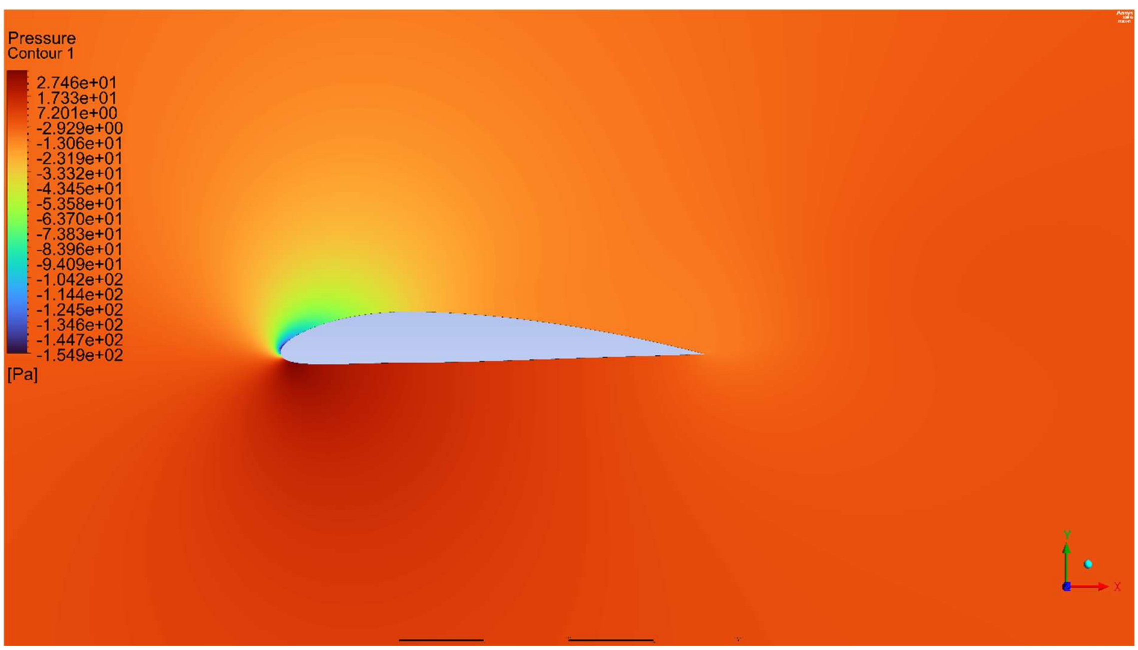

Figure 88 - Static pressure

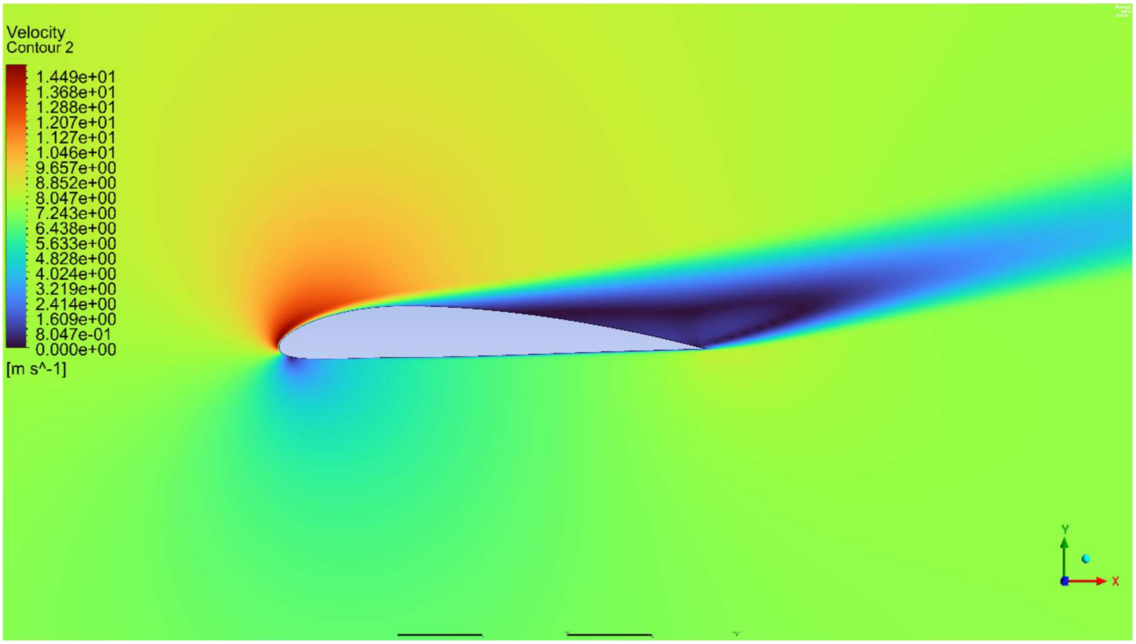

Figure 89 - Velocity magnitude

5.7.7 Results using SST KW model

5.7.7.1 Angle of attack 0

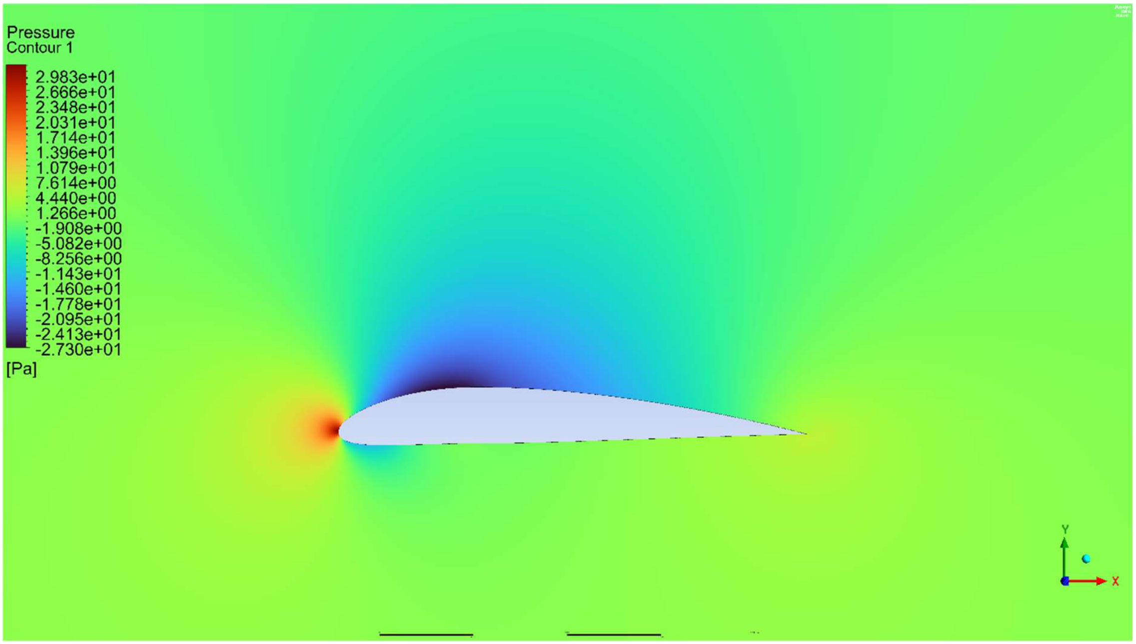

Figure 90- Static pressure

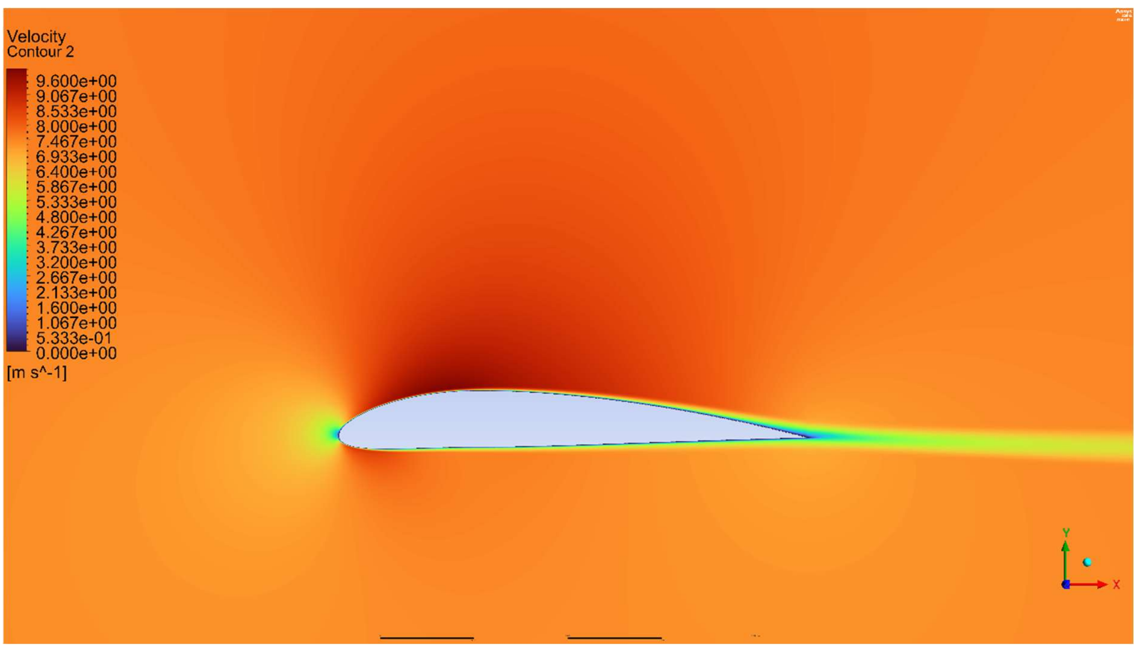

Figure 91 – Velocity magnitude

Angle of attack 2

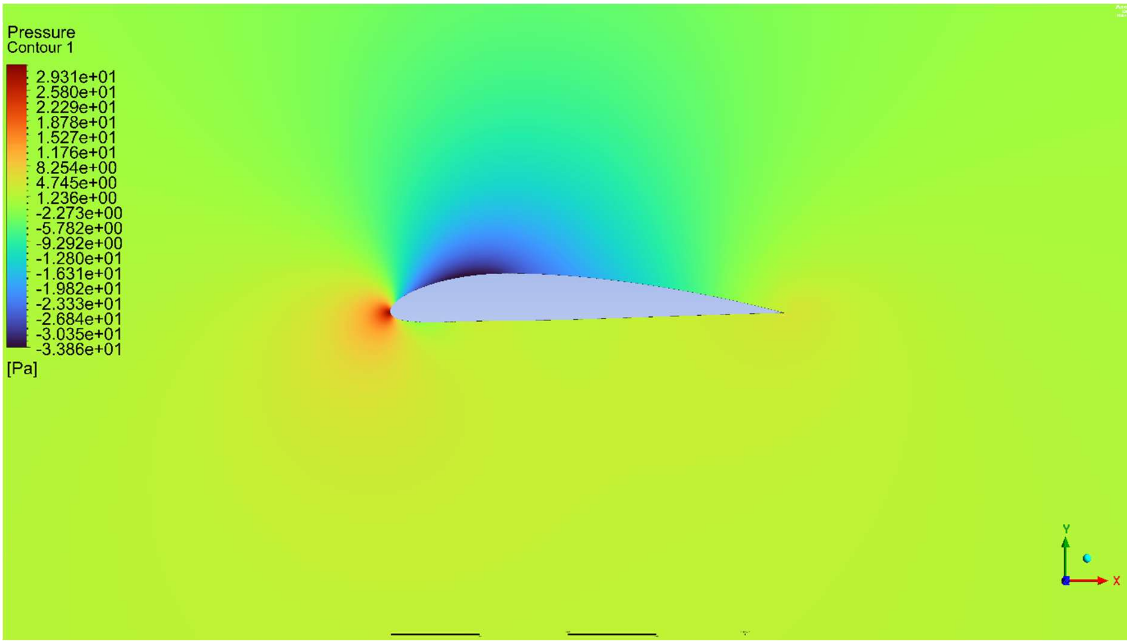

Figure 92-Static pressure

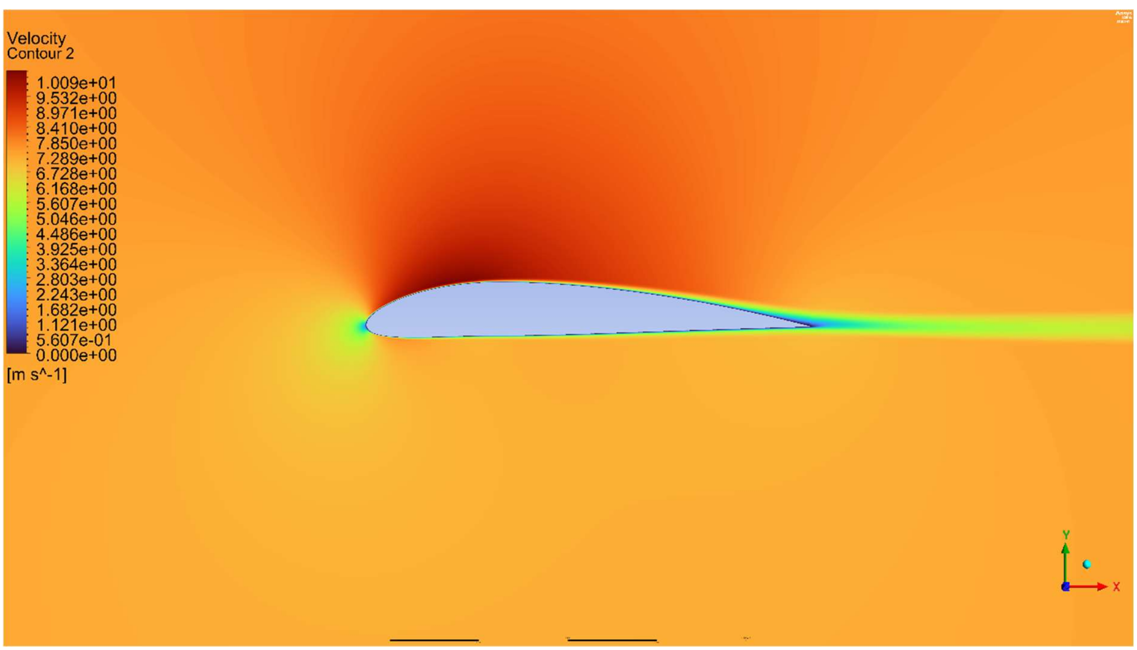

Figure 93-Velocity magnitude

Angle of attack 4

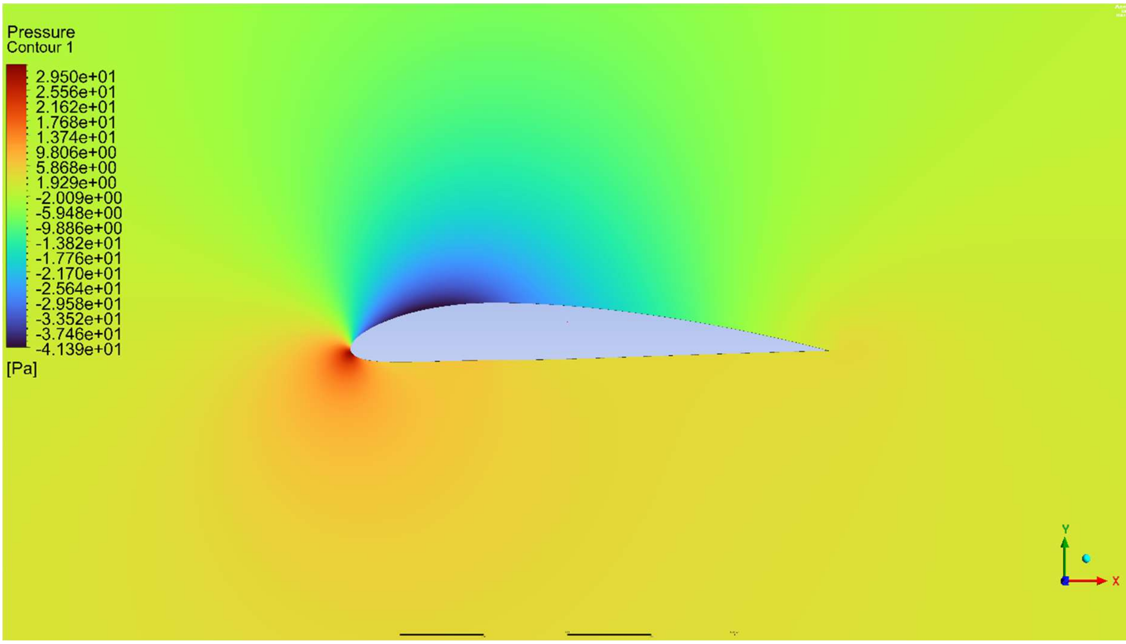

Figure 94-Static pressure

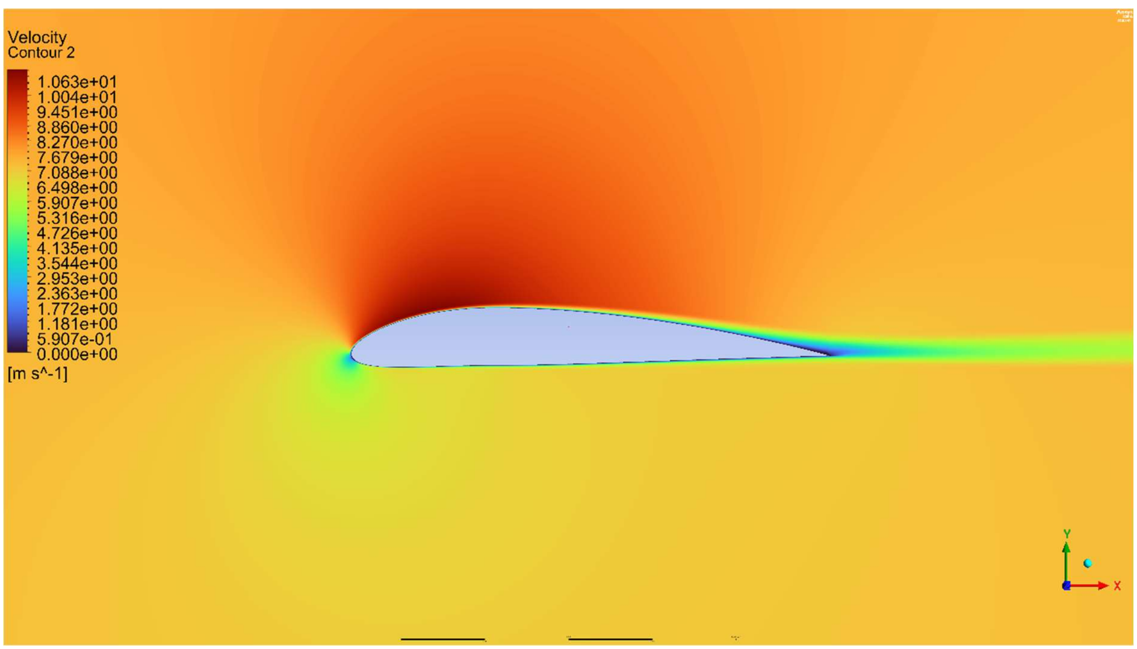

Figure 95 - Velocity magnitude

Angle of attack 6

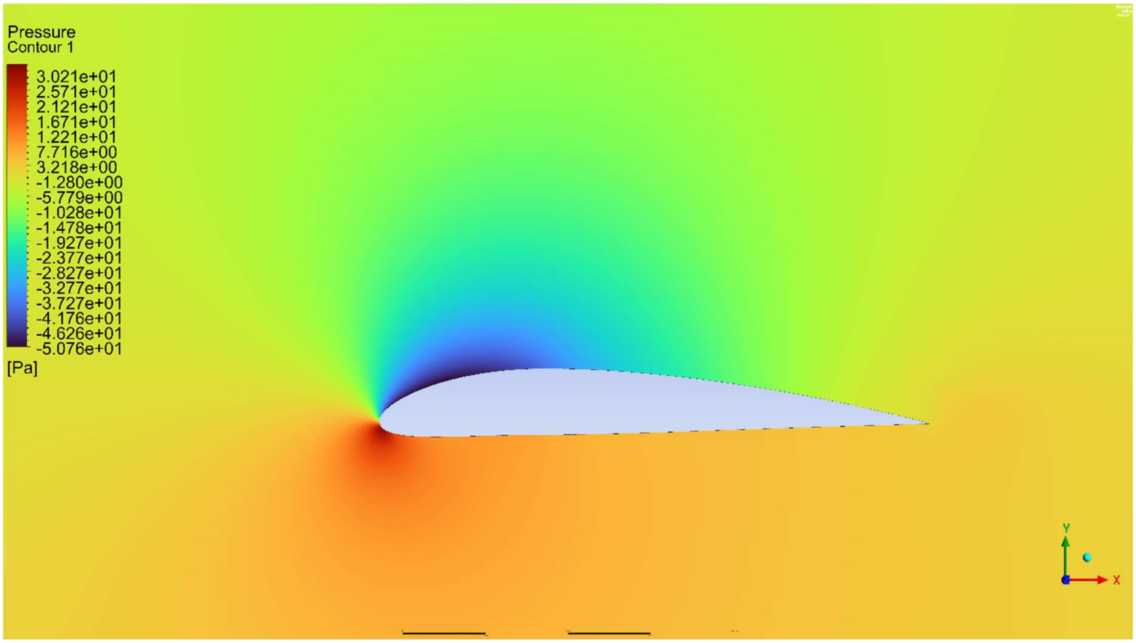

Figure 96 – Static pressure

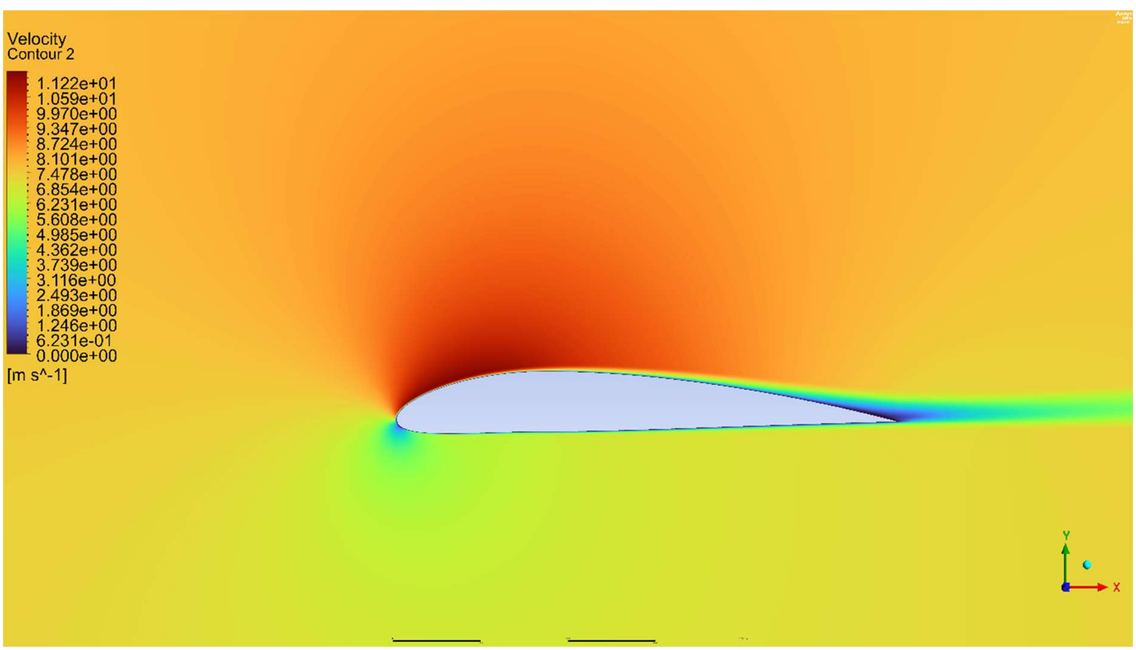

Figure 97 - Velocity magnitude

Angle of attack 8

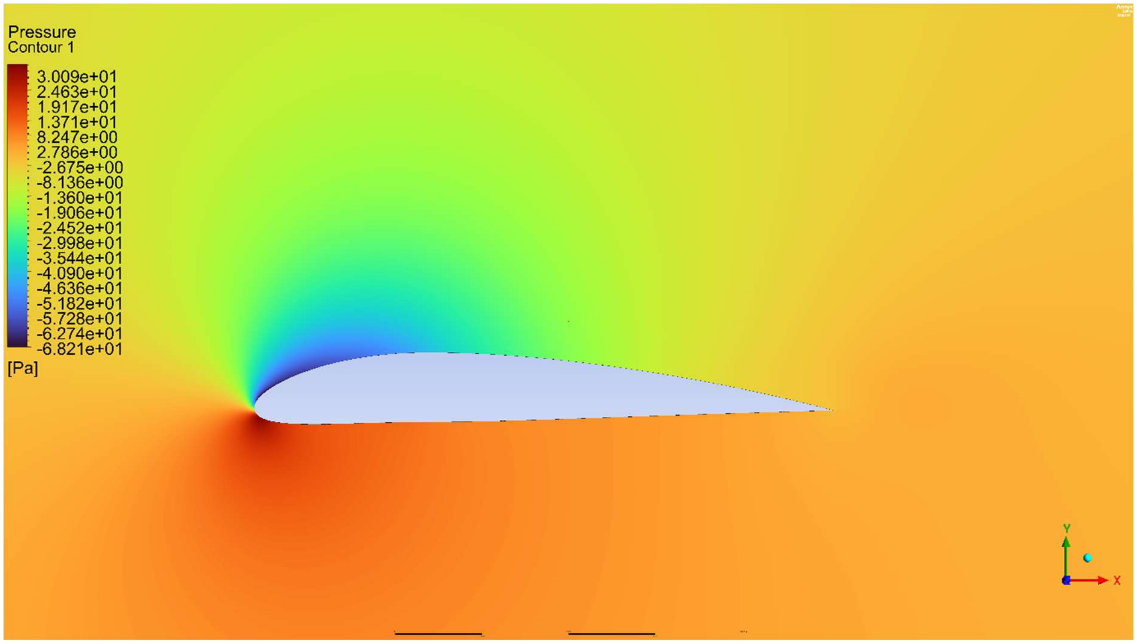

Figure 98 - Static pressure

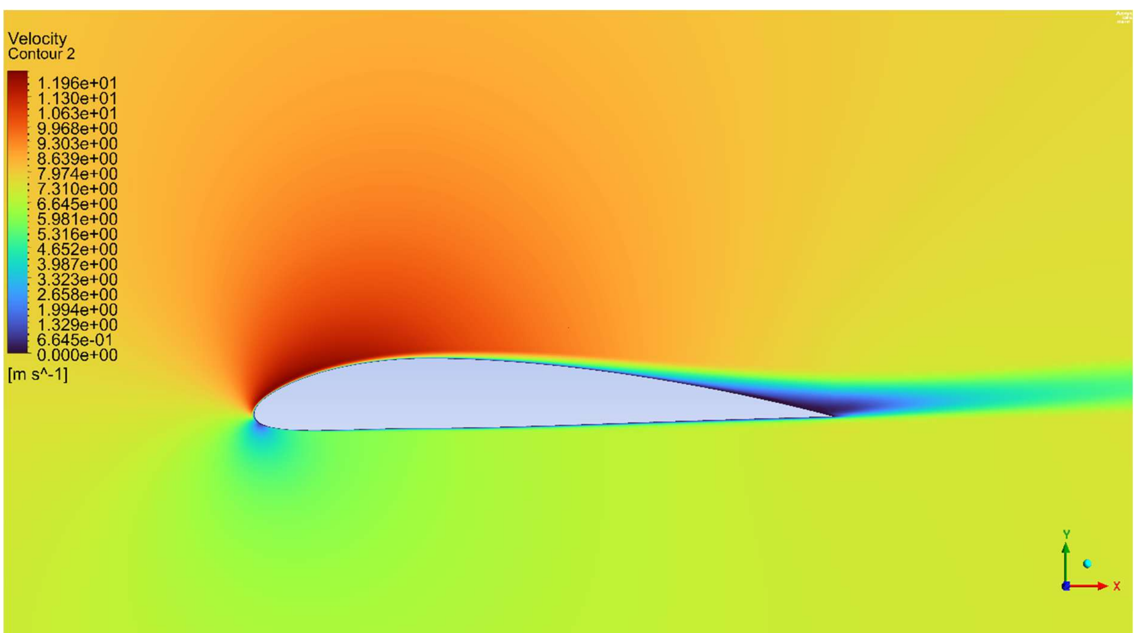

Figure 99 - Velocity magnitude

Angle of attack 10

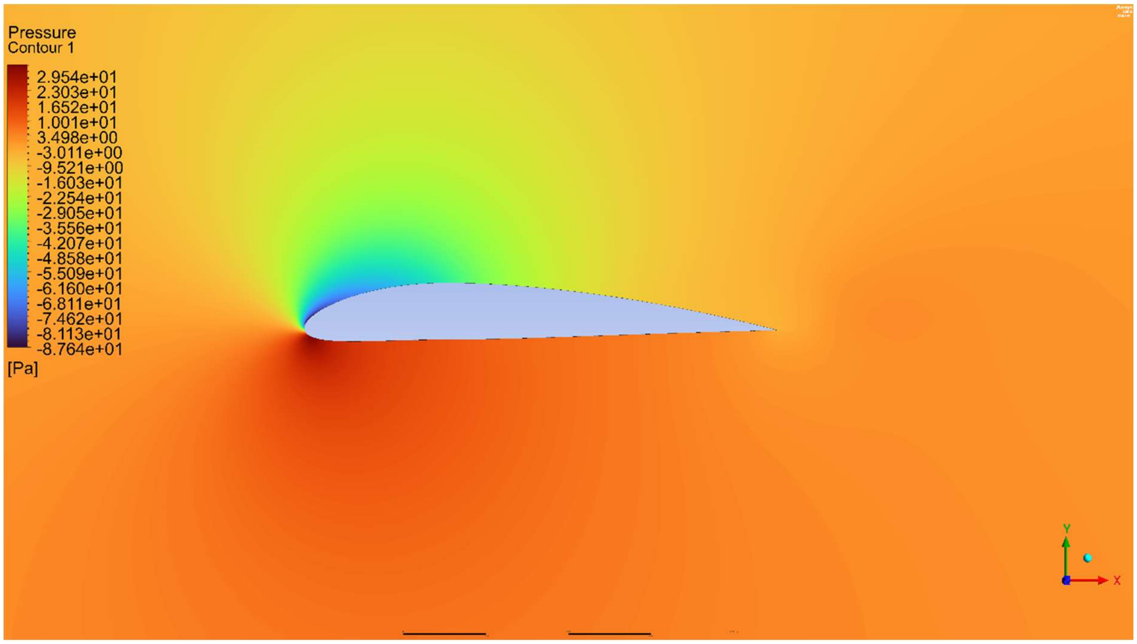

Figure 100 - Static pressure

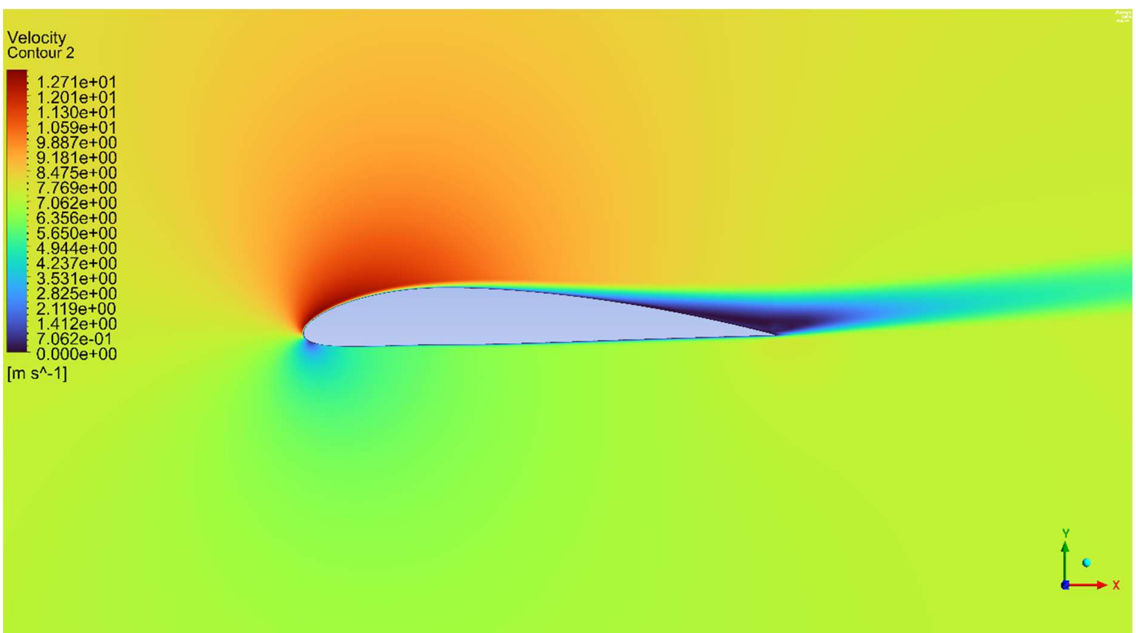

Figure 101 - Velocity magnitude

Angle of attack 12

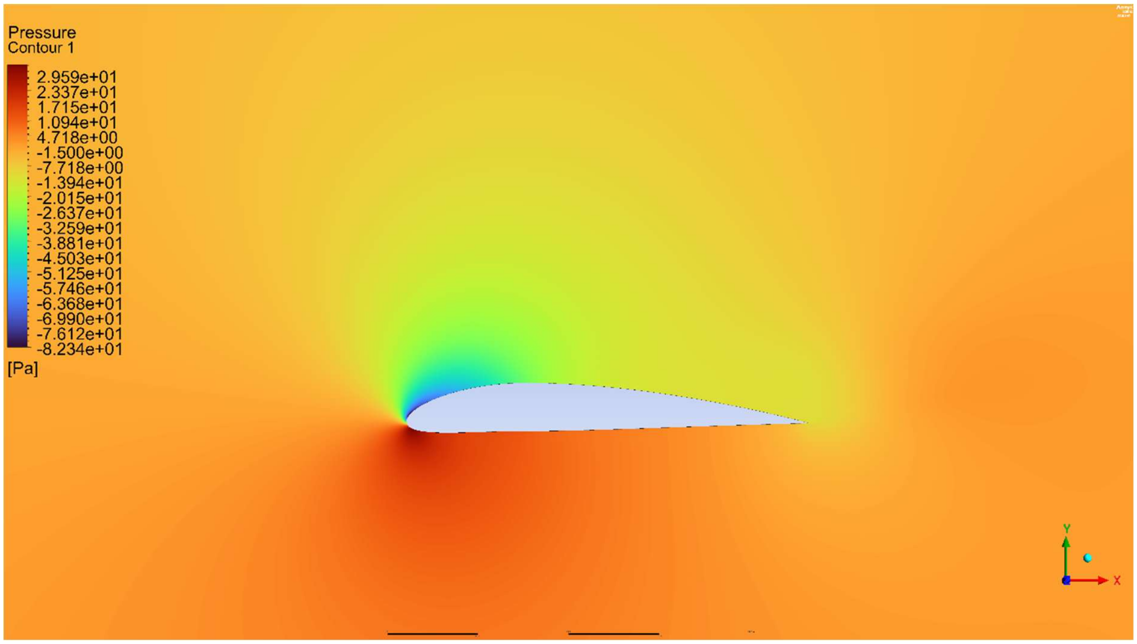

Figure 102 - Static pressure

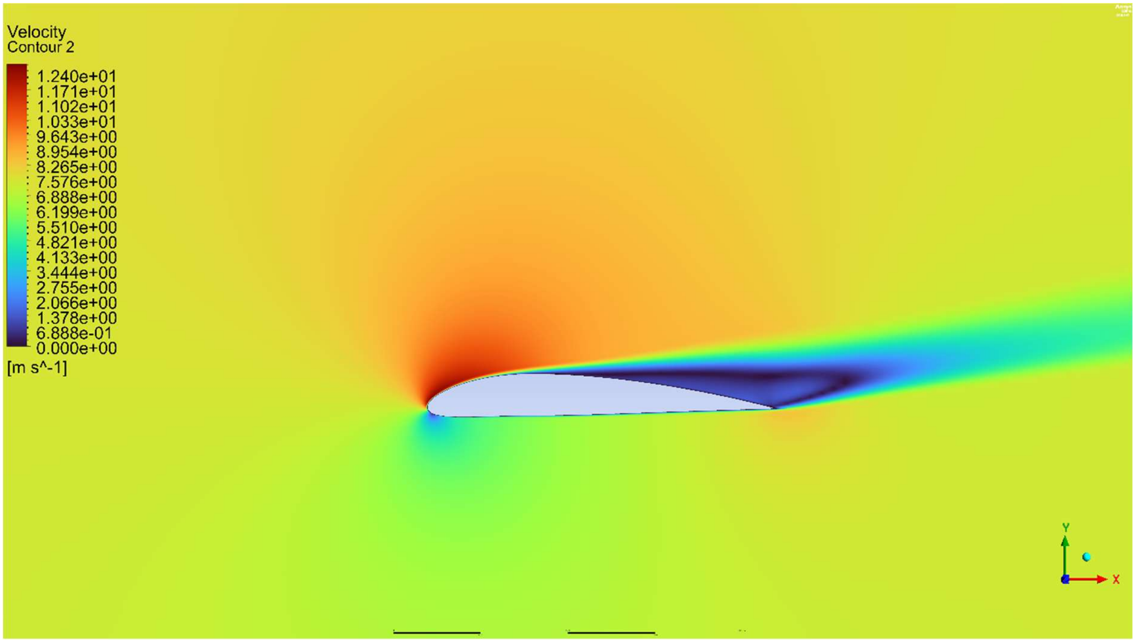

Figure 103 - Velocity magnitude

Angle of attack 14

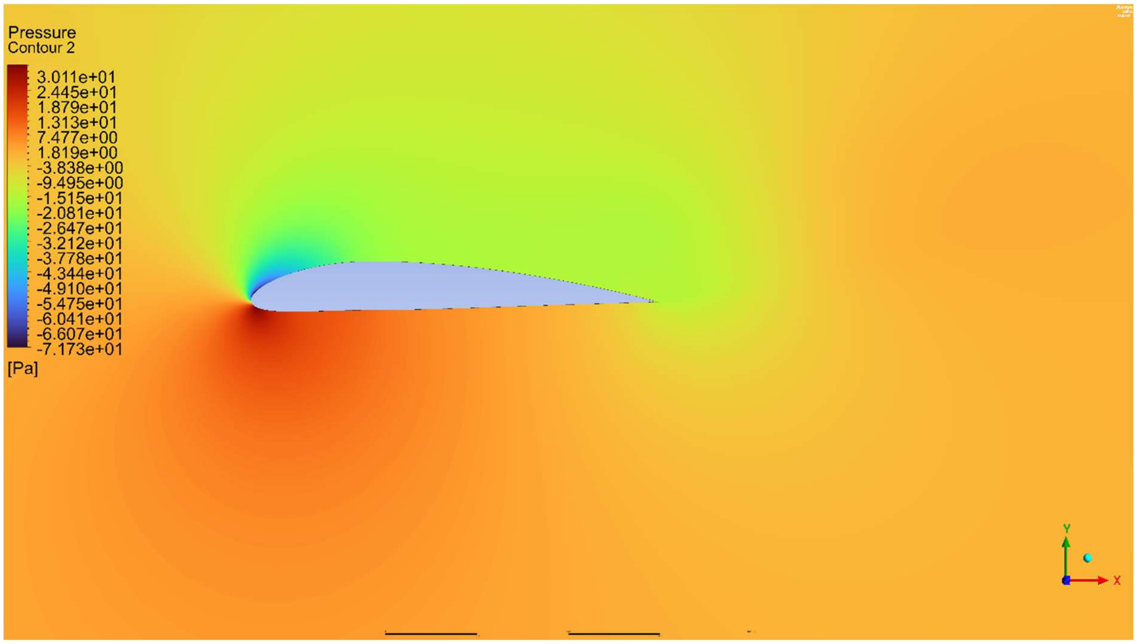

Figure 104 - Static pressure

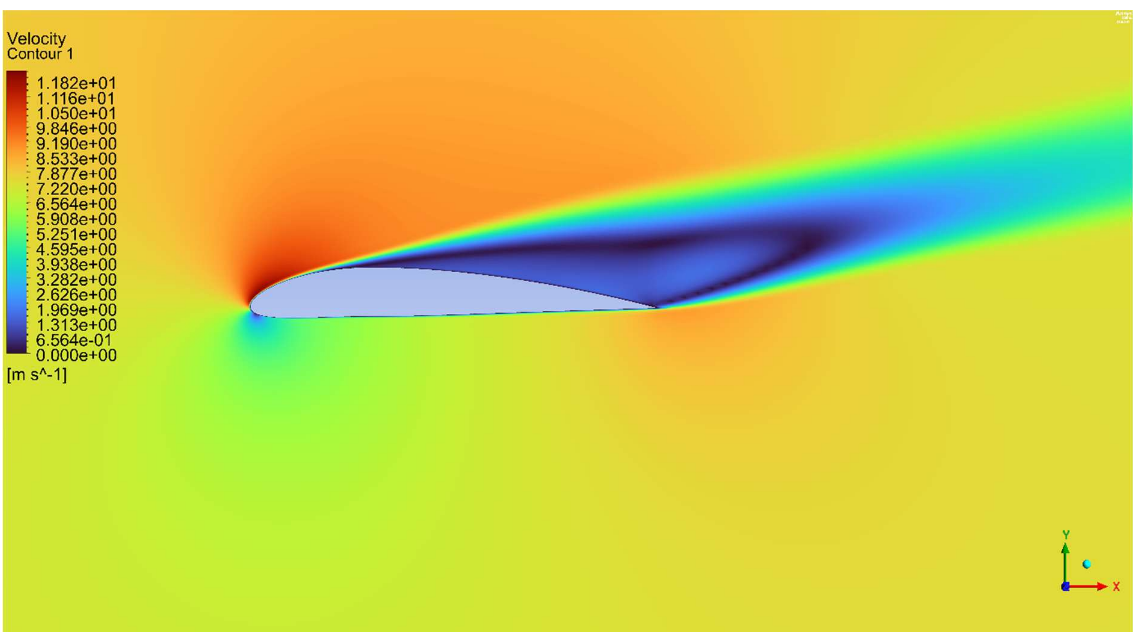

Figure 105 - Velocity magnitude

5.7.8 Angle of attack 16

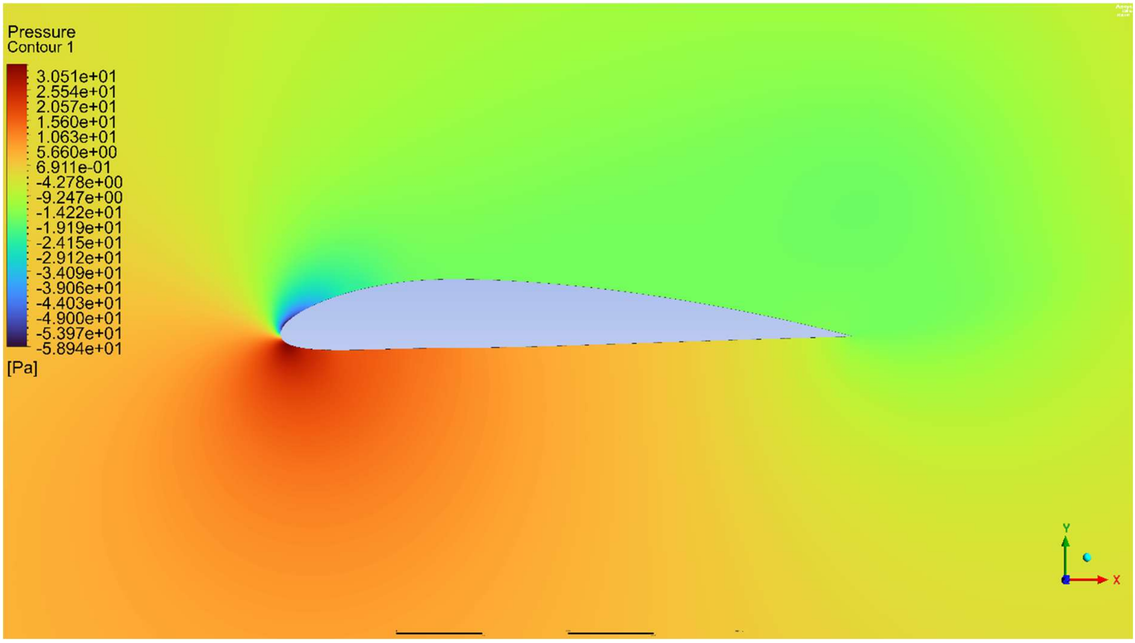

Figure 106 - Static pressure

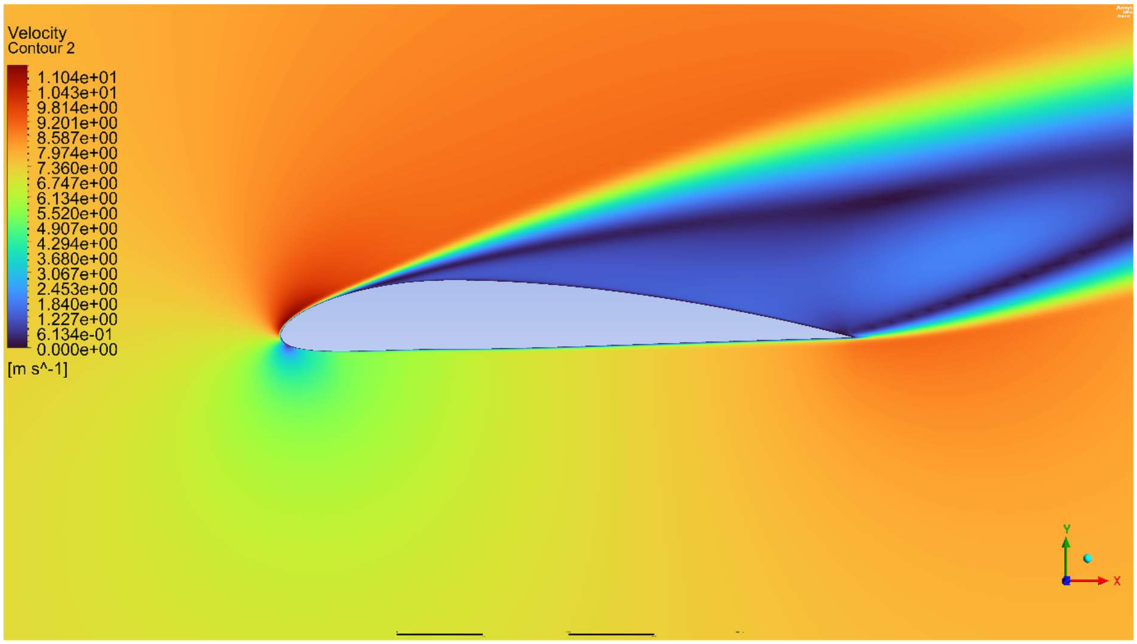

Figure 107 -Velocity magnitude

5.7.9 Standard KE model vs SST KW model result comparison

5.7.9.1 Lift coefficient

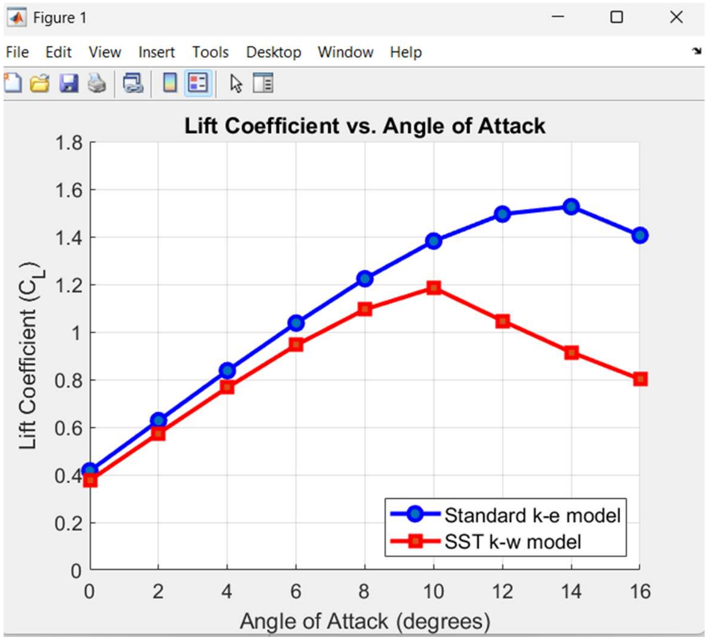

Figure 108 - Lift coefficient vs Angle of attack

This graph compares the lift coefficient (C_L) predicted by the Standard K-E model and the SST k-w model.

- **Lift Prediction:** The Standard k-e model consistently predicts a higher lift coefficient than the SST k-w model across all positive angles of attack.
- **SST k-w Stall:** The SST k-w model shows flow separation (stall) beginning after 10 degrees, reaching a maximum lift coefficient of approximately 1.19.
- **Standard k-e Stall:** The Standard k-e model predicts a delayed stall, reaching a much higher maximum lift coefficient of approximately 1.53 at 14 degrees before lift begins to decrease.

5.7.9.2 Drag coefficient

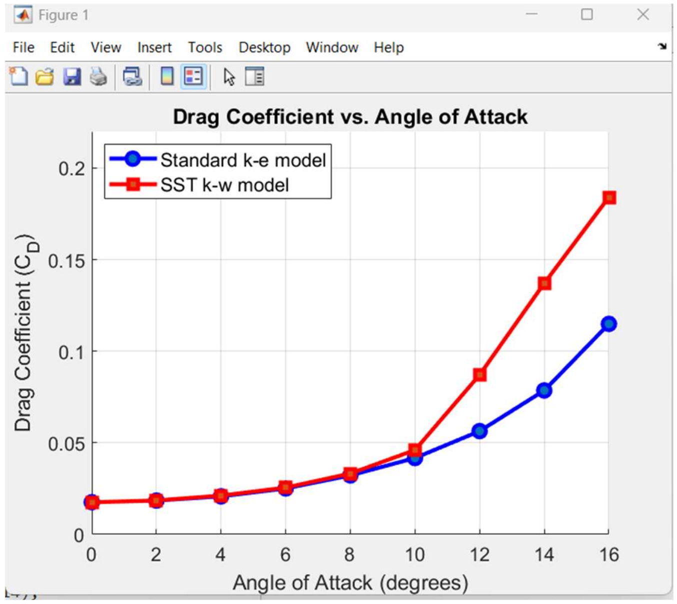

Figure 109 - Drag coefficient vs angle of attack

This graph compares the drag coefficient for the Standard K-E model and the SST k-w model.

- **Low Angles (0-8°):** Both models predict similar, low drag.
- **High Angles (Above 8°):** The models diverge. The SST k-w model shows a much faster increase in drag.
- **Correlation with Lift**: The drag for the SST k-w model increases sharply at 10 degrees, the same point where it stalled in the lift plot.
- **Final Values**: At 16 degrees, the SST k-w model's drag is much higher (around 0.185) than the Standard k-e model's (around 0.115).

5.8 Result validation

5.9 Simulation results vs research paper results

5.9.1 Lift coefficient comparison

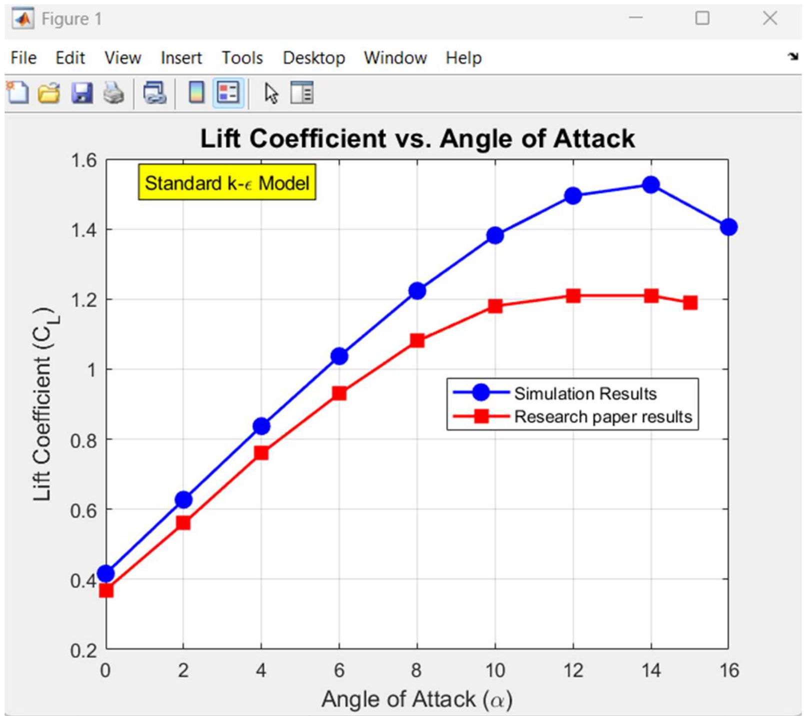

Figure 110 - Standard KE model

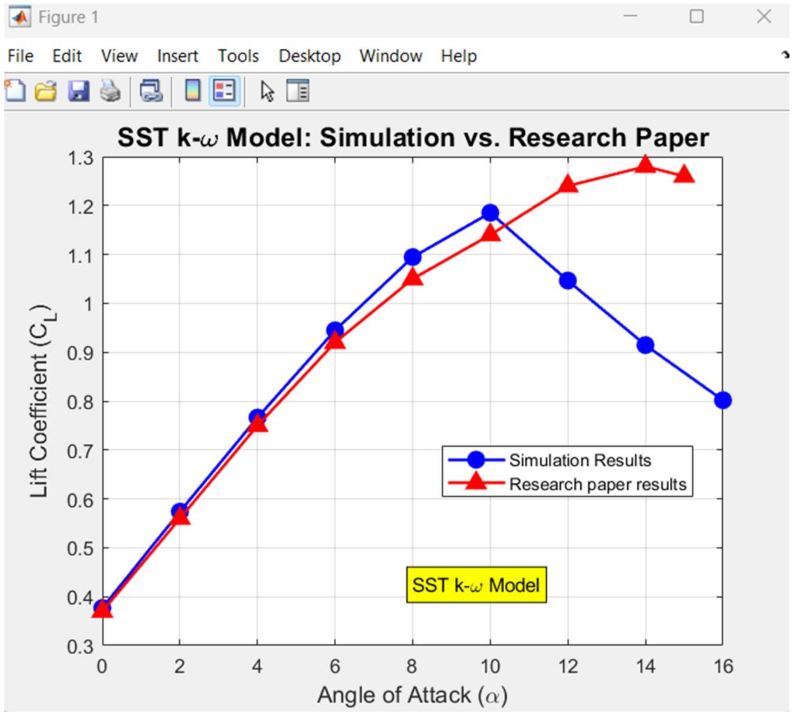

Figure 111 - SST KW model

5.9.2 Drag coefficient comparison

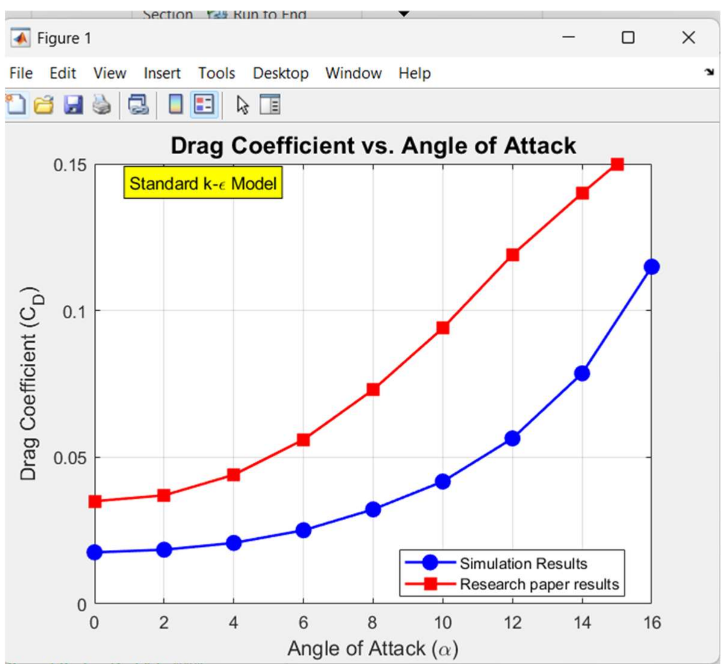

Figure 112 - Standard KE model

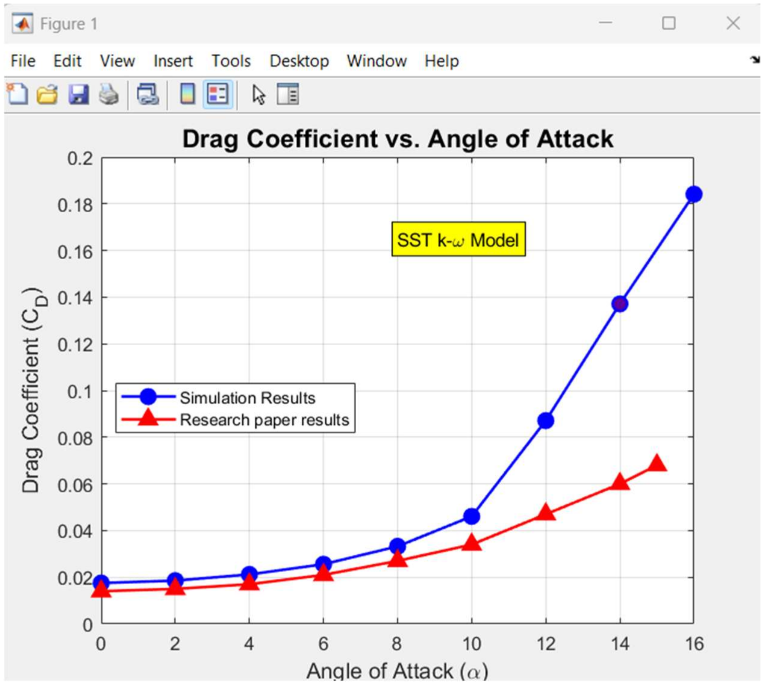

Figure 113 - SST KW model

6 Discussion

- **6.1 Accuracy of the Results**

The accuracy of the results was evaluated by comparing the simulation data against the findings of a reference research paper for both the Standard k-e and SST k-w models.

- **Standard KE Model:** The C_L vs $\propto$ plot (Figure 108) shows that the simulation results capture the linear trend of the research paper well. However, the simulation consistently over-predicts the lift coefficient. Furthermore, it predicts a delayed stall, with a C_{l_Max} of 1.53 at angle 14, whereas the reference paper indicates an earlier stall. The drag coefficient comparison (Figure 110) also shows a similar trend, but the simulation predicts significantly different drag values, especially at higher angles of attack.

Table 11 - Error percentages of lift coefficient for Standard kE model

Angle of Attack (α)	Simulation Results (C_l)	Research Paper Results (C_l)	Percent Error (%)
0	0.41647076	0.37	12.56%
2	0.62731829	0.56	12.02%
4	0.83717901	0.76	10.16%
6	1.0370278	0.93	11.51%
8	1.2233617	1.08	13.27%
10	1.381645	1.18	17.09%
12	1.4947617	1.21	23.53%
14	1.5263049	1.21	26.14%
15/16	1.4056102	1.19	18.12%

Table 12 - Error percentages of Drag coefficient for Standard kE model

AOA	C_d simulation	C_d research paper	Error (%)
0	0.017567	0.035	49.807509
2	0.018464	0.037	50.096197
4	0.020778	0.044	52.777411
6	0.025093	0.056	55.191252
8	0.032214	0.073	55.871499
10	0.041706	0.094	55.631398
12	0.056402	0.119	52.603494
14	0.078582	0.14	43.869814

15/16	0.114922	0.15	23.3853

- **SST KW Model:** The SST model (Figure 109) provided results that aligned more closely with the reference paper, particularly in predicting the stall. The simulation captured the stall onset after angle 10 and a C_l max of approximately 1.19, which is much closer to the expected aerodynamic behavior than the KE model's prediction. The drag coefficient (Figure 111) also showed a better correlation, capturing the sharp increase in drag that corresponds with the lift stall. However, at higher angles of attack (specifically after 12°), the simulation predicts a significantly higher drag coefficient than the values reported in the research paper.

Table 13 - Error percentages of lift coefficient for SST kw model

AOA	C_l simulation	C_l research paper	Error (%)
0	0.376753	0.37	1.825014
2	0.573826	0.56	2.46885
4	0.766208	0.75	2.161103
6	0.944725	0.92	2.687533
8	1.09451	0.105	4.239076
10	1.185169	1.14	3.962211
12	1.046456	1.24	15.608379
14	0.914563	1.28	28.549729
15/16	0.801763	1.26	36.367997

Table 14 - Error percentages of drag coefficient for SST kw model

AOA	C_d simulation	C_d research paper	Error (%)
0	0.017523	0.014	25.165521
2	0.018549	0.015	23.6607
4	0.021197	0.017	24.687806
6	0.025566	0.021	21.744781
8	0.03324	0.027	23.111822
10	0.046065	0.034	35.484376
12	0.087114	0.047	85.348036
14	0.137114	0.06	128.52268
15/16	0.18411	0.068	170.7496

- **6.2 Possible Errors in the Simulation and Reductions**

Several factors may have contributed to discrepancies between the simulation and reference data:

The deviations observed between the current simulation and the reference paper, particularly at higher angles of attack, can likely be attributed to differences in mesh configuration and near-wall treatment. The reference study successfully utilized a mesh of approximately 105,000 elements, whereas the present simulation employed a finer mesh of 210,000 cells, which was selected after performing a grid independence analysis.

This discrepancy in cell count, even with a grid-independent solution, suggests the reference paper may have employed a highly optimized mesh with a more efficient refinement strategy, concentrating cells in critical areas like the leading edge and boundary layer. Although the current mesh is denser, any differences in element distribution or quality (such as skewness or aspect ratio) compared to the reference study could still introduce small variations in the results.

A more significant factor is likely the difference in near-wall treatment. The reference paper reported using y+ values within the 30-60 range, which is the target region for standard wall functions. In the current study, while the y+ was targeted at near 30, it occasionally dropped to values of 27 or 28. This is a critical distinction, as y+ values below 30 fall into the "buffer layer," where standard wall functions are known to lose accuracy. This inconsistency in boundary layer resolution is a likely source of the discrepancies in the predicted aerodynamic coefficients.

To reduce these errors, the mesh could be further refined. More importantly, the first layer height should be carefully adjusted to ensure the y+ value remains consistently within the 30-60 range across all angles of attack, fully aligning with the reference paper's validated methodology. This would ensure a more accurate and consistent application of the wall function.

6.3 Difficulties Faced

- **Convergence at High AOA:** The most significant challenge was achieving a stable, converged solution at high angles of attack (12° to 16°). The convergence plots for both lift and drag showed significant oscillations (e.g., Figures 46, 47, 64, 65). This is not a numerical error but a physical reality: the flow in the stall region is inherently unsteady (transient). Forcing a steady-state solver to find a single "stable" solution for this phenomenon was difficult and time-consuming.

- **Near-Wall Meshing:** As noted in the report, applying an inflation layer was not possible with the chosen face meshing strategy. A bias factor had to be used instead. Manually tuning this bias to achieve the desired y+ range (30-60) across the entire airfoil surface was an iterative and non-trivial process.

6.4 What Can Be Learned from the Simulation Results?

- **Turbulence Model Selection Involves Trade-offs:** This study clearly demonstrated that the choice of turbulence model is critical and involves trade-offs. The standard k-ε model, for instance, provided a better prediction of the lift coefficient in the linear range but significantly over-predicted the drag coefficient.

- **SST k-ω Model Nuances**: The k-ω SST model offered a more physically realistic package, but with its own set of deviations. It predicted the drag coefficient's trend more accurately than the k-ε model. However, it also predicted an early stall, with C_l_max occurring at a lower angle of attack than shown in the reference paper.

- **Mesh Quality and y+ Consistency Outweigh Cell Count:** A primary lesson was that mesh quality is more influential than mere quantity. Despite the current simulation using a denser mesh (210,000 cells) than the reference paper (105,000 cells), discrepancies remained. This was potentially due to the critical y+ value. While the simulation aimed for a y+ near 30 (appropriate for wall functions), it was observed in a few simulation cases to drop to values of 27-28. This dip into the inaccurate buffer layer, even if infrequent, highlights the high sensitivity of wall functions and could be a source of the observed deviations.

- **Grid Independence is a Prerequisite for Trust:** The project reinforces the necessity of a grid independence study. Performing this analysis provided confidence that the 210,000-element mesh was sufficient and that the solution was not an artifact of the grid. This step is essential for ensuring that the final results are a reliable representation of the flow physics, not just numerical error.

- **Visualizing Flow Physics Confirms the Data:** The simulation was powerful for connecting quantitative data to physical phenomena. By correlating the C_l plots (which showed the stall) with the velocity and pressure contours at high angles of attack, the underlying physics became clear. The massive flow separation seen on the airfoil's upper surface in the contours provided a direct visual explanation for the C_l_max and the subsequent drop in lift.

- **Reproducibility in CFD is Highly Sensitive**: This reproduction attempt highlighted how sensitive CFD solutions are to the fine details of the setup. It demonstrated that slight variations in meshing strategy, y+ values, and turbulence model choice can lead to significant, and sometimes complex, differences in predicted forces. This underscores the challenge of verification and the importance of precise setup

7 Suggestions for Future Improvement

One of the key improvements for future work is the refinement of the turbulence model application. The current study confirmed that the k-ω SST (Shear Stress Transport) model is more appropriate for this aerodynamic application than the standard k-ε model, particularly in capturing stall. However, the simulation still relied on standard wall functions, targeting a y+ value near 30. This approach proved problematic, as y+ values sometimes fell into the 27-28 range, leading to known inaccuracies in the buffer layer. A more robust and high-fidelity approach would be to fully resolve the viscous sublayer. This would involve creating a significantly finer near-wall mesh to ensure the y+ value is consistently at or below 1. Although this method demands substantially higher computational resources, it allows the k-ω SST model to operate at its full potential, bypassing wall functions entirely. This would provide a much more accurate prediction of near-wall turbulence, skin friction, and flow separation, which is critical for correctly modeling both lift and drag, especially at high angles of attack where discrepancies were largest.

Another significant improvement relates to the simulation of the angle of attack. In the current study, a C-type domain was used, and the angle of attack was set by decomposing the inlet velocity vector into x and y components. This method is computationally efficient but can introduce non-physical flow behavior, particularly at high angles. For example, flow may be observed exiting the domain through the upper inlet boundary instead of entering, which compromises the solution's realism. A more physically accurate and reliable approach would be to rotate the airfoil geometry itself relative to a uniform, fixed-direction inlet flow. This would require parameterizing the angle of attack and generating a new, high-quality mesh for each angle simulated. While more complex, this method ensures that the flow physics, wake development, and boundary interactions are captured correctly, leading to a more dependable and accurate set of results across the entire flight envelope.

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DMS link: https://dms.uom.lk/s/SAbZx3JsfEaZecr